>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

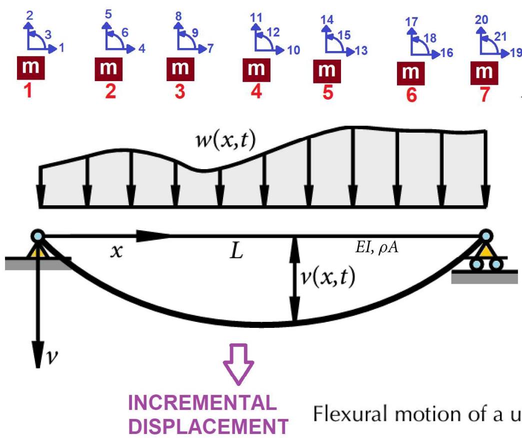
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL A SIMPLY SUPPORTED BEAM ELEMENT: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

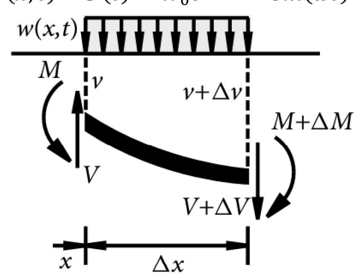
$$W(x,t) = P(t) = W_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

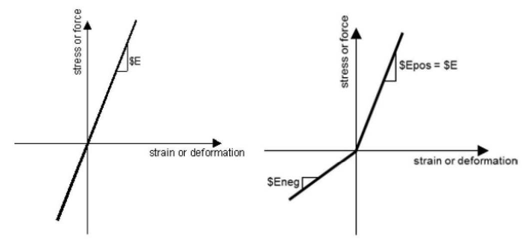
$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$



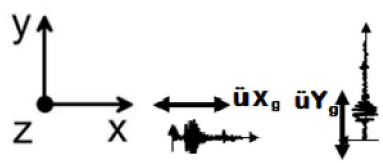
$$W(x,t) = P(t) = W_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

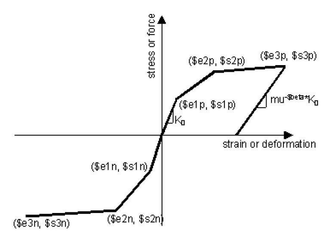


Flexural motion of a uniform elastic beam



ELEMENT ELASTIC MATERIAL



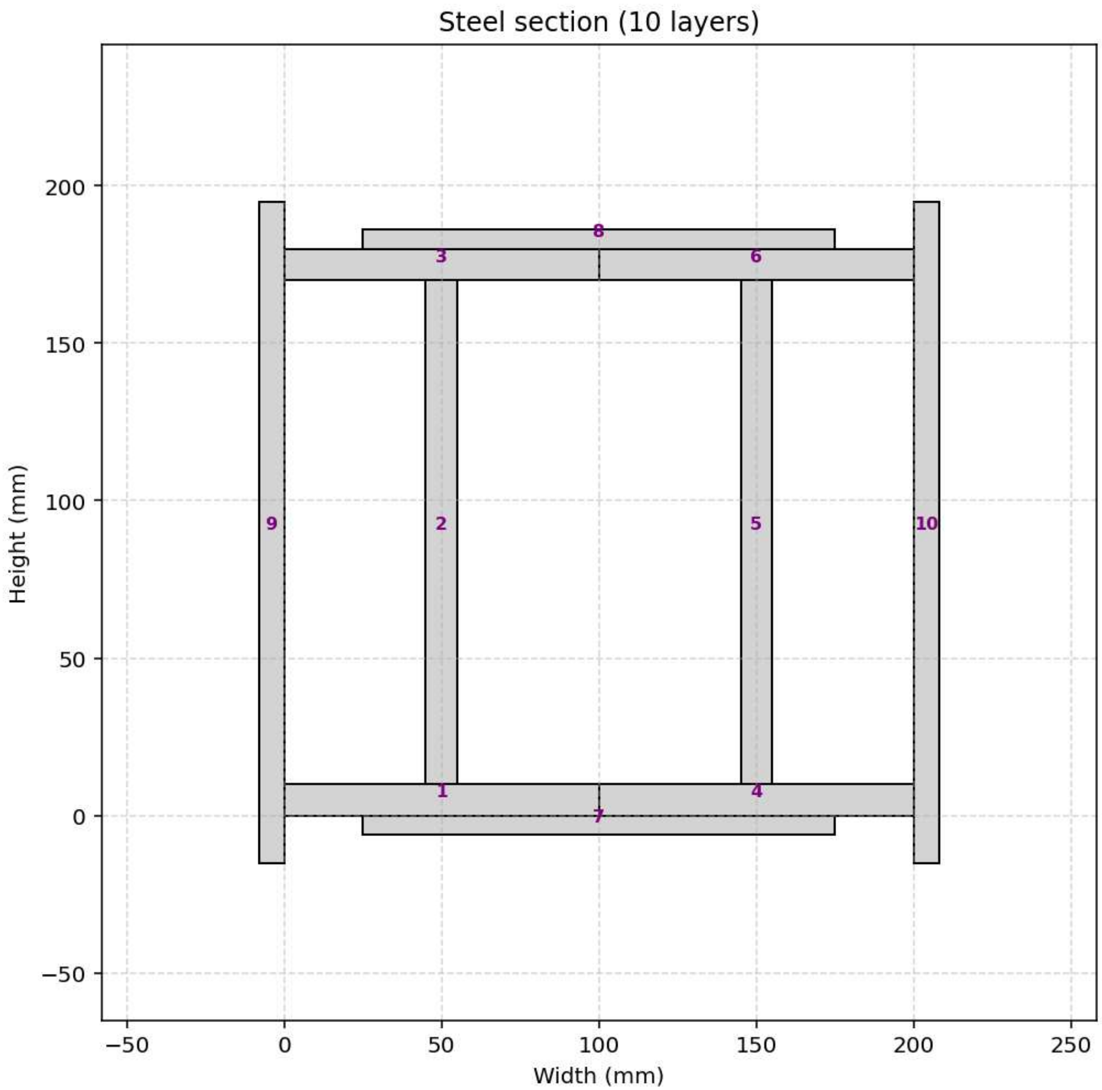


ELEMENT INELASTIC MATERIAL

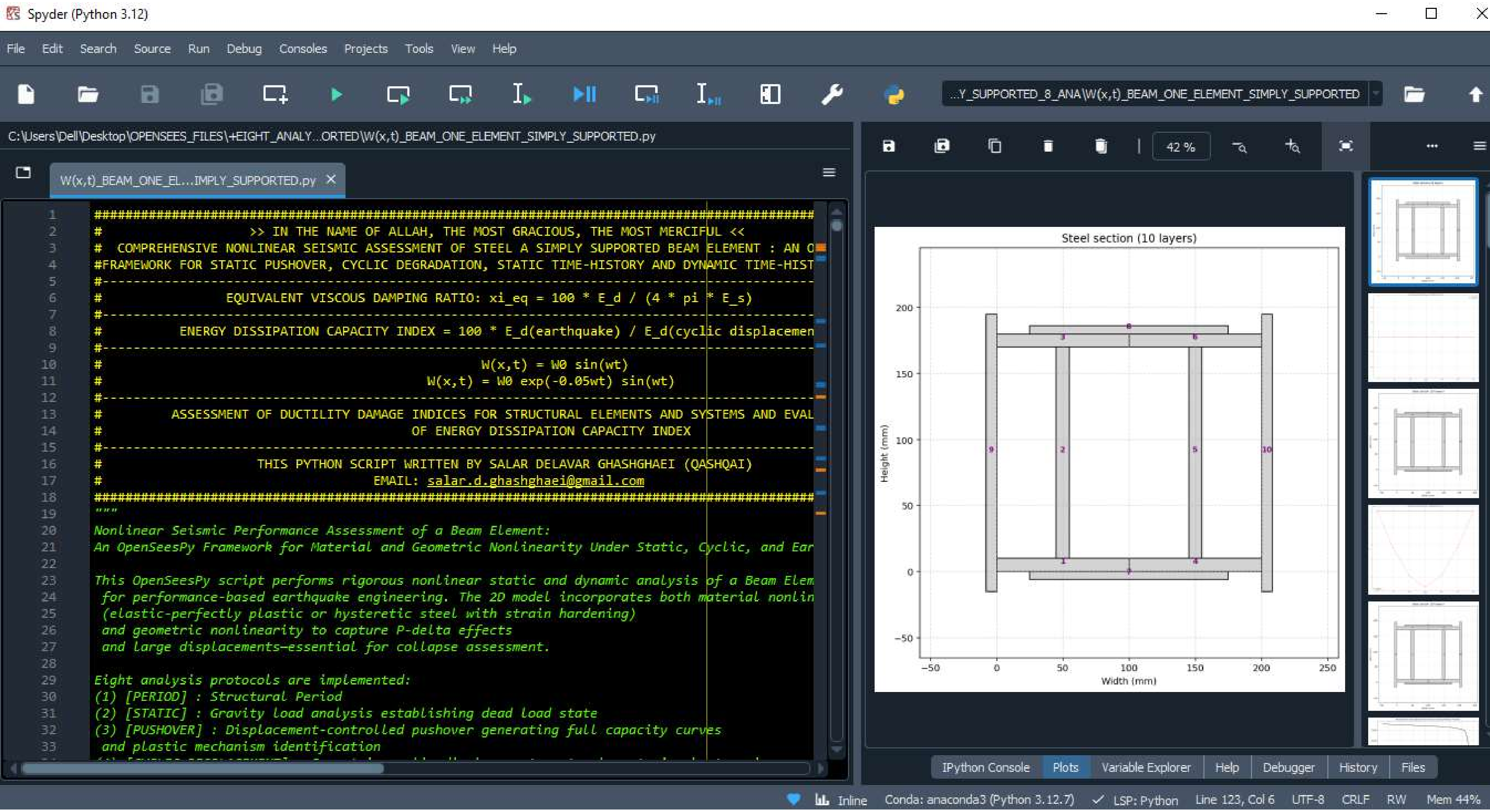
$$\rho A \frac{\partial^2 v}{\partial t^2} + EI \frac{\partial^4 v}{\partial x^4} = w(x,t).$$

boundary conditions

$$v(0,t) = v(L,t) = 0, \quad \left.\frac{\partial^2 v(x,t)}{\partial x^2}\right|_{x=0} = \left.\frac{\partial^2 v(x,t)}{\partial x^2}\right|_{x=L} = 0.$$



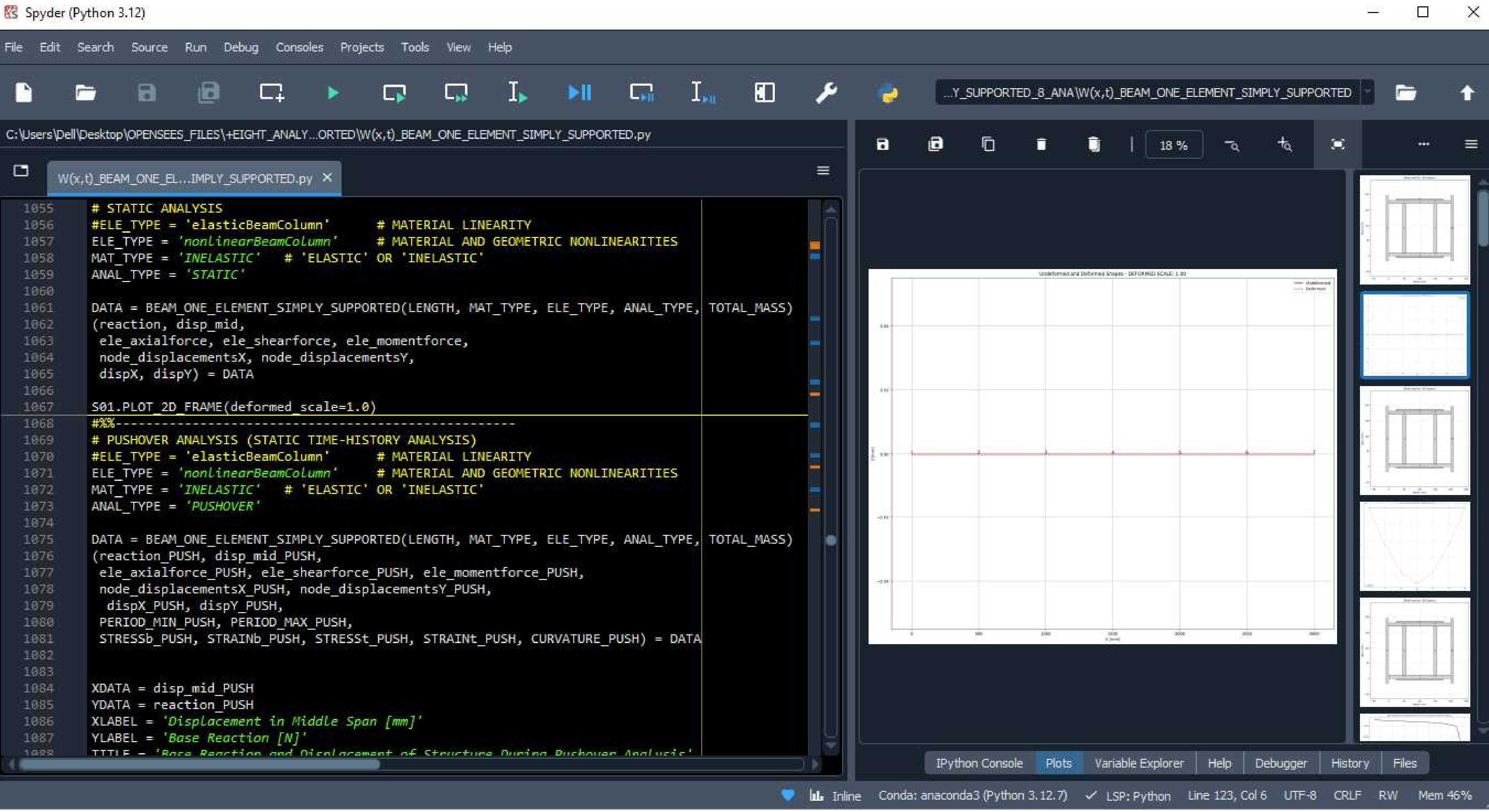
Steel section (10 layers)
Height (mm)
Width (mm)
200
150
100
50
0
−50
−50
0
50
100
150
200
250
1
2
3
4
5
6
7
8
9
10

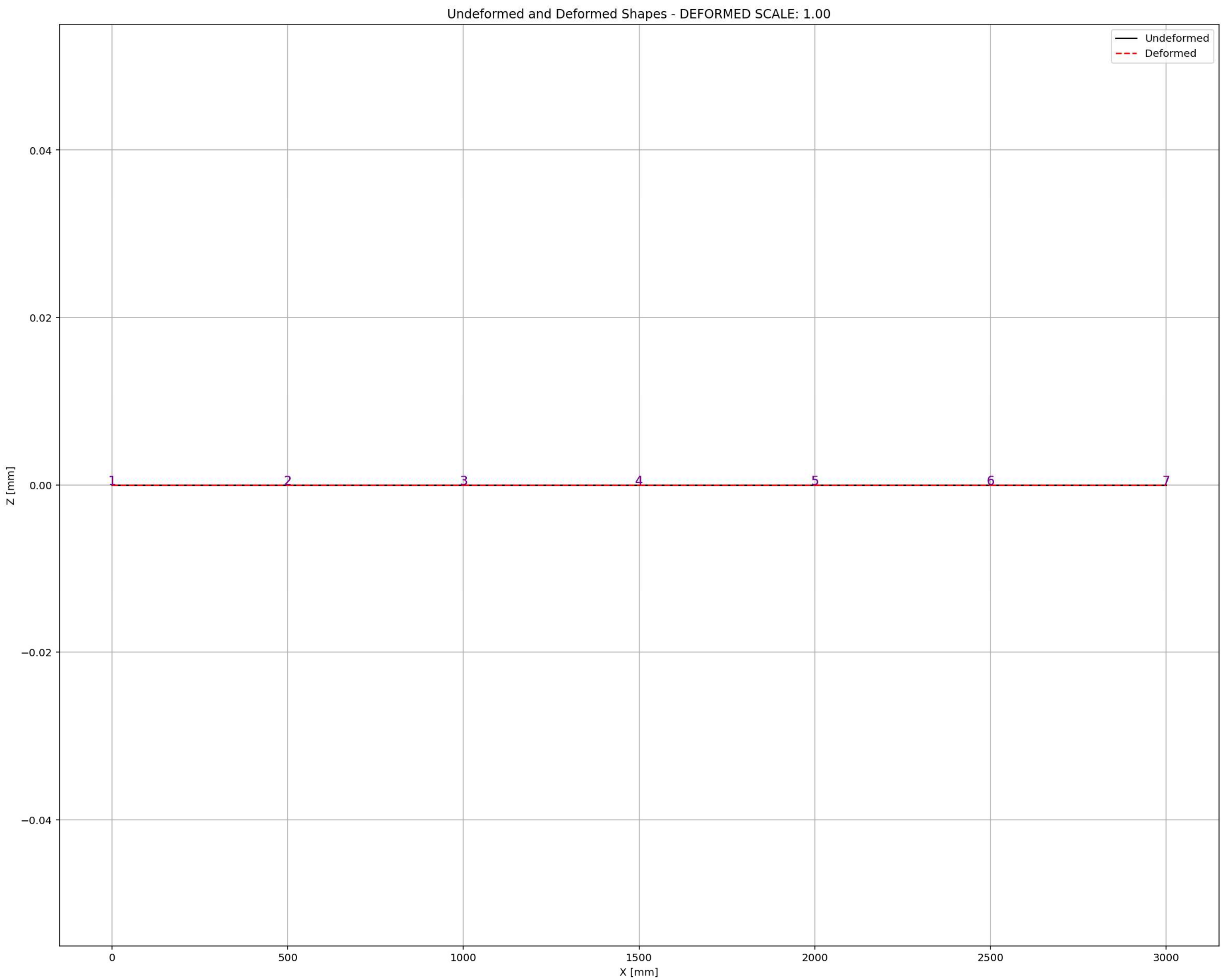

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...Y_SUPPORTED_8_ANA\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...ORTED\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED.py
W(x,t)_BEAM_ONE_EL...IMPLY_SUPPORTED.py
##
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL A SIMPLY SUPPORTED BEAM ELEMENT : AN O
#FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HIST
#--
EQUIVALENT VISCOUS DAMPING RATIO: xi_eq = 100 * E_d / (4 * pi * E_s)
#--
ENERGY DISSIPATION CAPACITY INDEX = 100 * E_d(earthquake) / E_d(cyclic displacemen
#--
W(x,t) = W0 sin(wt)
W(x,t) = W0 exp(-0.05wt) sin(wt)
#--
ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVAL
OF ENERGY DISSIPATION CAPACITY INDEX
#--
THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
EMAIL: salar.d.ghashghaei@gmail.com
##
"""
Nonlinear Seismic Performance Assessment of a Beam Element:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Ear

This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a Beam Elem
 for performance-based earthquake engineering. The 2D model incorporates both material nonlin
 (elastic-perfectly plastic or hysteretic steel with strain hardening)
 and geometric nonlinearity to capture P-delta effects
 and large displacements—essential for collapse assessment.

Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
(2) [STATIC] : Gravity load analysis establishing dead load state
(3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
 and plastic mechanism identification
42 %
Steel section (10 layers)
Height (mm)
Width (mm)
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 123, Col 6 UTF-8 CRLF RW Mem 44%

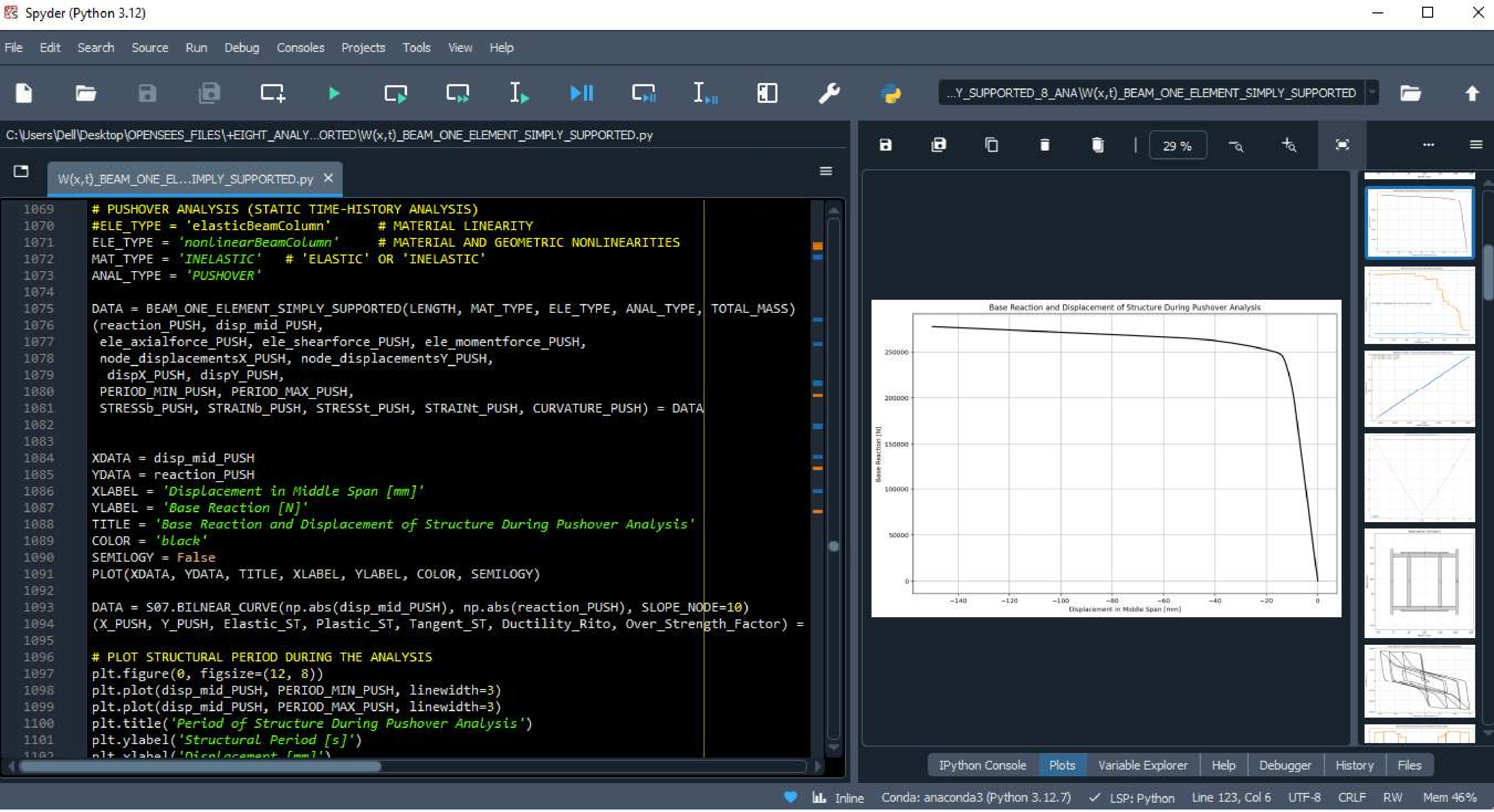
STATIC ANALYSIS

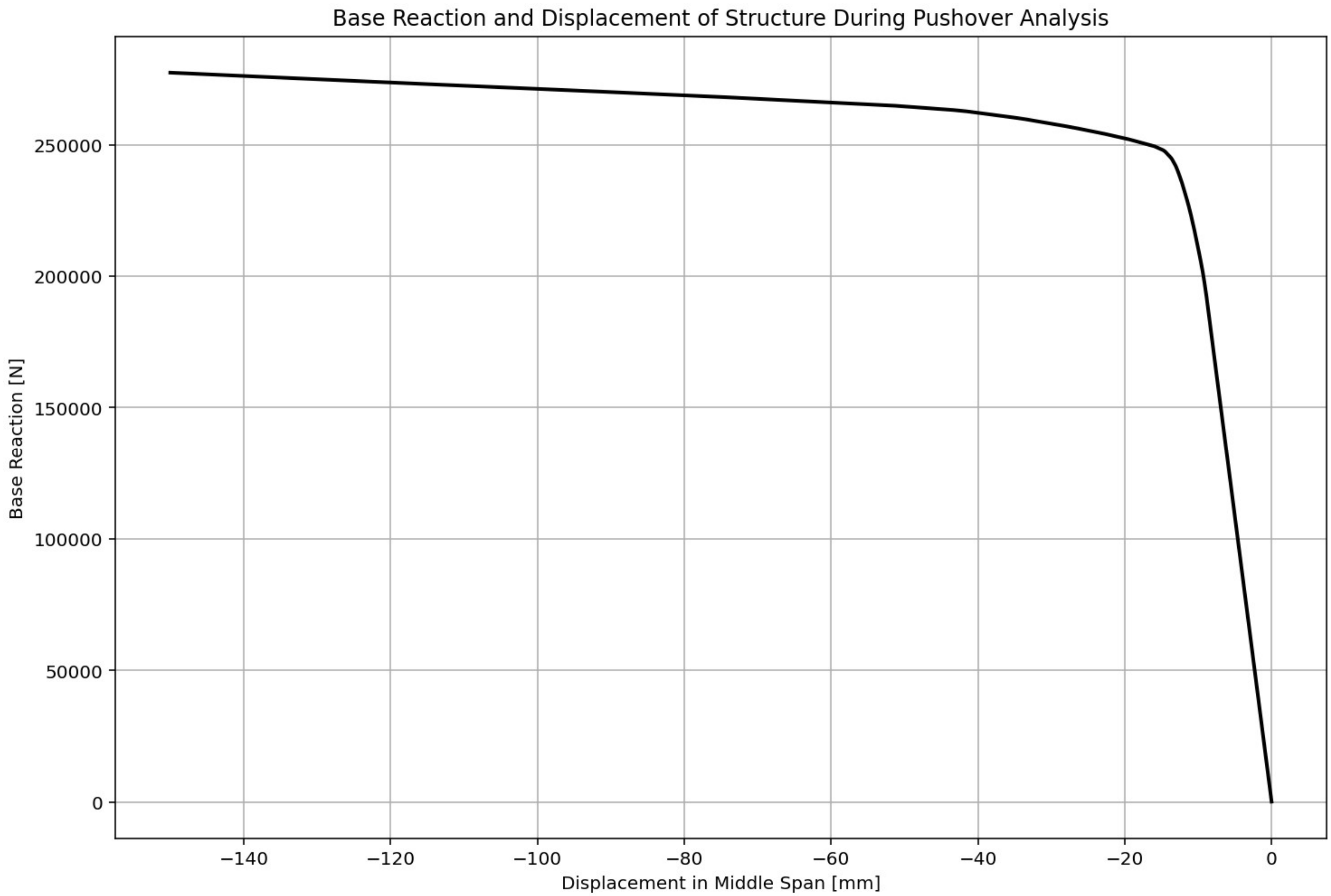

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...Y_SUPPORTED_8_ANA\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...ORTED\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED.py
W(x,t)_BEAM_ONE_EL...IMPLY_SUPPORTED.py
STATIC ANALYSIS
#ELE_TYPE = 'elasticBeamColumn'   # MATERIAL LINEARITY
ELE_TYPE = 'nonlinearBeamColumn'   # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC'
DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(reaction, disp_mid,
 ele_axialforce, ele_shearforce, ele_momentforce,
 node_displacementsX, node_displacementsY,
 dispX, dispY) = DATA
S01.PLOT_2D_FRAME(deformed_scale=1.0)
#%%--
PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'elasticBeamColumn'   # MATERIAL LINEARITY
ELE_TYPE = 'nonlinearBeamColumn'   # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'PUSHOVER'
DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(reaction_PUSH, disp_mid_PUSH,
 ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
 node_displacementsX_PUSH, node_displacementsY_PUSH,
 dispX_PUSH, dispY_PUSH,
 PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
 STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINt_PUSH, CURVATURE_PUSH) = DATA
XDATA = disp_mid_PUSH
YDATA = reaction_PUSH
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
18 %
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 123, Col 6 UTF-8 CRLF RW Mem 46%



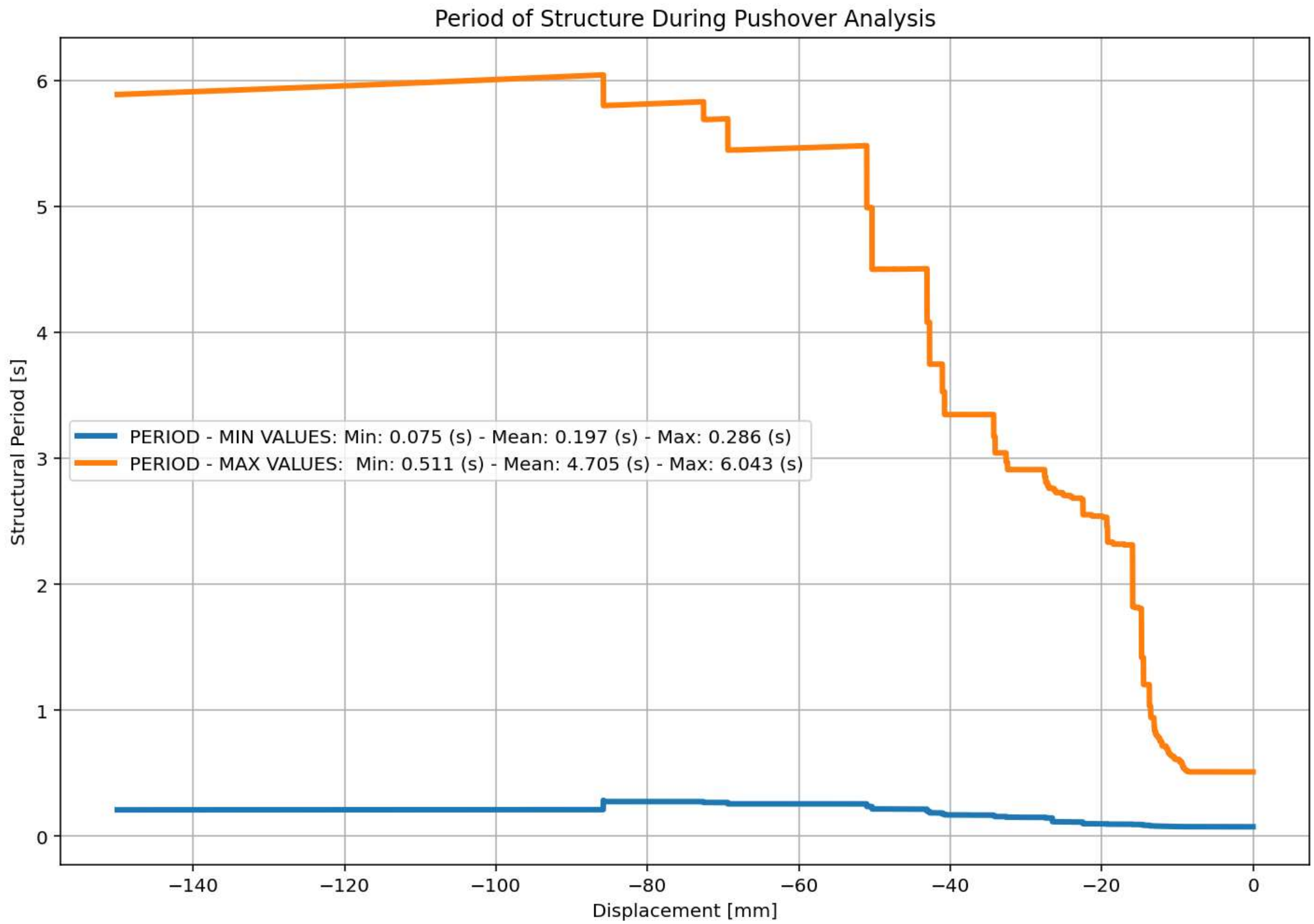
Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0.04
0.02
0.00
−0.02
−0.04
0
500
1000
1500
2000
2500
3000
1
2
3
4
5
6
7

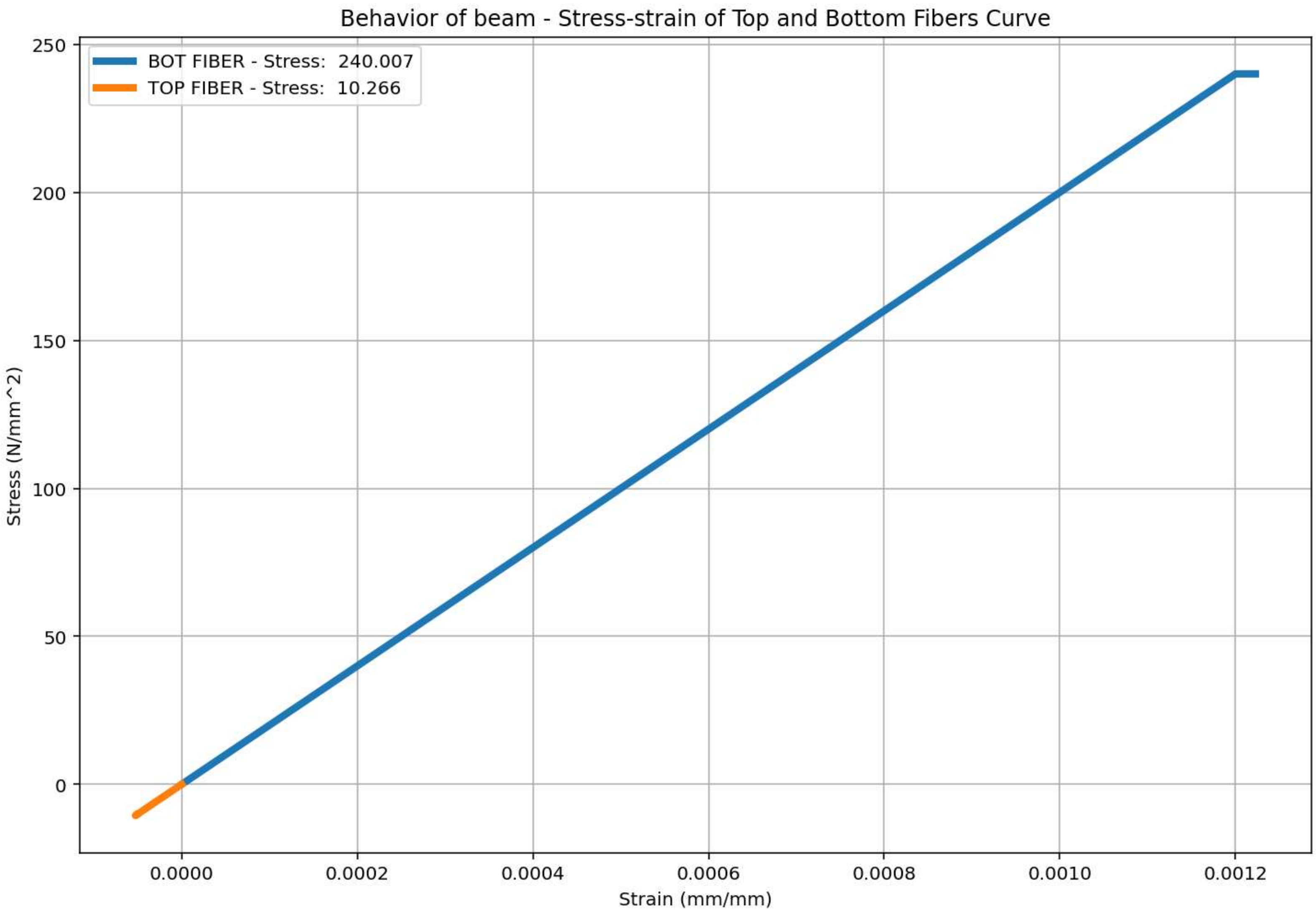
PUSHOVER ANALYSIS


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...Y_SUPPORTED_8_ANA\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...ORTED\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED.py
W(x,t)_BEAM_ONE_EL...IMPLY_SUPPORTED.py
PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'elasticBeamColumn'   # MATERIAL LINEARITY
ELE_TYPE = 'nonlinearBeamColumn'   # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'PUSHOVER'
DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(reaction_PUSH, disp_mid_PUSH,
 ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
 node_displacementsX_PUSH, node_displacementsY_PUSH,
  dispX_PUSH, dispY_PUSH,
 PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
 STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINt_PUSH, CURVATURE_PUSH) = DATA
XDATA = disp_mid_PUSH
YDATA = reaction_PUSH
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
DATA = S07.BILNEAR_CURVE(np.abs(disp_mid_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
(X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) =
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_PUSH, PERIOD_MIN_PUSH, linewidth=3)
plt.plot(disp_mid_PUSH, PERIOD_MAX_PUSH, linewidth=3)
plt.title('Period of Structure During Pushover Analysis')
plt.ylabel('Structural Period [s]')
29 %
Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
250000
200000
150000
100000
50000
0
−140 −120 −100 −80 −60 −40 −20 0
Displacement in Middle Span [mm]
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 123, Col 6 UTF-8 CRLF RW Mem 46%

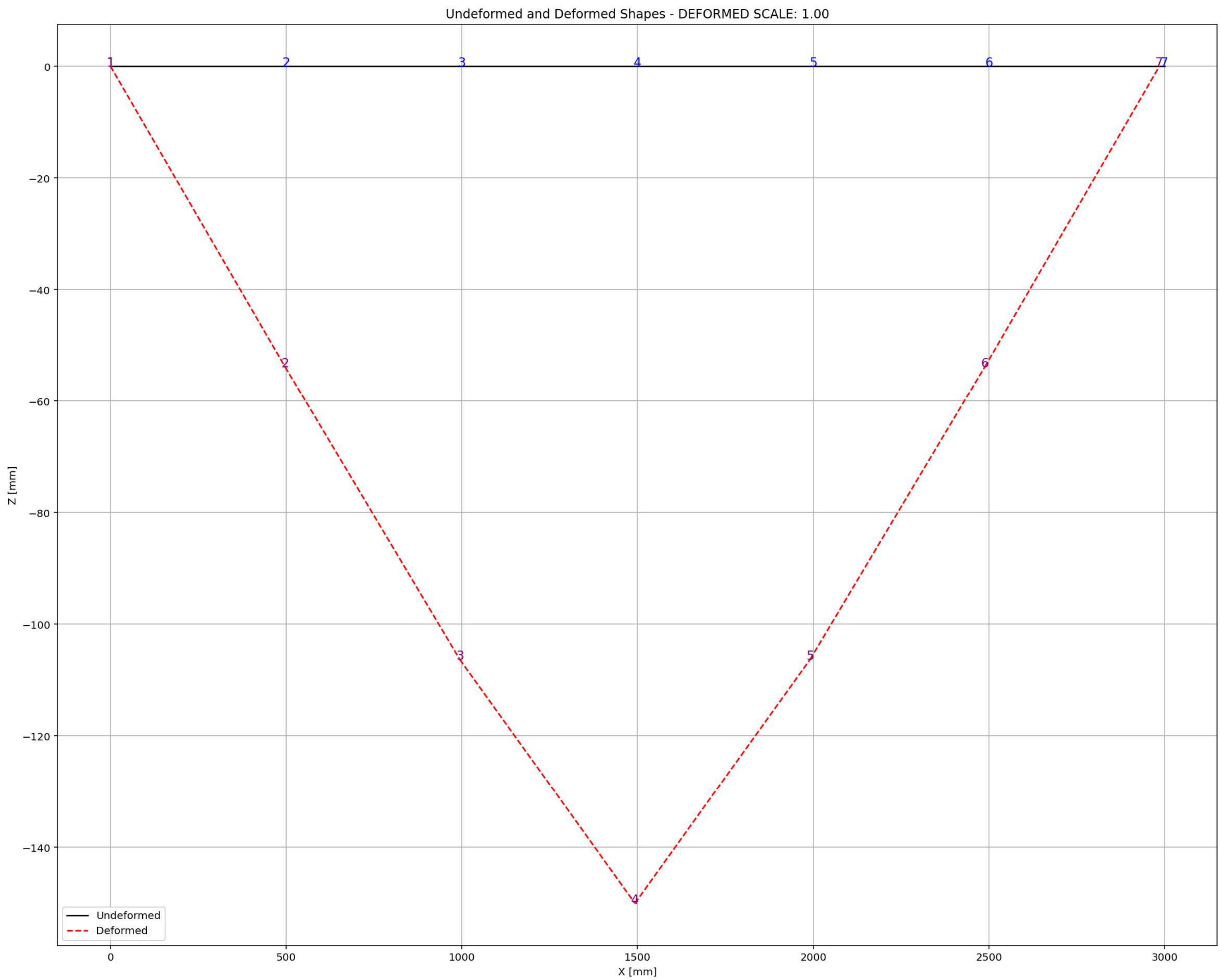


Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
250000
200000
150000
100000
50000
0
−140
−120
−100
−80
−60
−40
−20
0
Displacement in Middle Span [mm]


Period of Structure During Pushover Analysis
Structural Period [s]
Displacement [mm]
PERIOD - MIN VALUES: Min: 0.075 (s) - Mean: 0.197 (s) - Max: 0.286 (s)
PERIOD - MAX VALUES: Min: 0.511 (s) - Mean: 4.705 (s) - Max: 6.043 (s)
0
1
2
3
4
5
6
−140
−120
−100
−80
−60
−40
−20
0

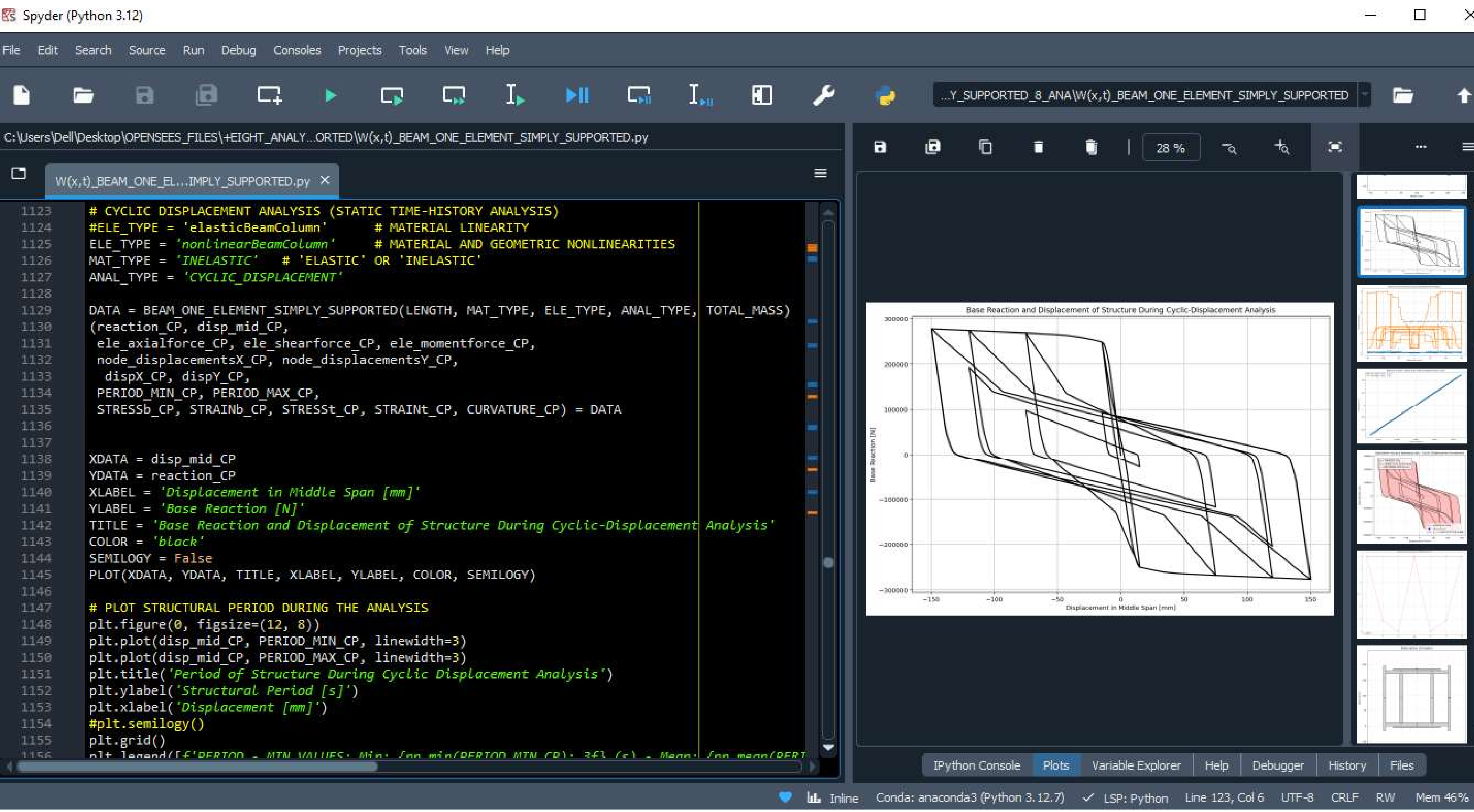


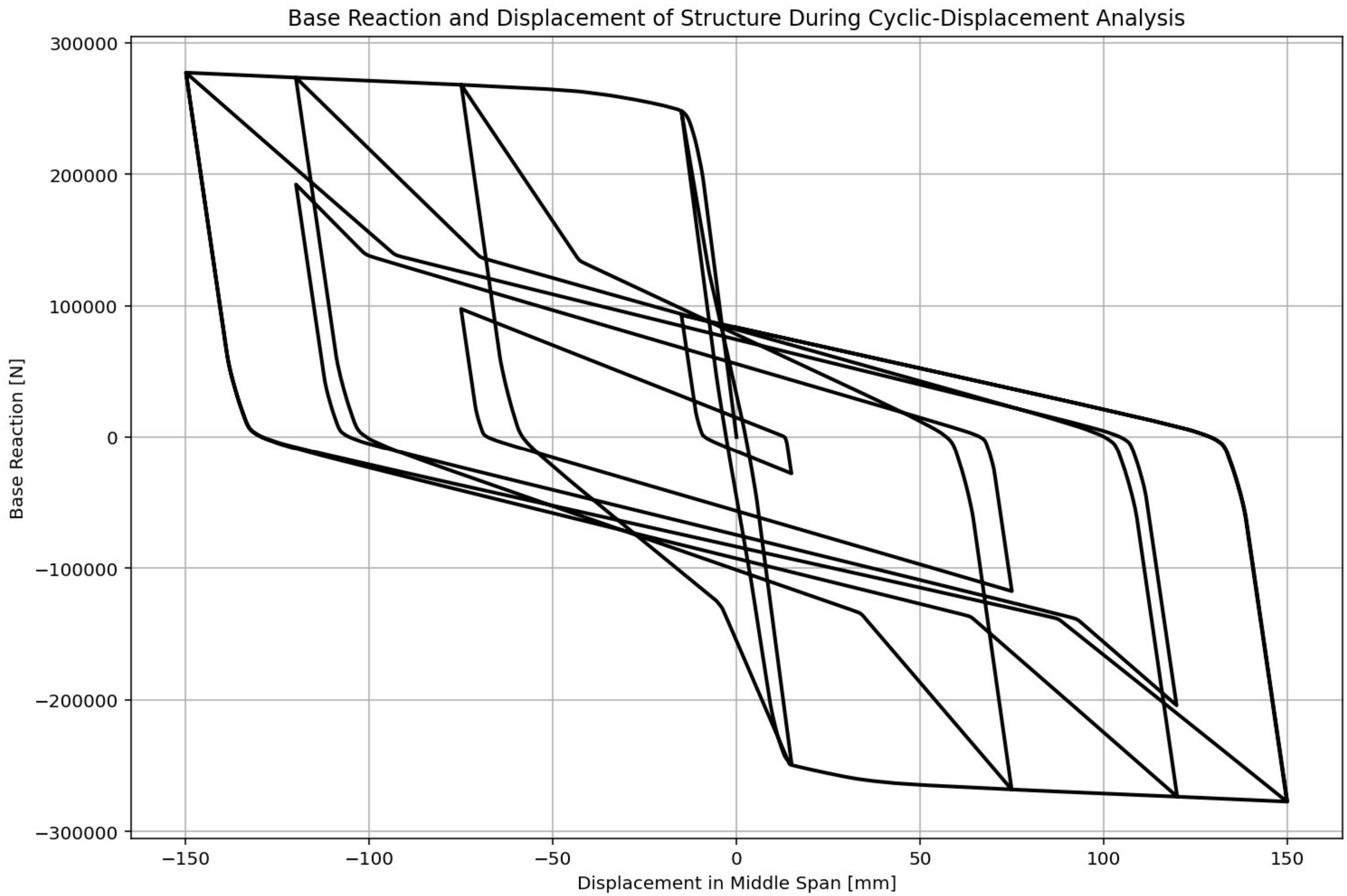
Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 240.007
TOP FIBER - Stress: 10.266
250
200
150
100
50
0
Stress (N/mm^2)
0.0000
0.0002
0.0004
0.0006
0.0008
0.0010
0.0012
Strain (mm/mm)

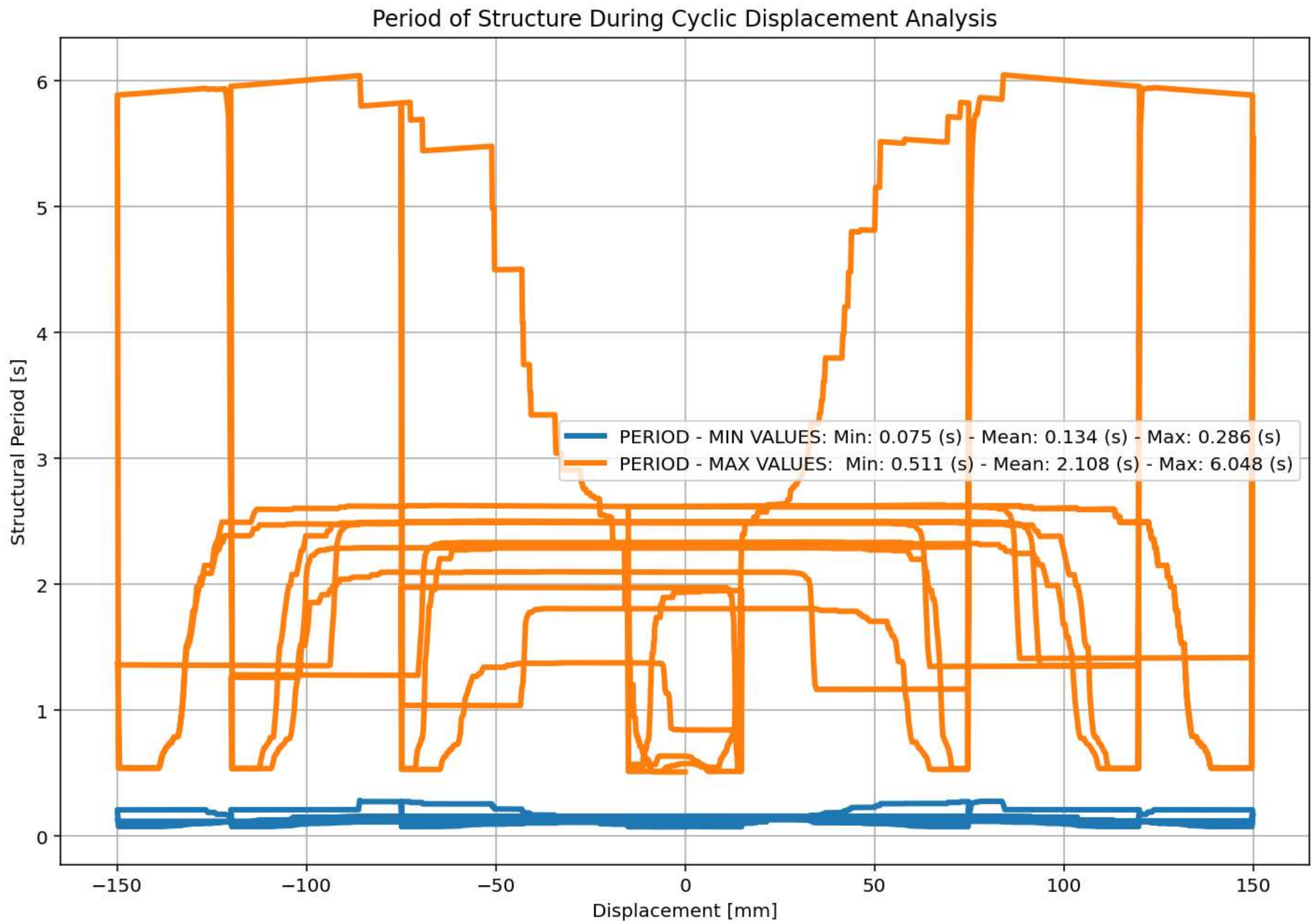


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
−20
−40
−60
−80
−100
−120
−140
0
500
1000
1500
2000
2500
3000
1
2
3
4
5
6
7
Undeformed
Deformed

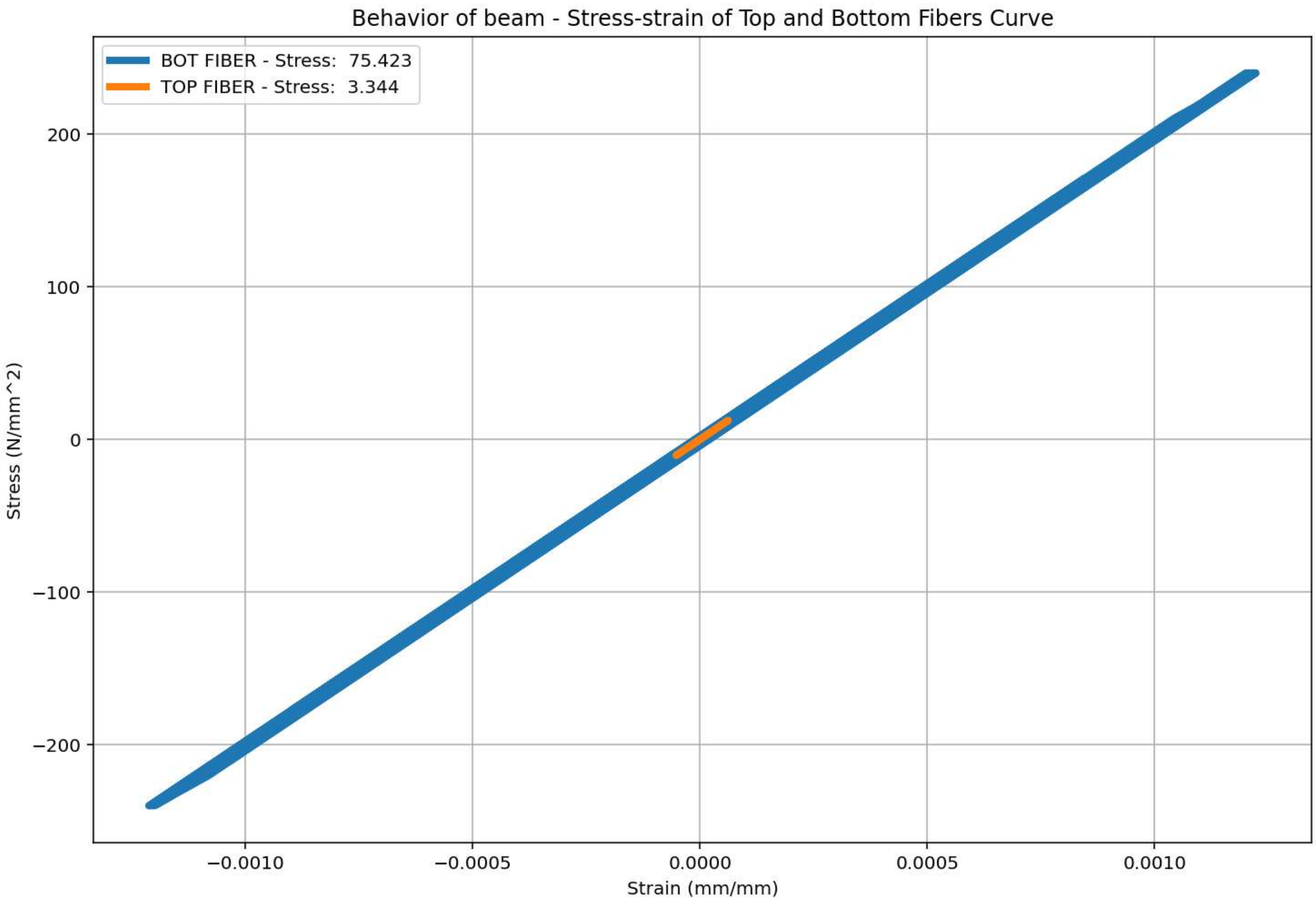
CYCLIC DISPLACEMENT ANALYSIS

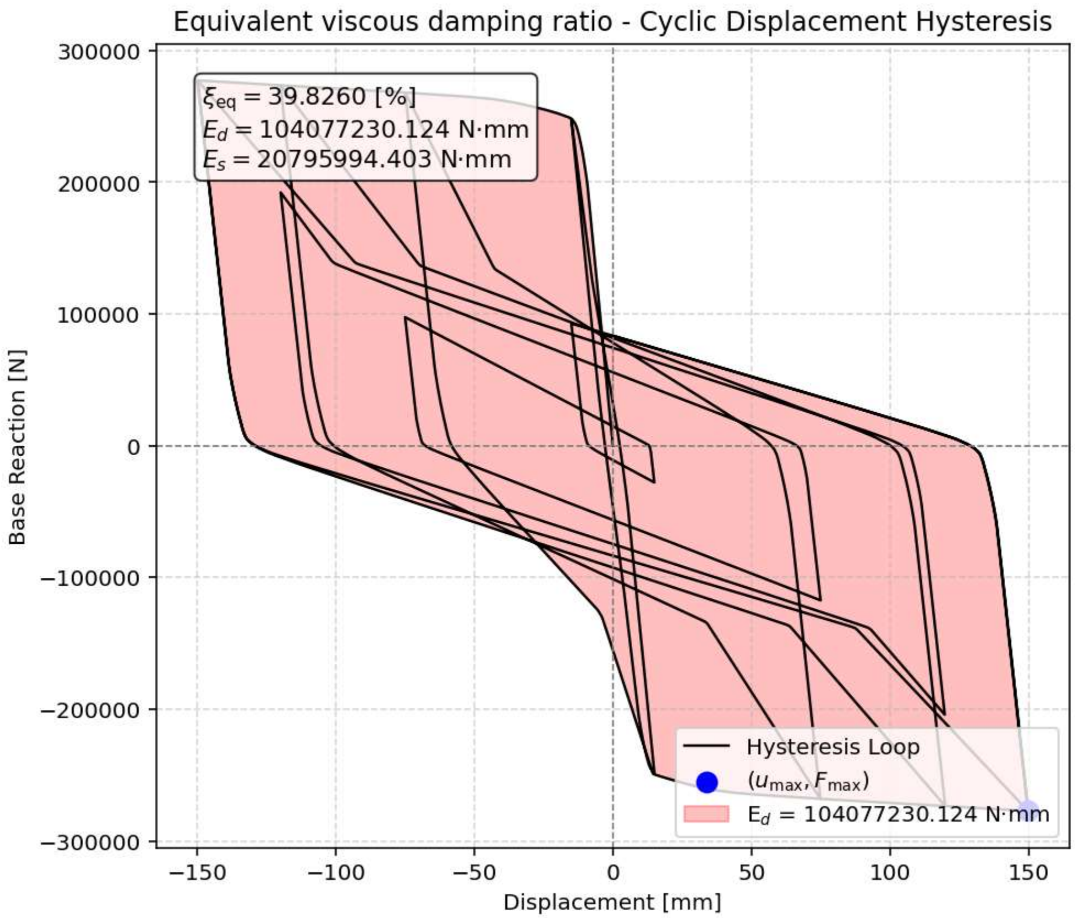

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...Y_SUPPORTED_8_ANA\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...ORTED\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED.py
W(x,t)_BEAM_ONE_EL...IMPLY_SUPPORTED.py
CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'elasticBeamColumn'   # MATERIAL LINEARITY
ELE_TYPE = 'nonlinearBeamColumn'   # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(reaction_CP, disp_mid_CP,
ele_axialforce_CP, ele_shearforce_CP, ele_momentforce_CP,
node_displacementsX_CP, node_displacementsY_CP,
dispX_CP, dispY_CP,
PERIOD_MIN_CP, PERIOD_MAX_CP,
STRESSb_CP, STRAINb_CP, STRESSt_CP, STRAINt_CP, CURVATURE_CP) = DATA
XDATA = disp_mid_CP
YDATA = reaction_CP
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_CP, PERIOD_MIN_CP, linewidth=3)
plt.plot(disp_mid_CP, PERIOD_MAX_CP, linewidth=3)
plt.title('Period of Structure During Cyclic Displacement Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [mm]')
#plt.semilogy()
plt.grid()
28 %
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 123, Col 6 UTF-8 CRLF RW Mem 46%


Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
300000
200000
100000
0
−100000
−200000
−300000
Base Reaction [N]
−150
−100
−50
0
50
100
150
Displacement in Middle Span [mm]

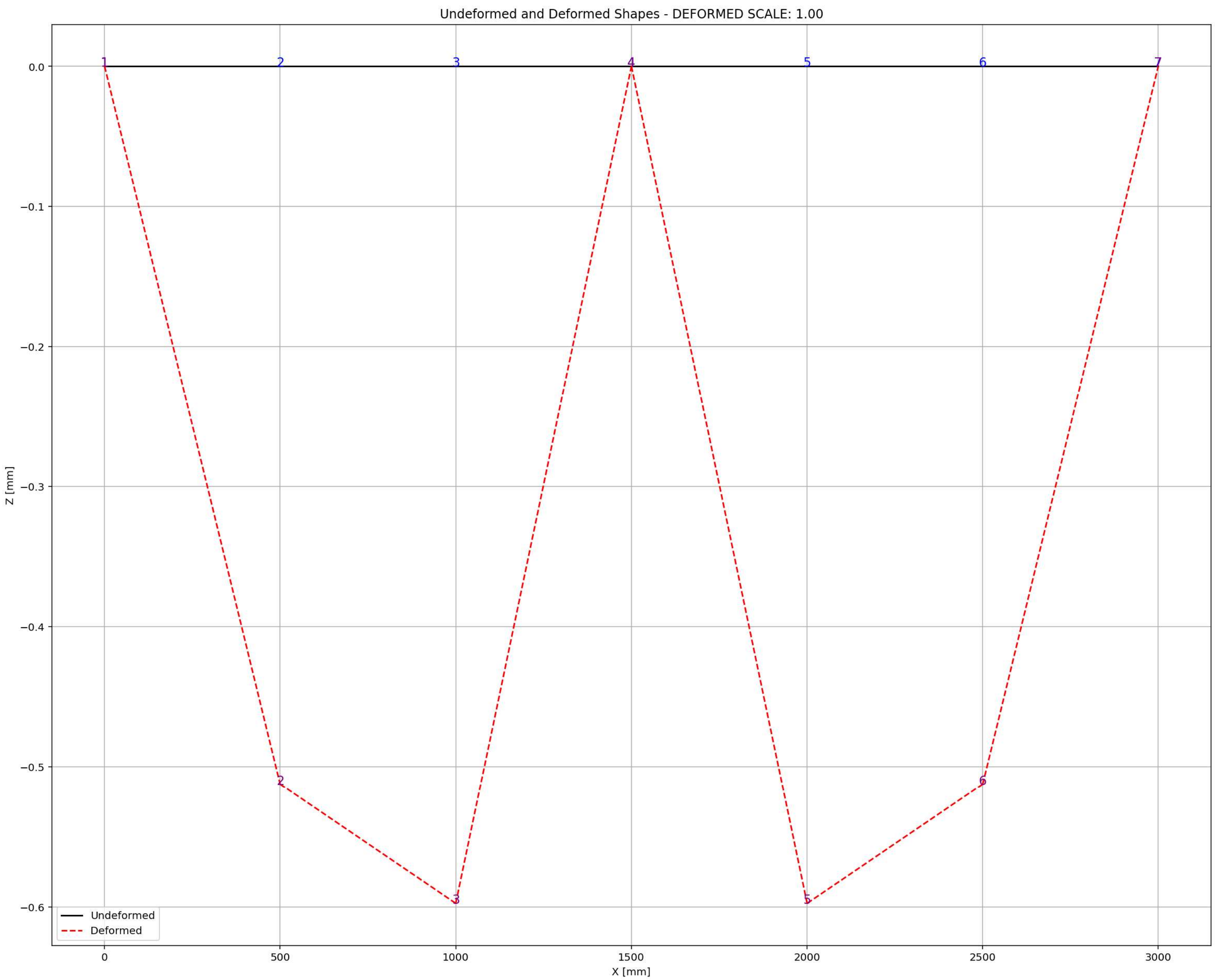


Period of Structure During Cyclic Displacement Analysis
PERIOD - MIN VALUES: Min: 0.075 (s) - Mean: 0.134 (s) - Max: 0.286 (s)
PERIOD - MAX VALUES: Min: 0.511 (s) - Mean: 2.108 (s) - Max: 6.048 (s)
Structural Period [s]
Displacement [mm]
−150
−100
−50
0
50
100
150
0
1
2
3
4
5
6


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 75.423
TOP FIBER - Stress: 3.344
Stress (N/mm^2)
200
100
0
−100
−200
−0.0010
−0.0005
0.0000
0.0005
0.0010
Strain (mm/mm)

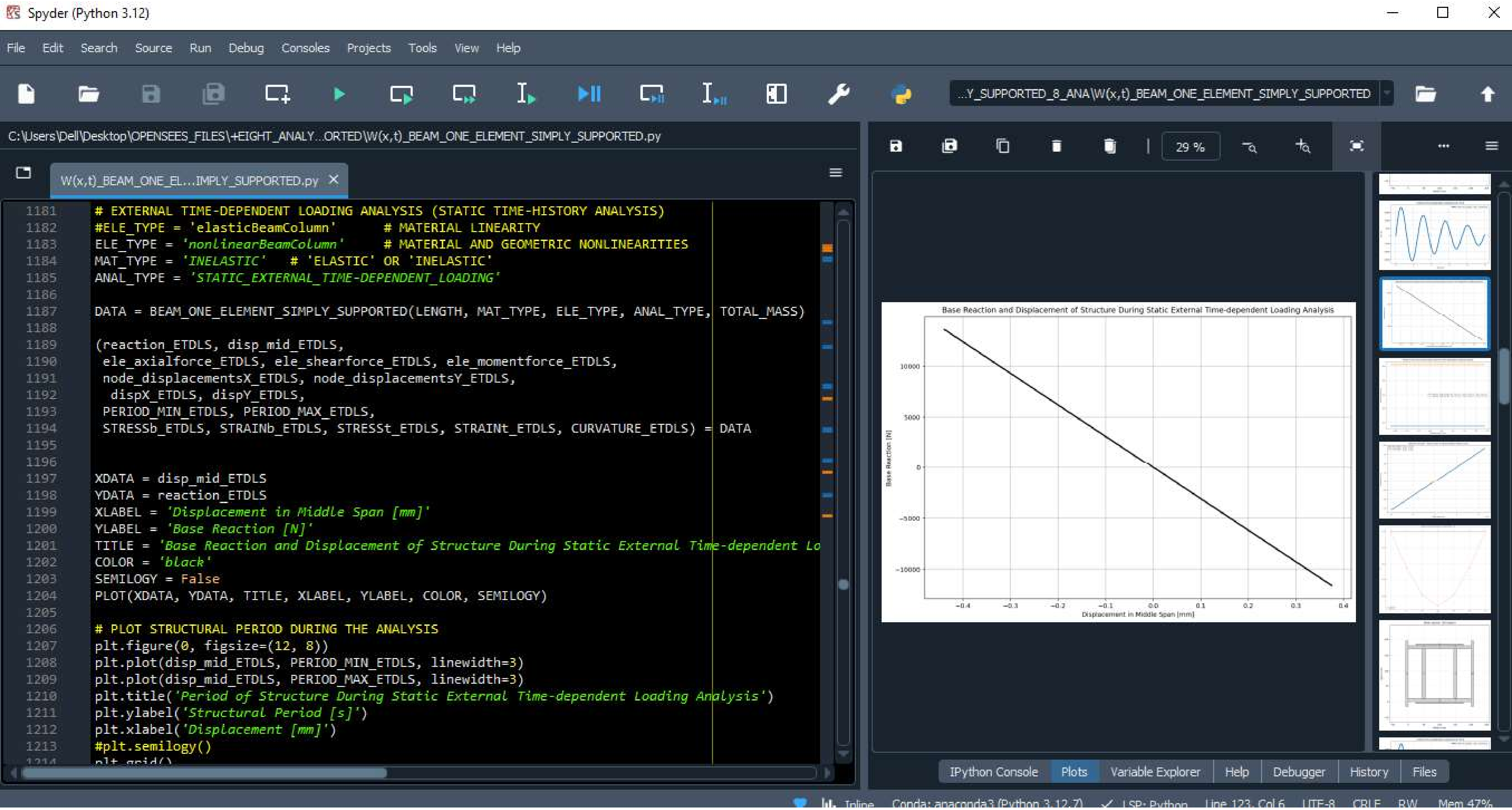


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
$\xi_{eq} = 39.8260$ [%]
$E_d = 104077230.124$ N·mm
$E_s = 20795994.403$ N·mm
Hysteresis Loop
(u_{max}, F_{max})
E_d = 104077230.124 N·mm
Base Reaction [N]
Displacement [mm]
300000
200000
100000
0
−100000
−200000
−300000
−150
−100
−50
0
50
100
150

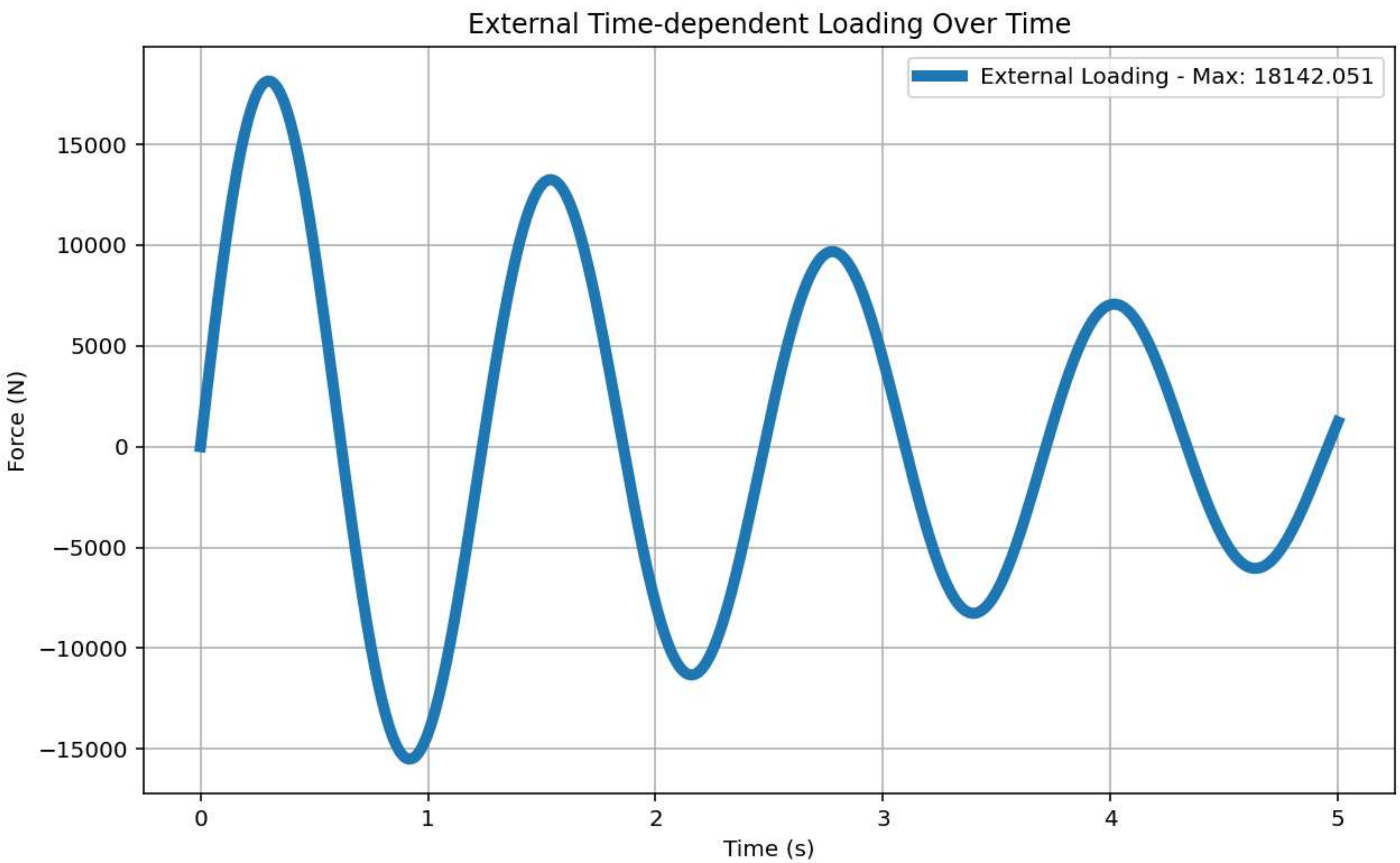


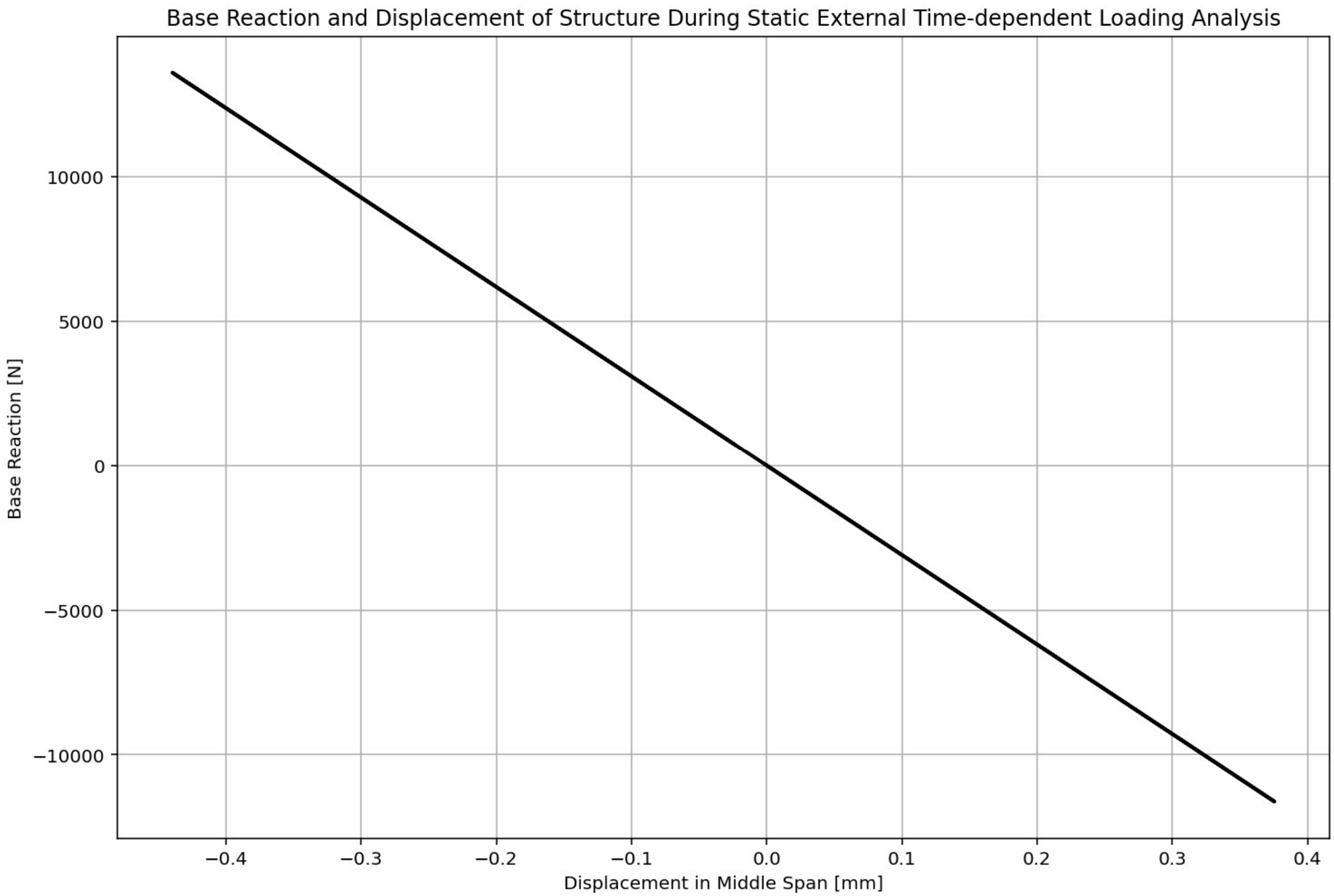
Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
1
2
3
4
5
6
7
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
Z [mm]
0
500
1000
1500
2000
2500
3000
X [mm]
Undeformed
Deformed

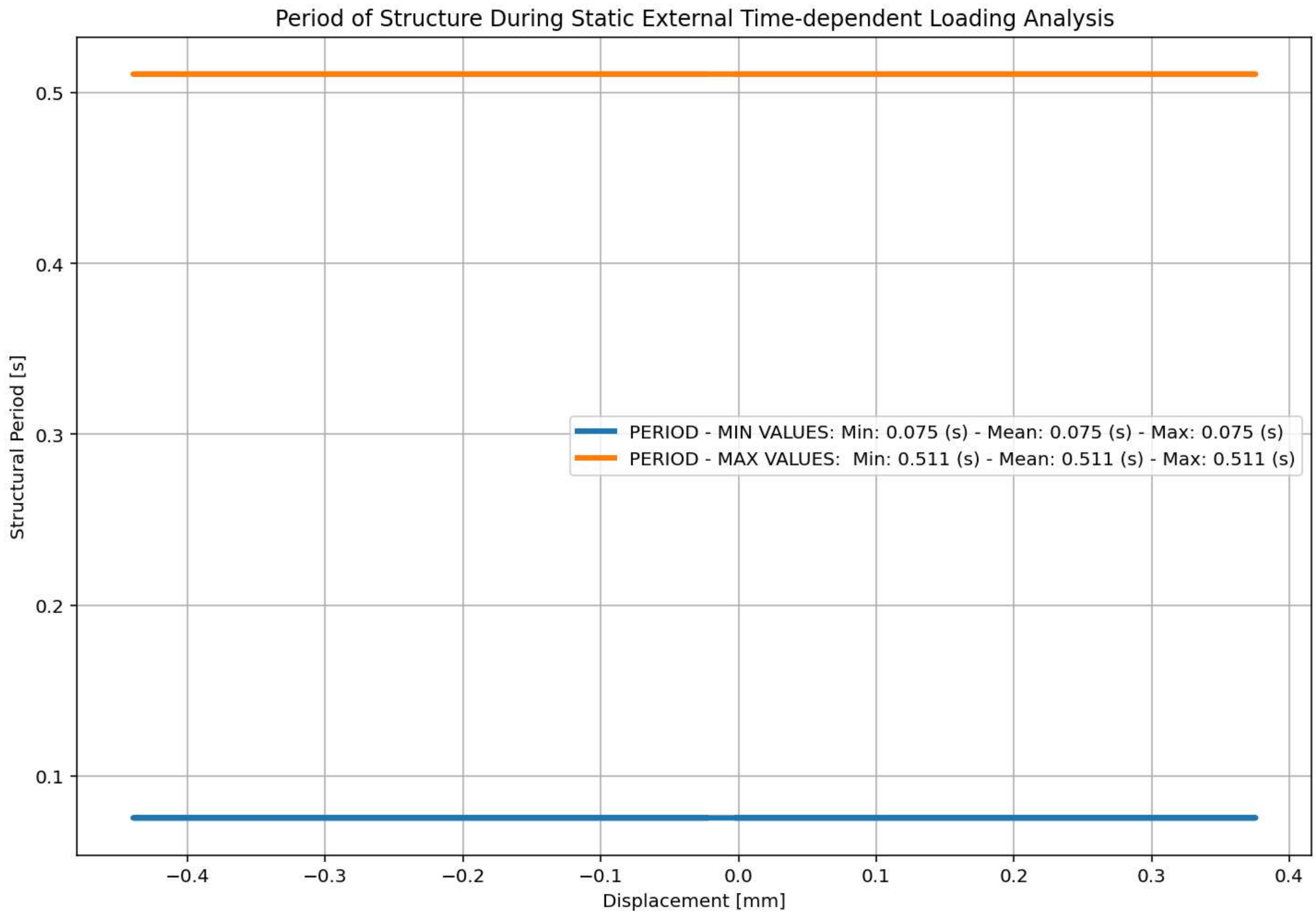
STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS



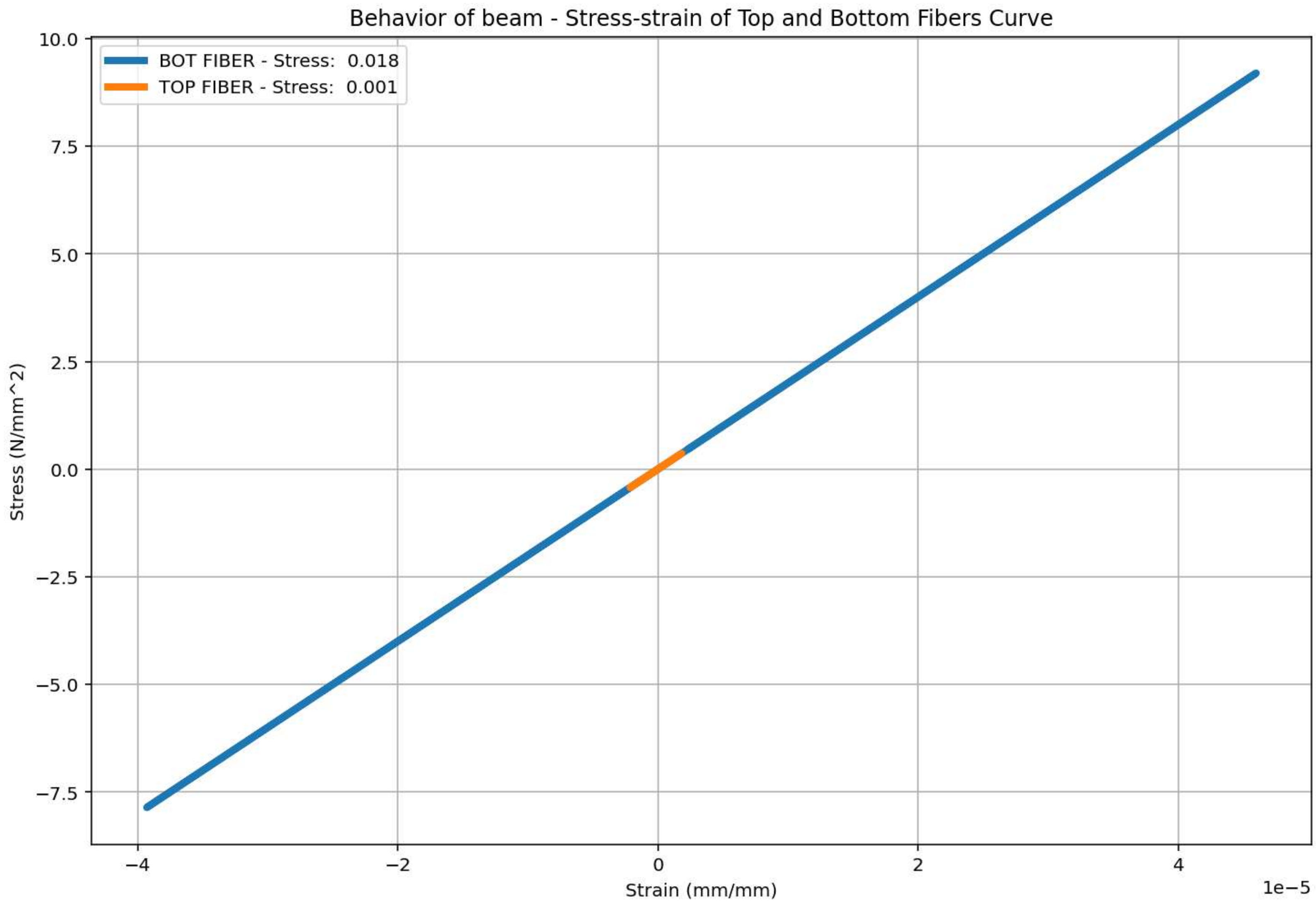
Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...Y_SUPPORTED_8_ANA\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...ORTED\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED.py
W(x,t)_BEAM_ONE_EL...IMPLY_SUPPORTED.py
EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'elasticBeamColumn'   # MATERIAL LINEARITY
ELE_TYPE = 'nonlinearBeamColumn'   # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(reaction_ETDLS, disp_mid_ETDLS,
 ele_axialforce_ETDLS, ele_shearforce_ETDLS, ele_momentforce_ETDLS,
 node_displacementsX_ETDLS, node_displacementsY_ETDLS,
  dispX_ETDLS, dispY_ETDLS,
 PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS,
 STRESSb_ETDLS, STRAINb_ETDLS, STRESSt_ETDLS, STRAINt_ETDLS, CURVATURE_ETDLS) = DATA
XDATA = disp_mid_ETDLS
YDATA = reaction_ETDLS
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent Lo
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [mm]')
#plt.semilogy()
29 %
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
IPython Console
Plots
Variable Explorer
Help
Debugger
History
Files


External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
15000
10000
5000
0
−5000
−10000
−15000
0
1
2
3
4
5
Time (s)

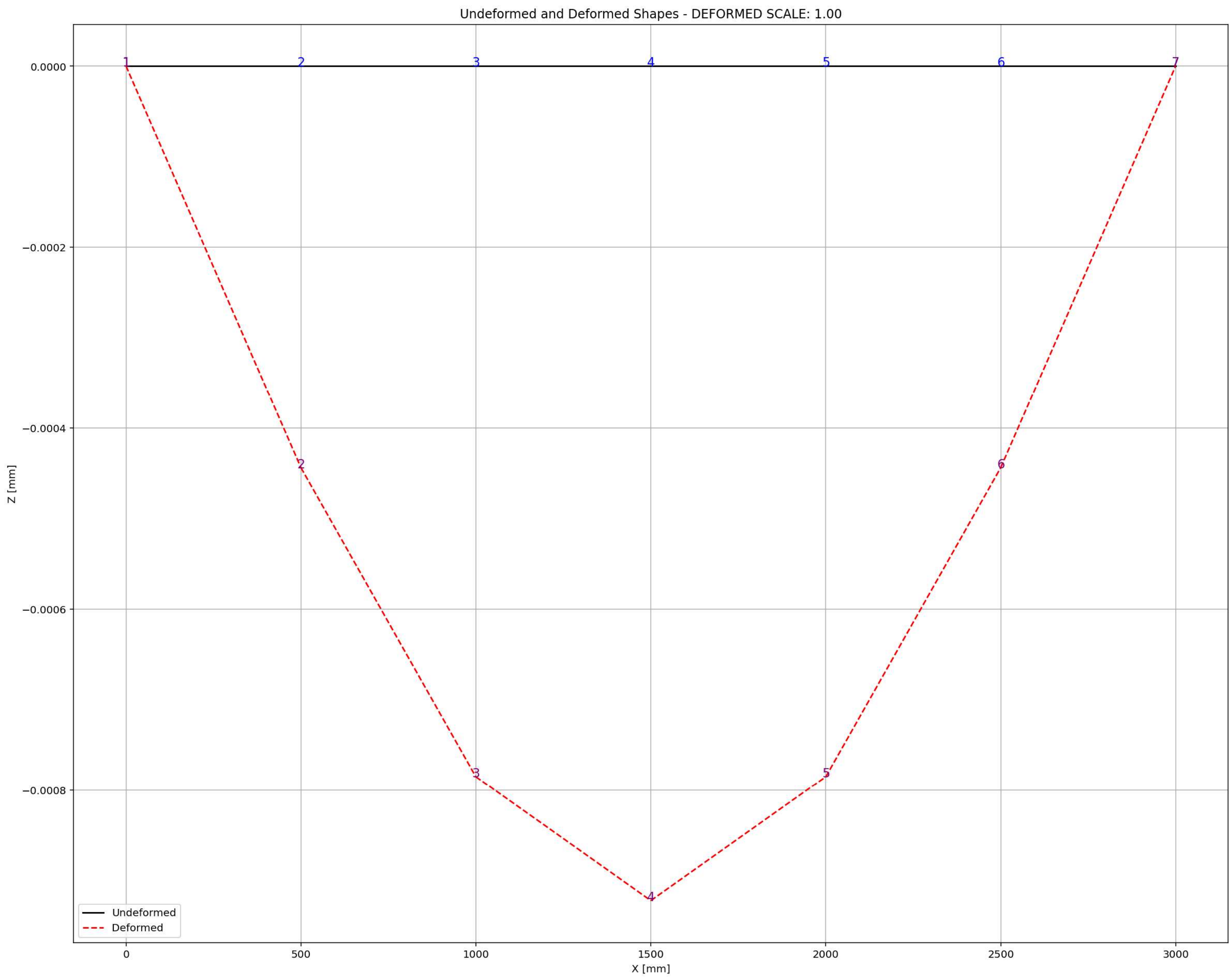

Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
10000
5000
0
−5000
−10000
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
Displacement in Middle Span [mm]



Period of Structure During Static External Time-dependent Loading Analysis
Structural Period [s]
0.1
0.2
0.3
0.4
0.5
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
Displacement [mm]
PERIOD - MIN VALUES: Min: 0.075 (s) - Mean: 0.075 (s) - Max: 0.075 (s)
PERIOD - MAX VALUES: Min: 0.511 (s) - Mean: 0.511 (s) - Max: 0.511 (s)

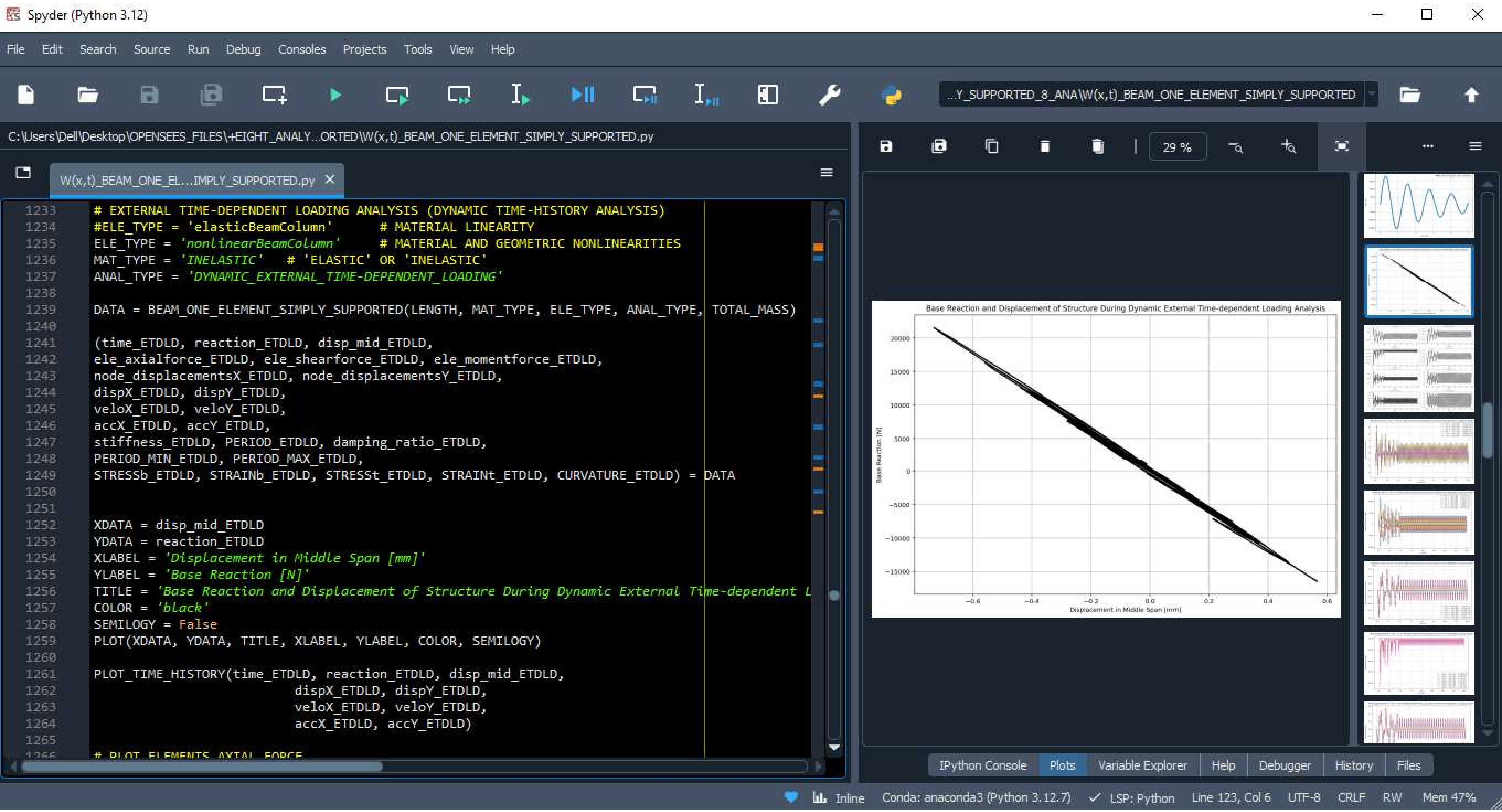


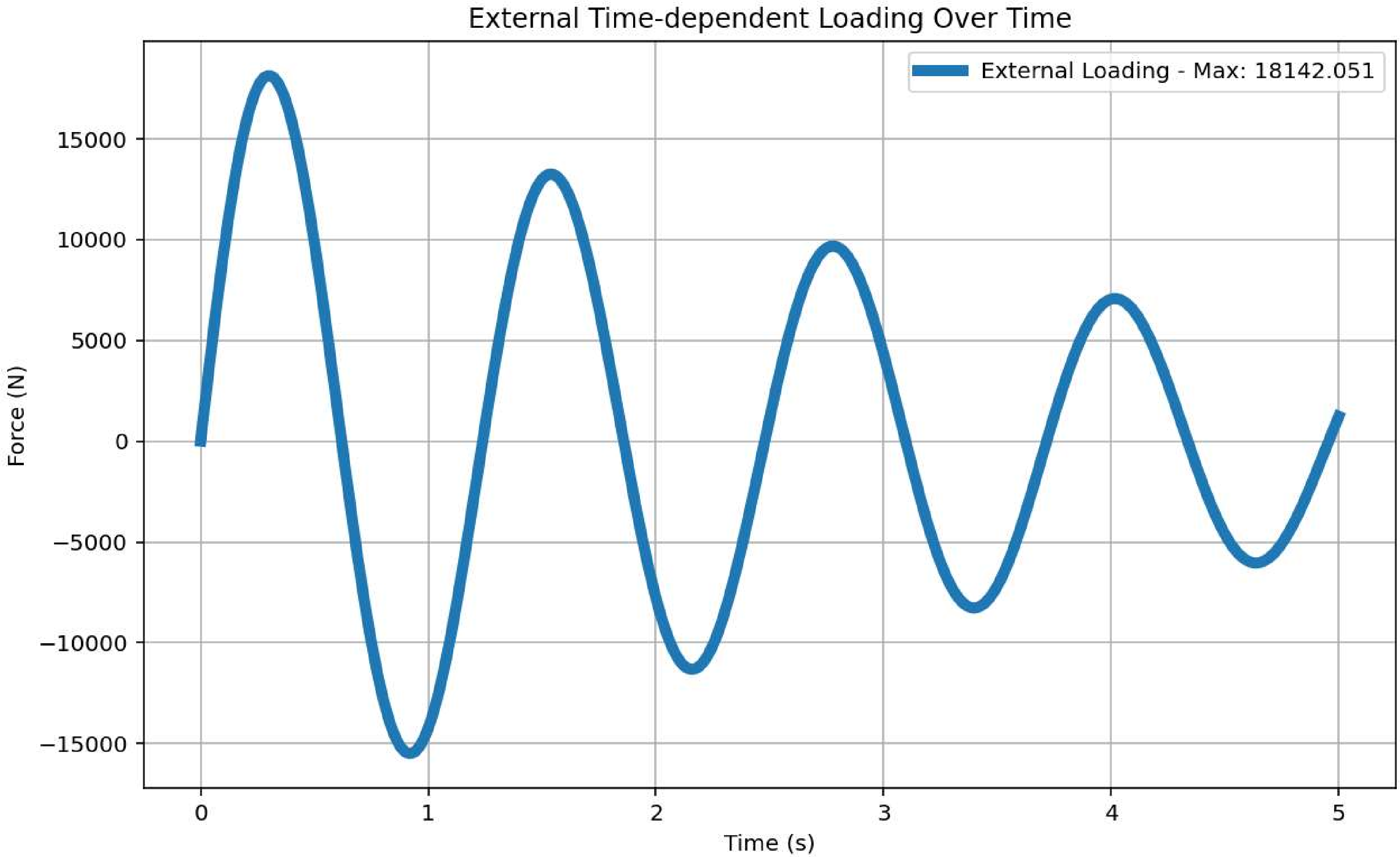
Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.018
TOP FIBER - Stress: 0.001
Stress (N/mm^2)
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−4
−2
0
2
4
1e−5
Strain (mm/mm)

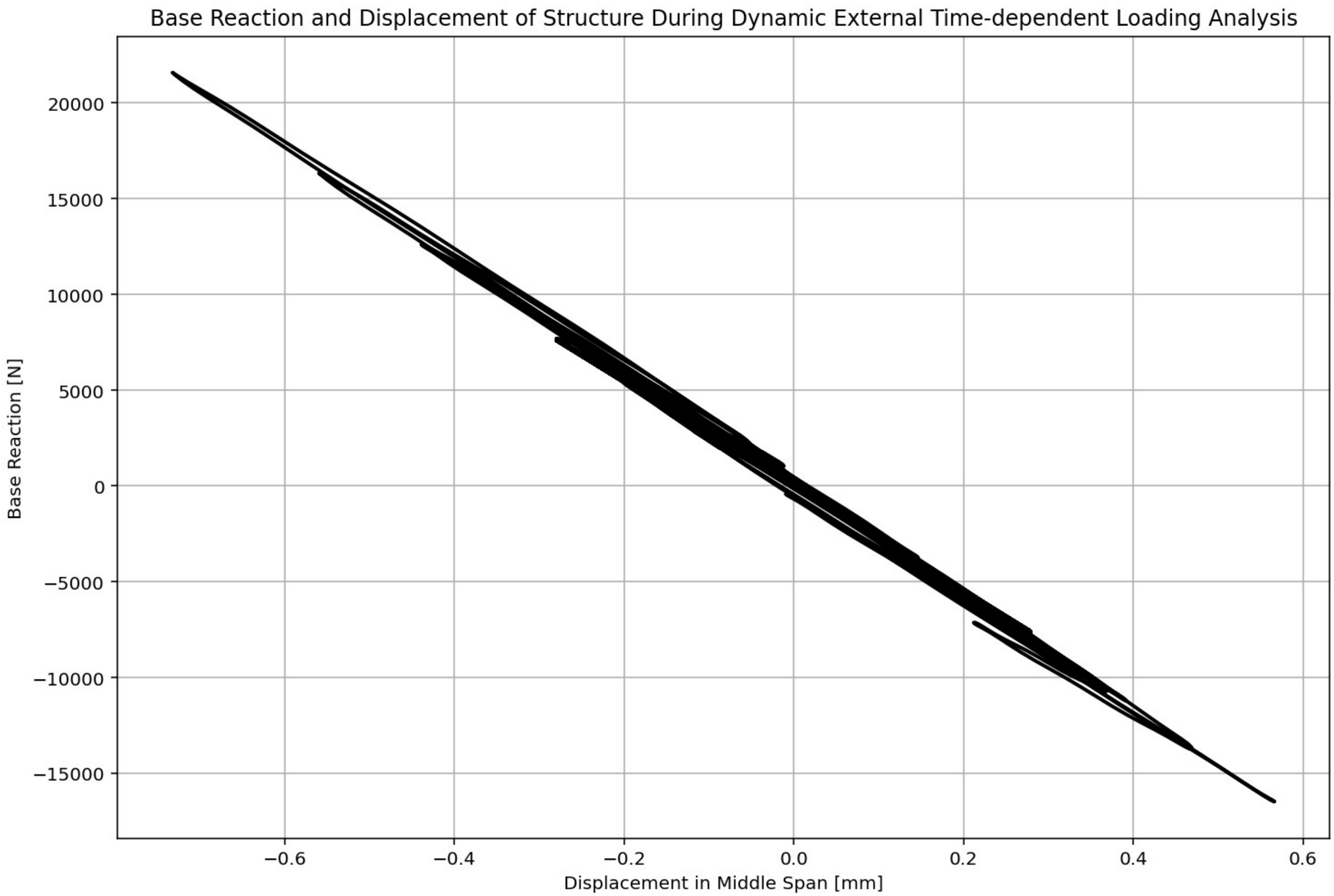


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
0.0000
−0.0002
−0.0004
−0.0006
−0.0008
Z [mm]
0
500
1000
1500
2000
2500
3000
X [mm]
1
2
3
4
5
6
7
Undeformed
Deformed

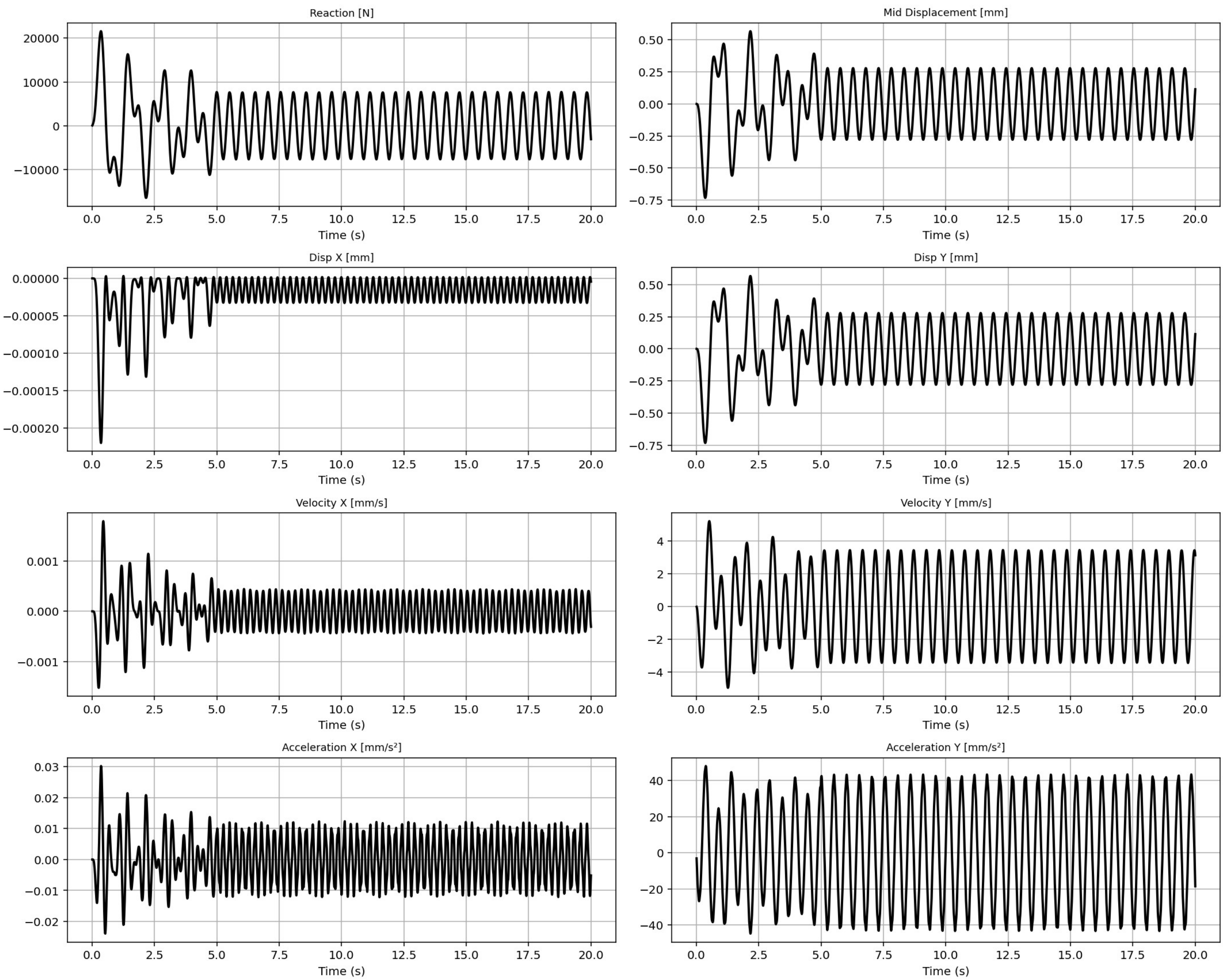
DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

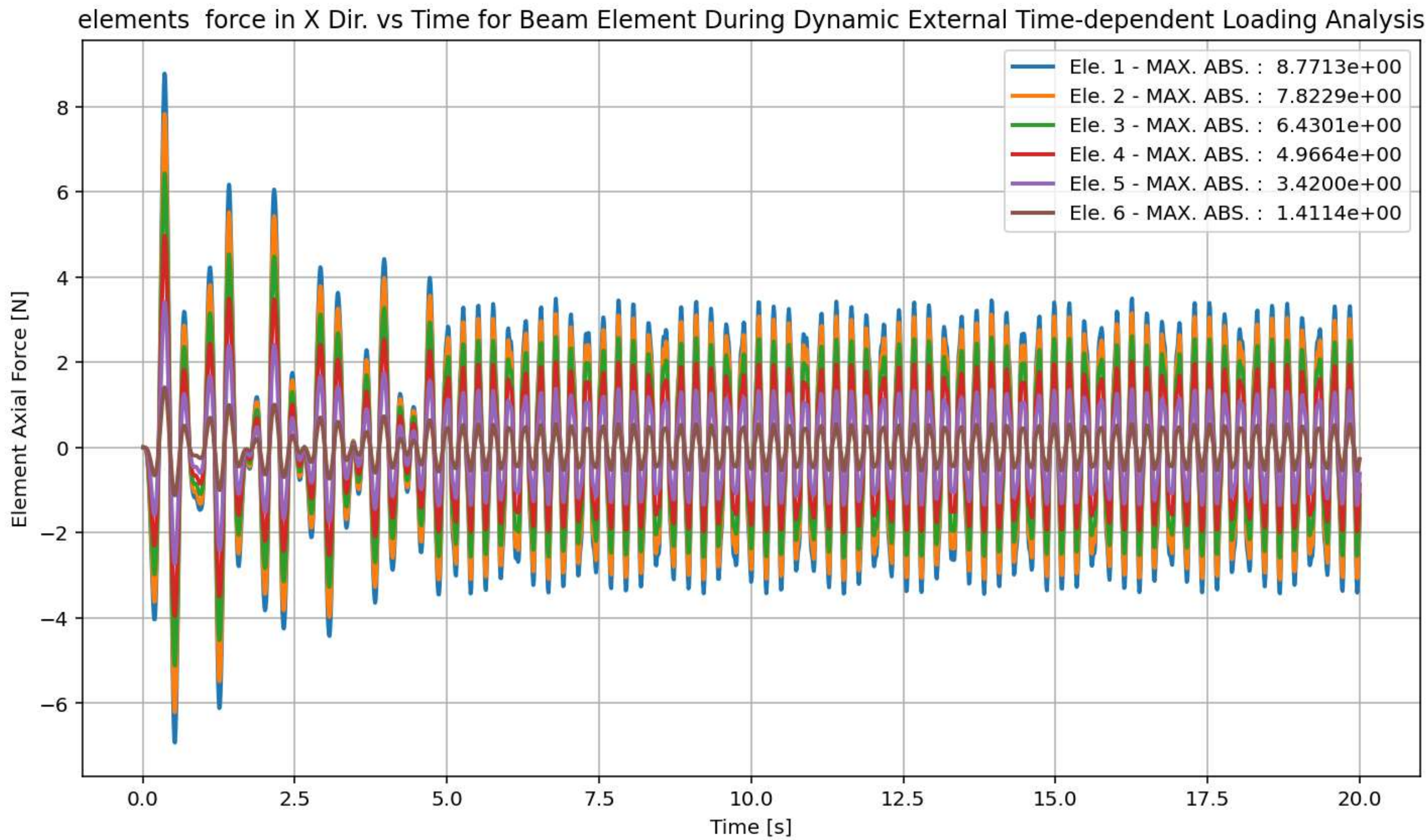

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...Y_SUPPORTED_8_ANA\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...ORTED\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED.py
W(x,t)_BEAM_ONE_EL...IMPLY_SUPPORTED.py
EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'elasticBeamColumn'   # MATERIAL LINEARITY
ELE_TYPE = 'nonlinearBeamColumn'   # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
ele_axialforce_ETDLD, ele_shearforce_ETDLD, ele_momentforce_ETDLD,
node_displacementsX_ETDLD, node_displacementsY_ETDLD,
dispX_ETDLD, dispY_ETDLD,
veloX_ETDLD, veloY_ETDLD,
accX_ETDLD, accY_ETDLD,
stiffness_ETDLD, PERIOD_ETDLD, damping_ratio_ETDLD,
PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD,
STRESSb_ETDLD, STRAINb_ETDLD, STRESSt_ETDLD, STRAINt_ETDLD, CURVATURE_ETDLD) = DATA
XDATA = disp_mid_ETDLD
YDATA = reaction_ETDLD
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependent L
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
dispX_ETDLD, dispY_ETDLD,
veloX_ETDLD, veloY_ETDLD,
accX_ETDLD, accY_ETDLD)
PLOT ELEMENTS AXIAL FORCE
29 %
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 123, Col 6 UTF-8 CRLF RW Mem 47%

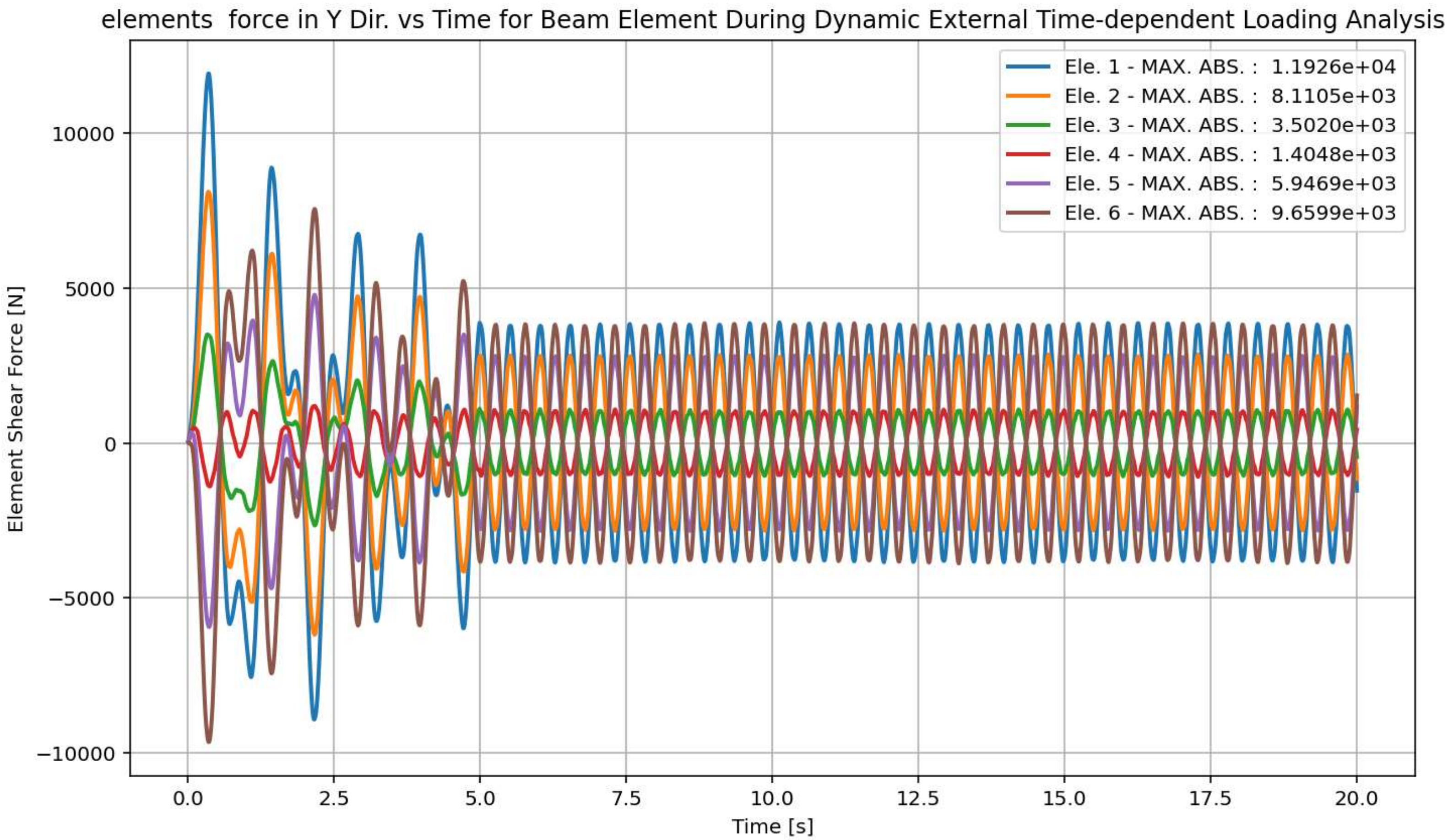

External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
Time (s)
15000
10000
5000
0
−5000
−10000
−15000
0
1
2
3
4
5

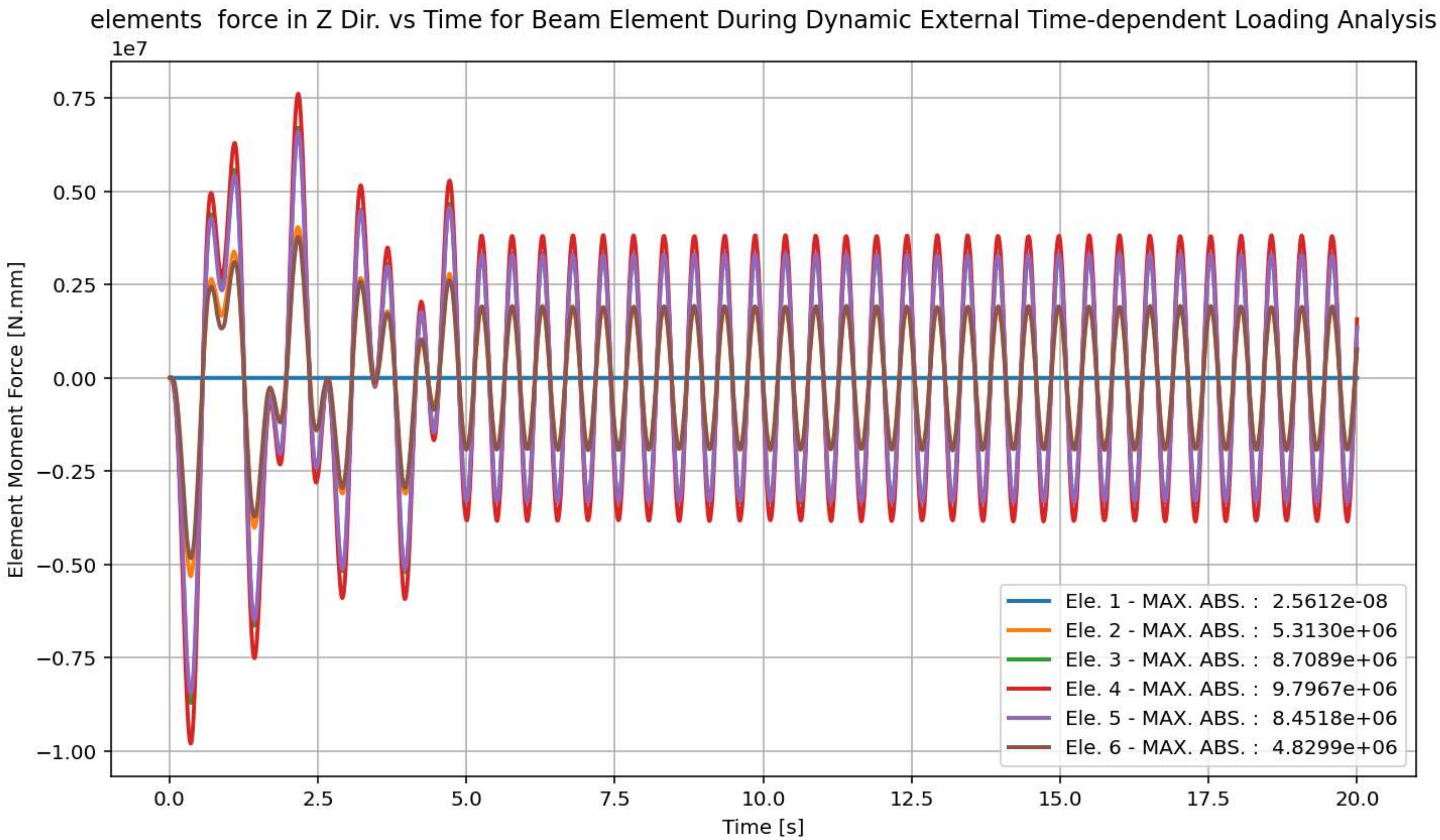


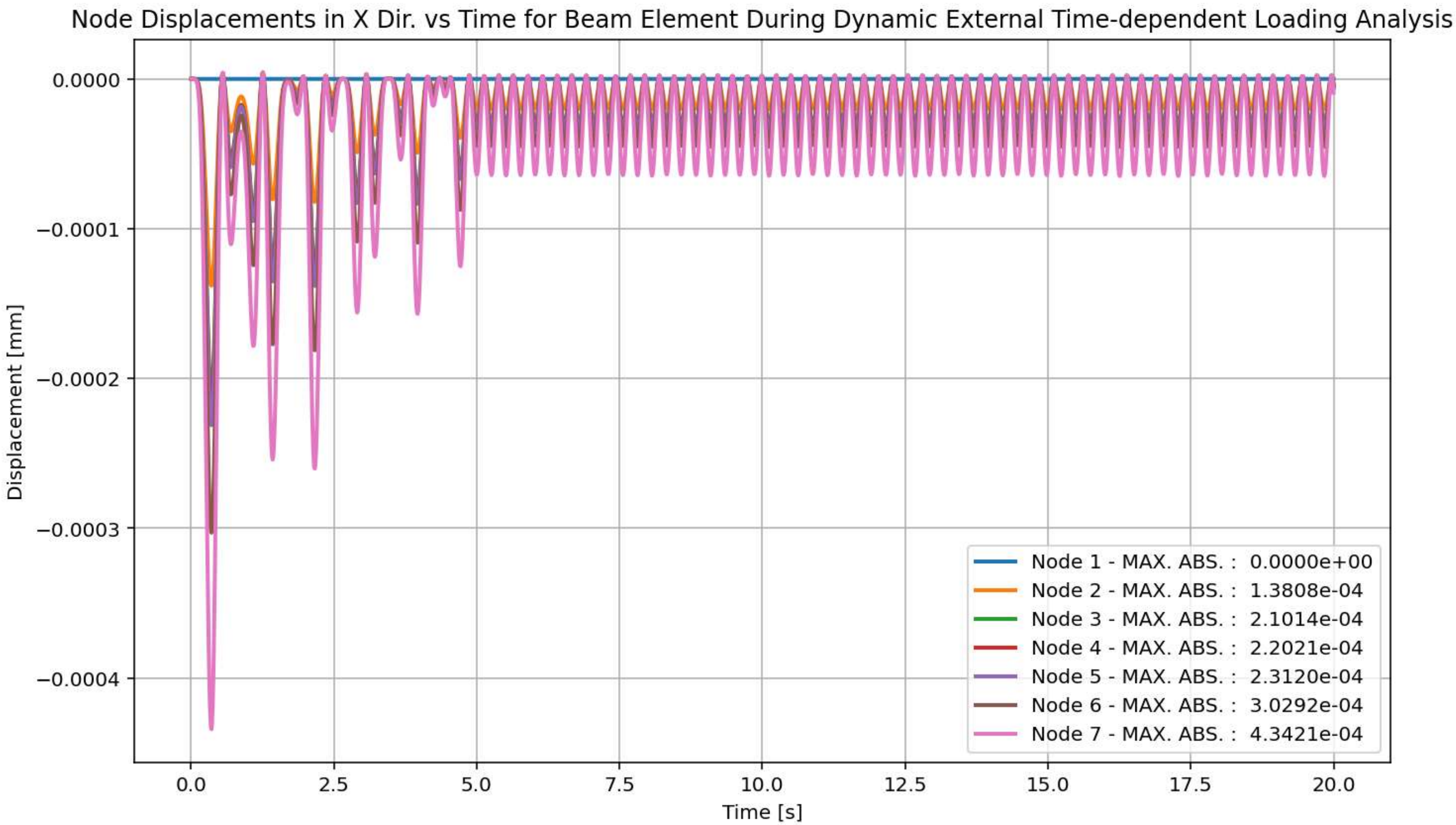
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
20000
15000
10000
5000
0
−5000
−10000
−15000
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
Displacement in Middle Span [mm]


Reaction [N]
Mid Displacement [mm]
Disp X [mm]
Disp Y [mm]
Velocity X [mm/s]
Velocity Y [mm/s]
Acceleration X [mm/s²]
Acceleration Y [mm/s²]
Time (s)

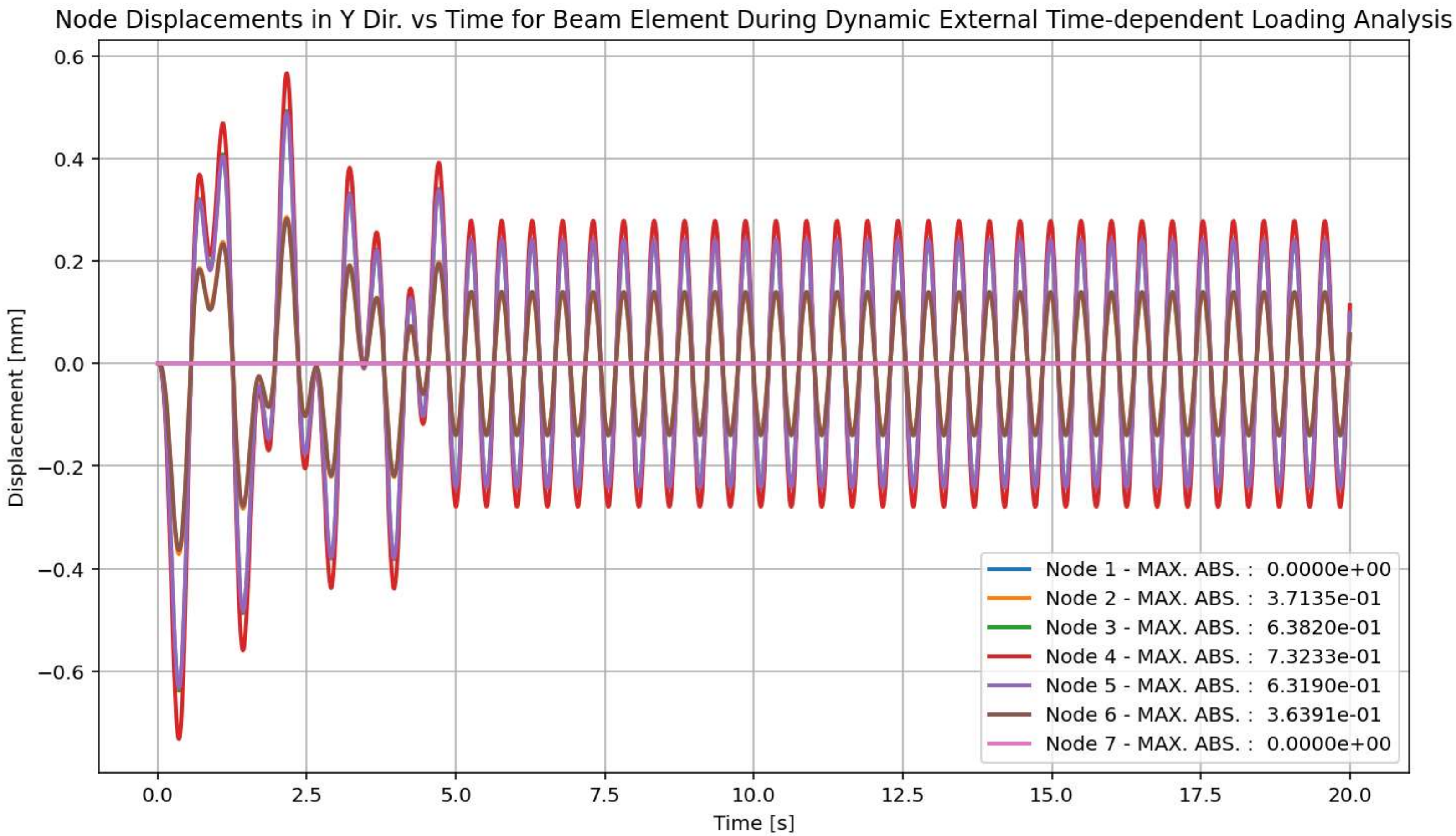

elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 8.7713e+00
Ele. 2 - MAX. ABS. : 7.8229e+00
Ele. 3 - MAX. ABS. : 6.4301e+00
Ele. 4 - MAX. ABS. : 4.9664e+00
Ele. 5 - MAX. ABS. : 3.4200e+00
Ele. 6 - MAX. ABS. : 1.4114e+00
Element Axial Force [N]
8
6
4
2
0
−2
−4
−6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


elements force in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 1.1926e+04
Ele. 2 - MAX. ABS. : 8.1105e+03
Ele. 3 - MAX. ABS. : 3.5020e+03
Ele. 4 - MAX. ABS. : 1.4048e+03
Ele. 5 - MAX. ABS. : 5.9469e+03
Ele. 6 - MAX. ABS. : 9.6599e+03
Element Shear Force [N]
10000
5000
0
−5000
−10000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

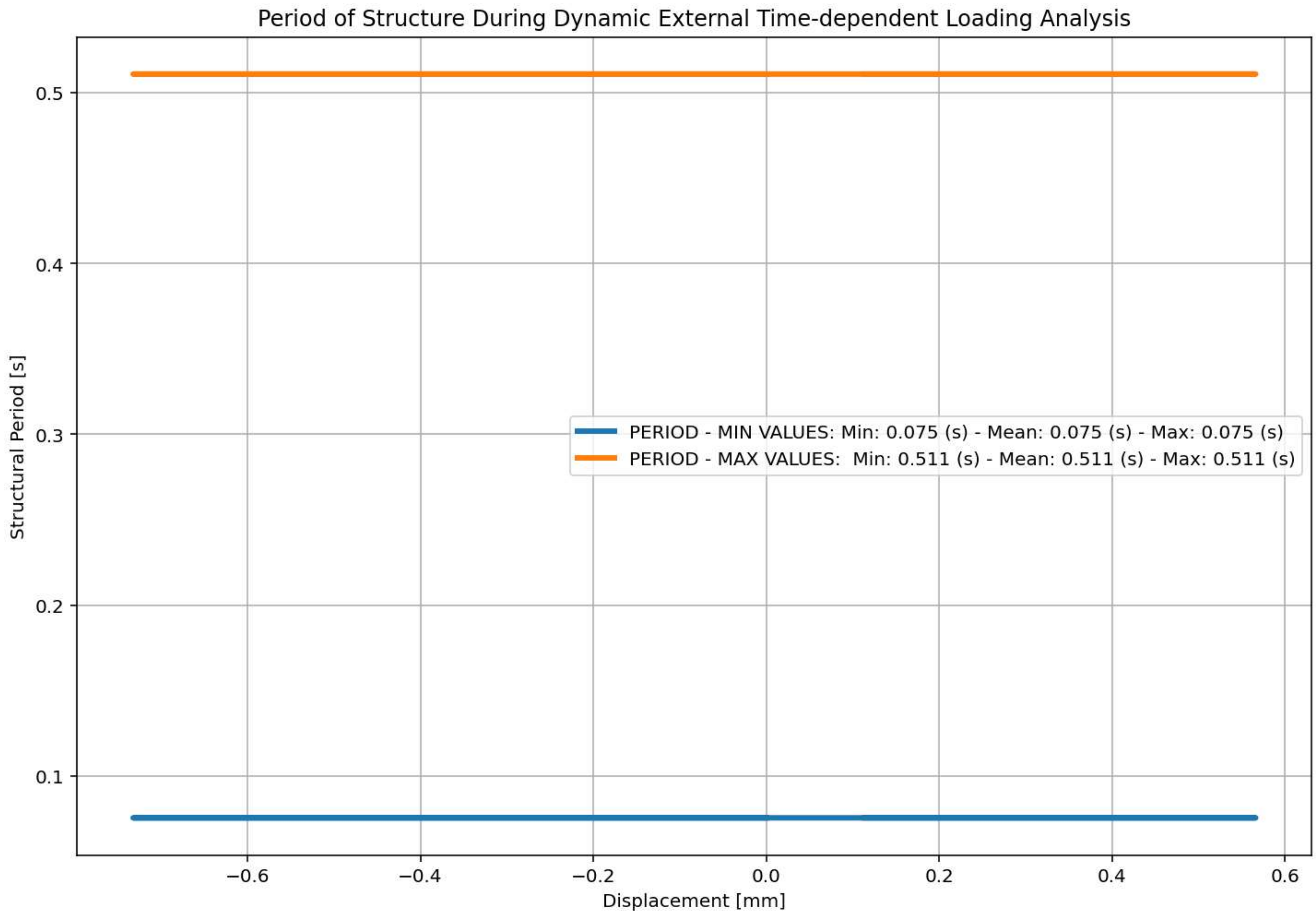

elements force in Z Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
1e7
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Element Moment Force [N.mm]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
Ele. 1 - MAX. ABS. : 2.5612e-08
Ele. 2 - MAX. ABS. : 5.3130e+06
Ele. 3 - MAX. ABS. : 8.7089e+06
Ele. 4 - MAX. ABS. : 9.7967e+06
Ele. 5 - MAX. ABS. : 8.4518e+06
Ele. 6 - MAX. ABS. : 4.8299e+06

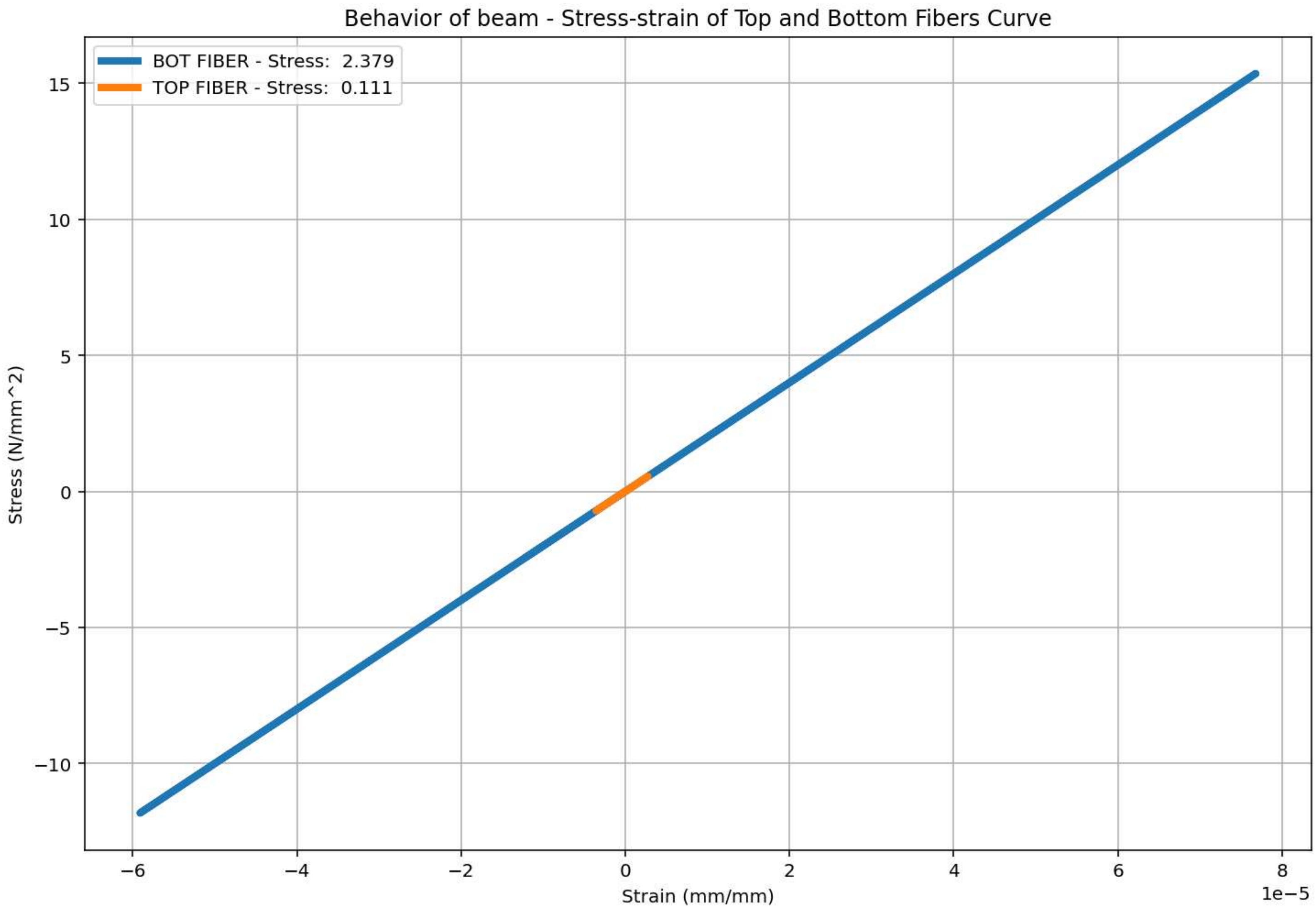


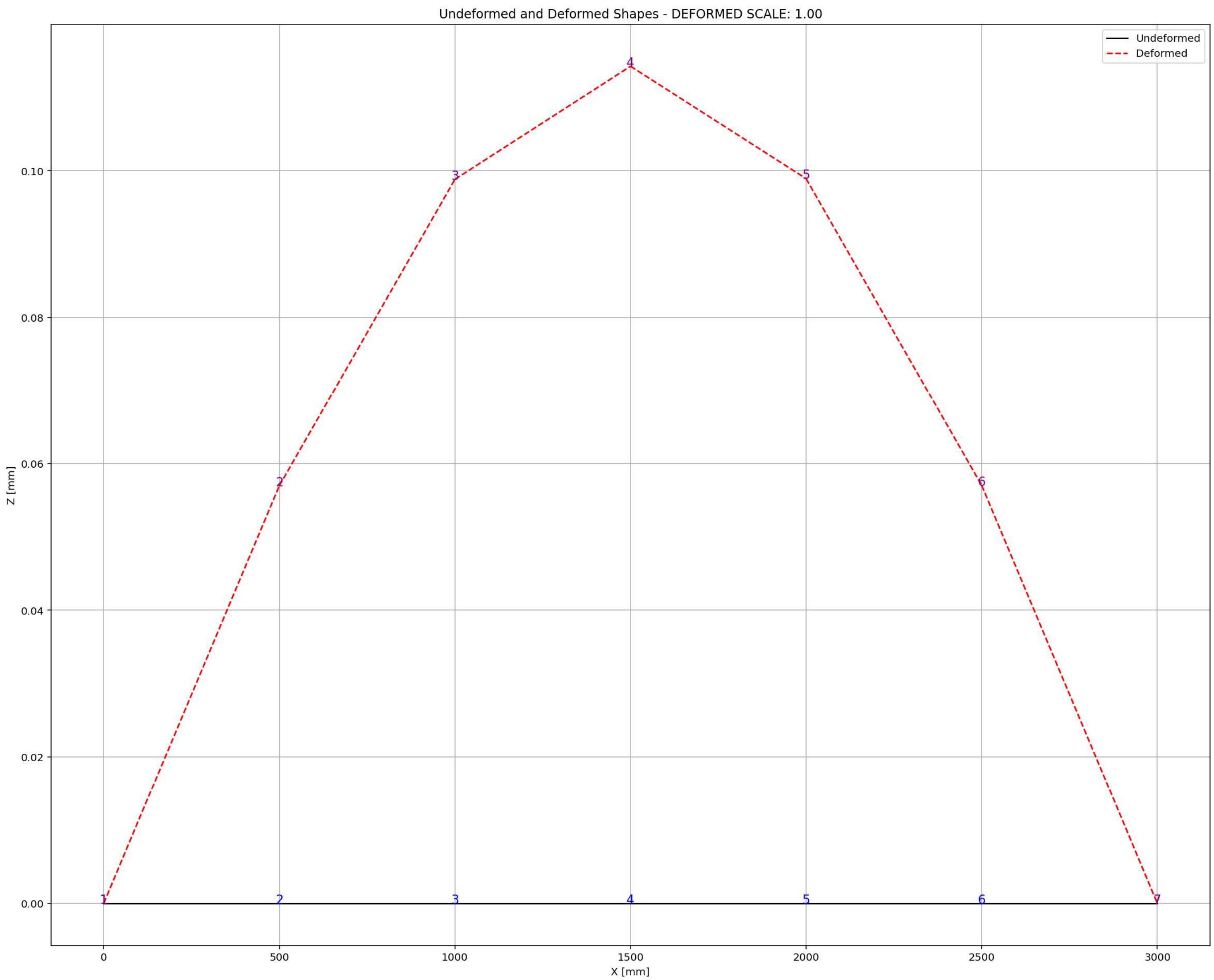
Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
0.0000
−0.0001
−0.0002
−0.0003
−0.0004
Displacement [mm]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 1.3808e-04
Node 3 - MAX. ABS. : 2.1014e-04
Node 4 - MAX. ABS. : 2.2021e-04
Node 5 - MAX. ABS. : 2.3120e-04
Node 6 - MAX. ABS. : 3.0292e-04
Node 7 - MAX. ABS. : 4.3421e-04



Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Displacement [mm]
Time [s]
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.7135e-01
Node 3 - MAX. ABS. : 6.3820e-01
Node 4 - MAX. ABS. : 7.3233e-01
Node 5 - MAX. ABS. : 6.3190e-01
Node 6 - MAX. ABS. : 3.6391e-01
Node 7 - MAX. ABS. : 0.0000e+00

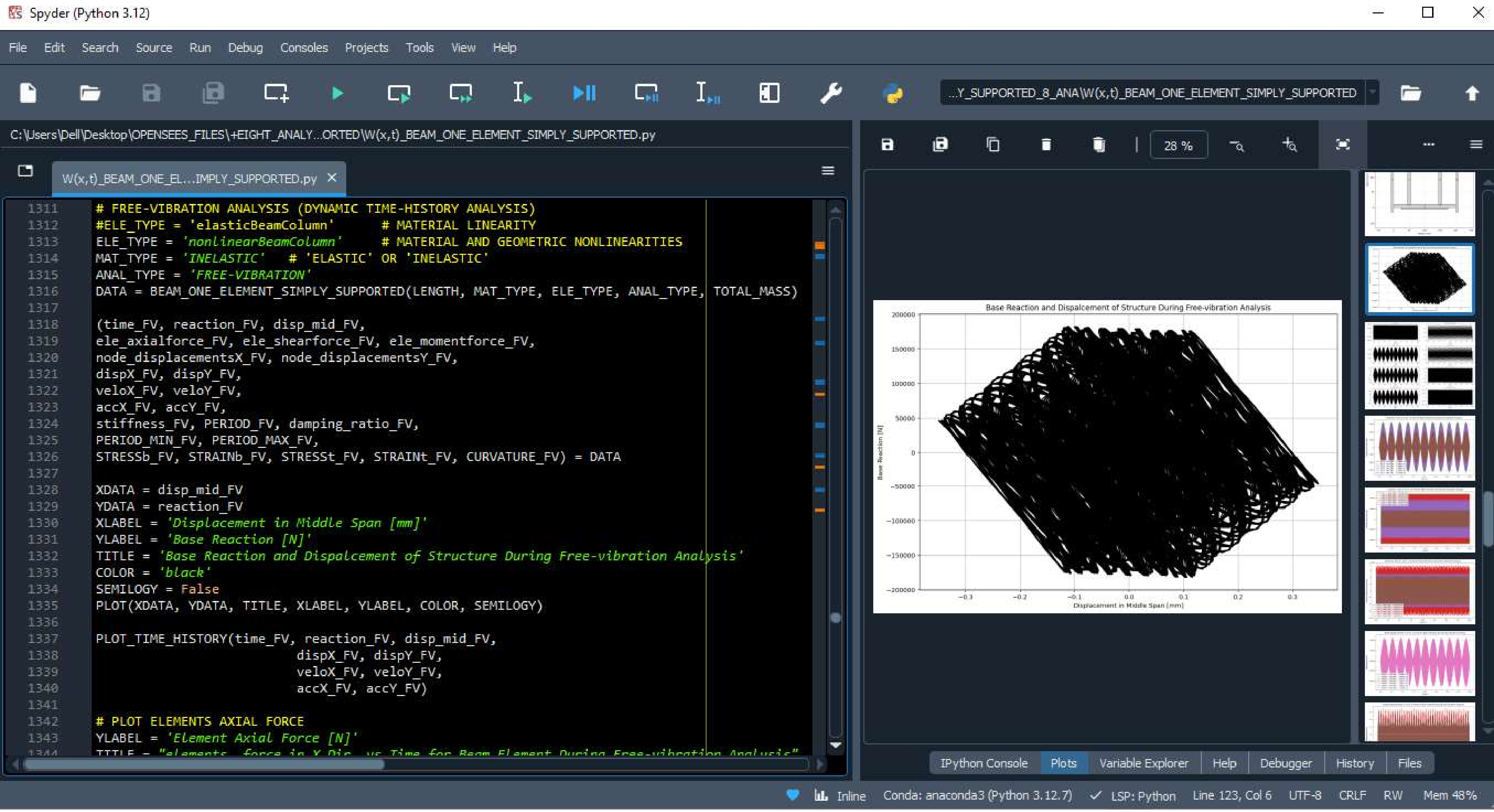

Period of Structure During Dynamic External Time-dependent Loading Analysis
Structural Period [s]
Displacement [mm]
PERIOD - MIN VALUES: Min: 0.075 (s) - Mean: 0.075 (s) - Max: 0.075 (s)
PERIOD - MAX VALUES: Min: 0.511 (s) - Mean: 0.511 (s) - Max: 0.511 (s)
0.1
0.2
0.3
0.4
0.5
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6

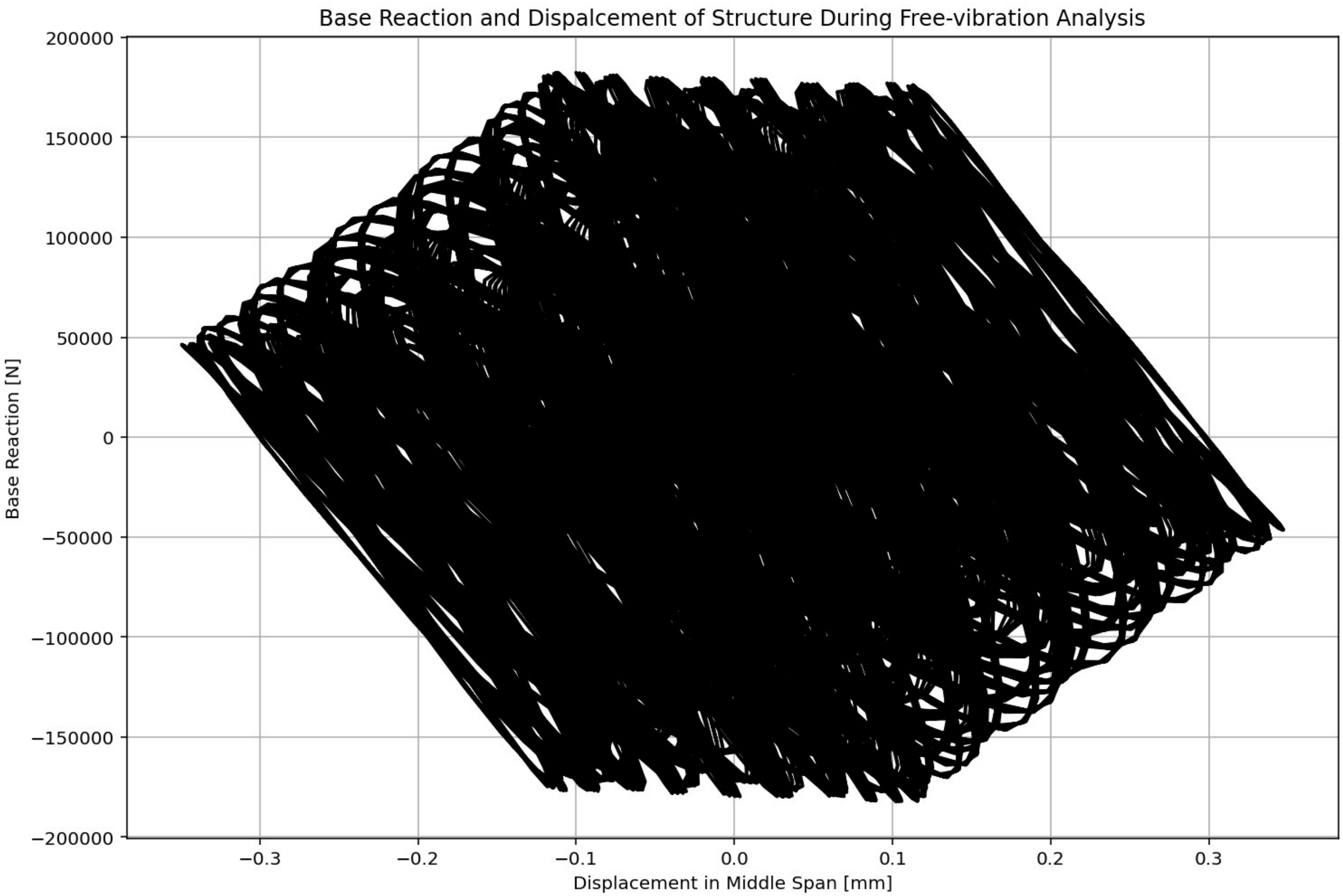

Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 2.379
TOP FIBER - Stress: 0.111
Stress (N/mm^2)
Strain (mm/mm)
1e−5
15
10
5
0
−5
−10
−6
−4
−2
0
2
4
6
8



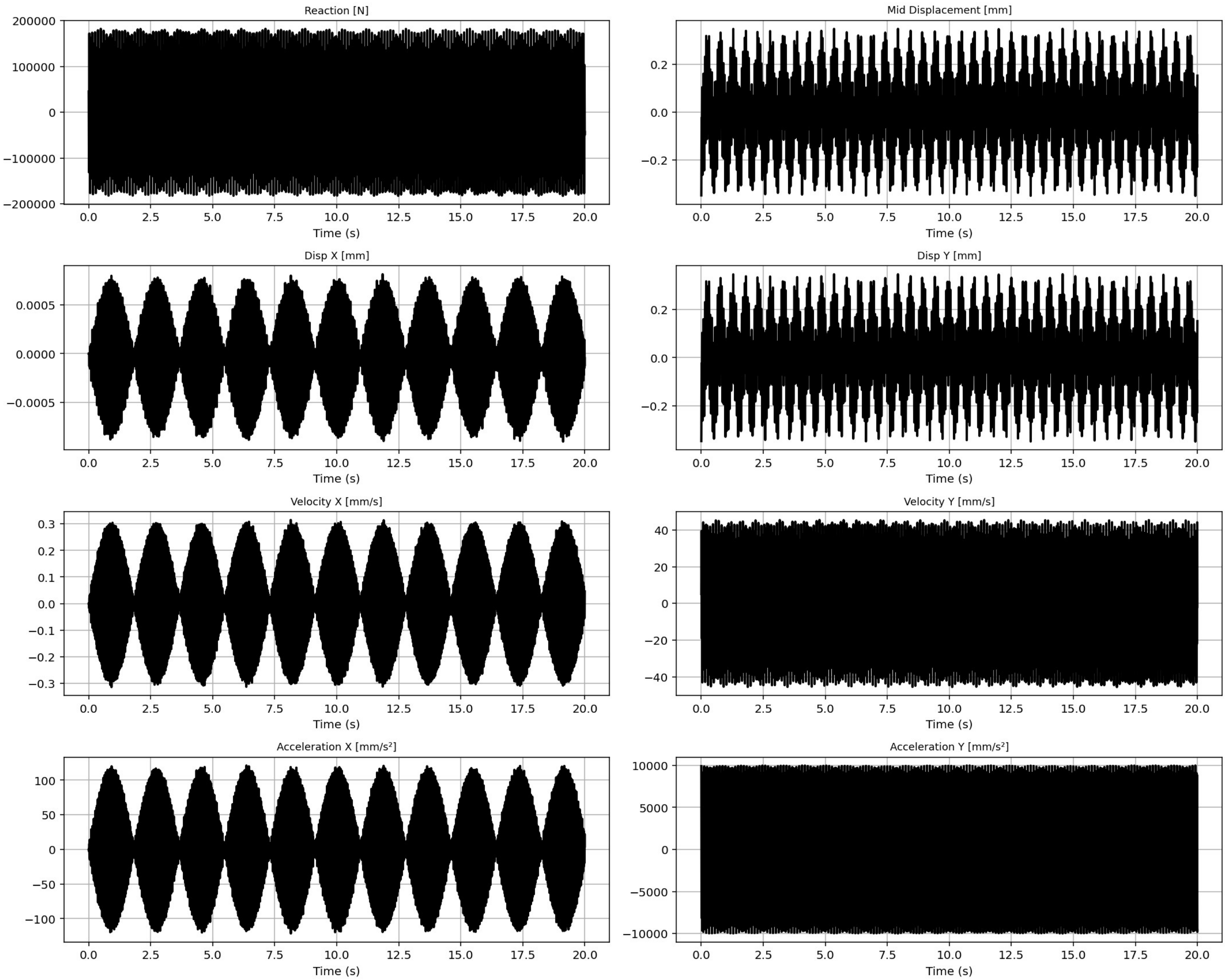
Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0.00
0.02
0.04
0.06
0.08
0.10
0
500
1000
1500
2000
2500
3000
1
2
3
4
5
6
7

FREE-VIBRATION ANALYSIS

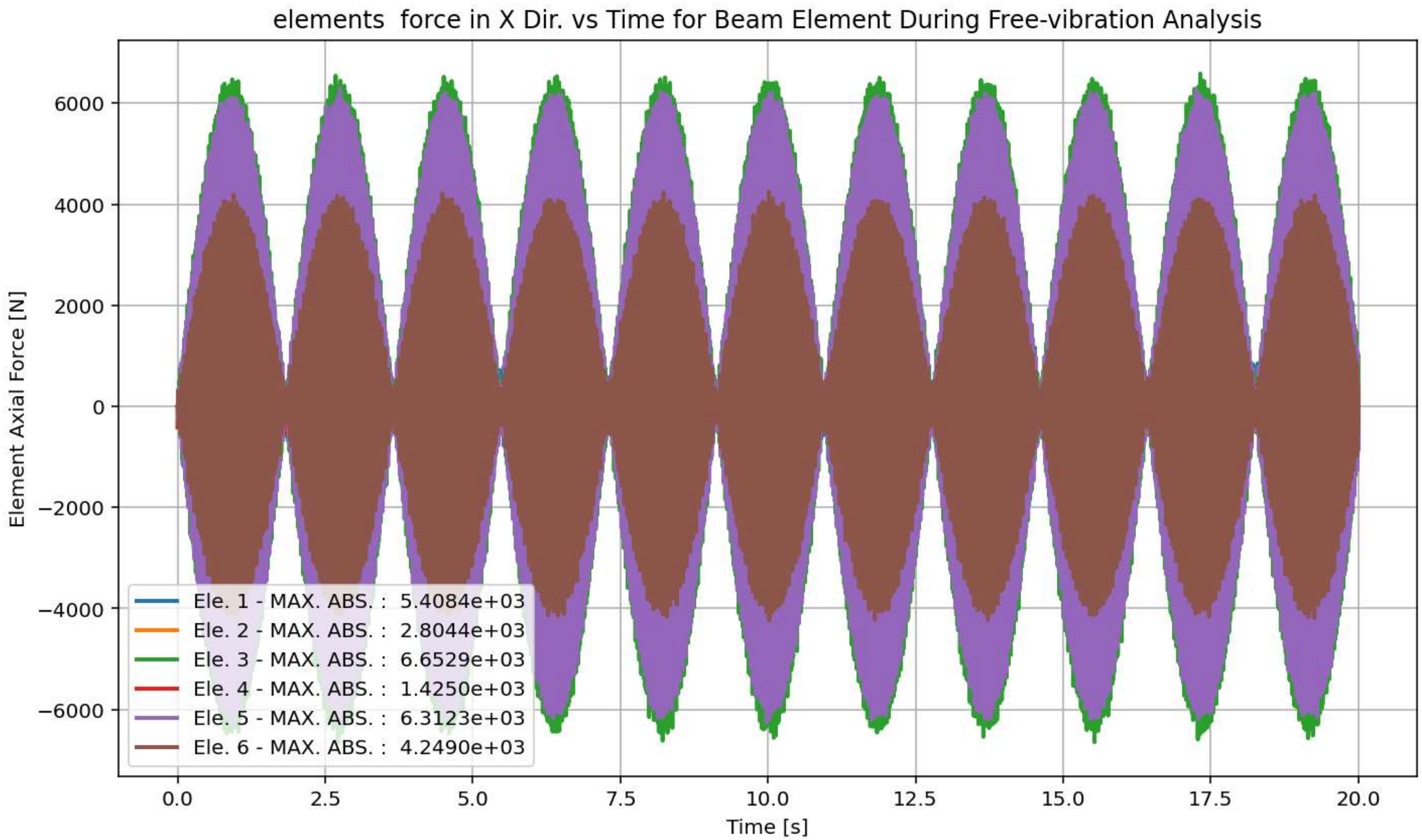

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...Y_SUPPORTED_8_ANA\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...ORTED\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED.py
W(x,t)_BEAM_ONE_EL...IMPLY_SUPPORTED.py
FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'elasticBeamColumn'   # MATERIAL LINEARITY
ELE_TYPE = 'nonlinearBeamColumn'   # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'FREE-VIBRATION'
DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(time_FV, reaction_FV, disp_mid_FV,
ele_axialforce_FV, ele_shearforce_FV, ele_momentforce_FV,
node_displacementsX_FV, node_displacementsY_FV,
dispX_FV, dispY_FV,
veloX_FV, veloY_FV,
accX_FV, accY_FV,
stiffness_FV, PERIOD_FV, damping_ratio_FV,
PERIOD_MIN_FV, PERIOD_MAX_FV,
STRESSb_FV, STRAINb_FV, STRESSt_FV, STRAINt_FV, CURVATURE_FV) = DATA
XDATA = disp_mid_FV
YDATA = reaction_FV
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Dispalcement of Structure During Free-vibration Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_FV, reaction_FV, disp_mid_FV,
                  dispX_FV, dispY_FV,
                  veloX_FV, veloY_FV,
                  accX_FV, accY_FV)
# PLOT ELEMENTS AXIAL FORCE
YLABEL = 'Element Axial Force [N]'
28 %
Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 123, Col 6 UTF-8 CRLF RW Mem 48%

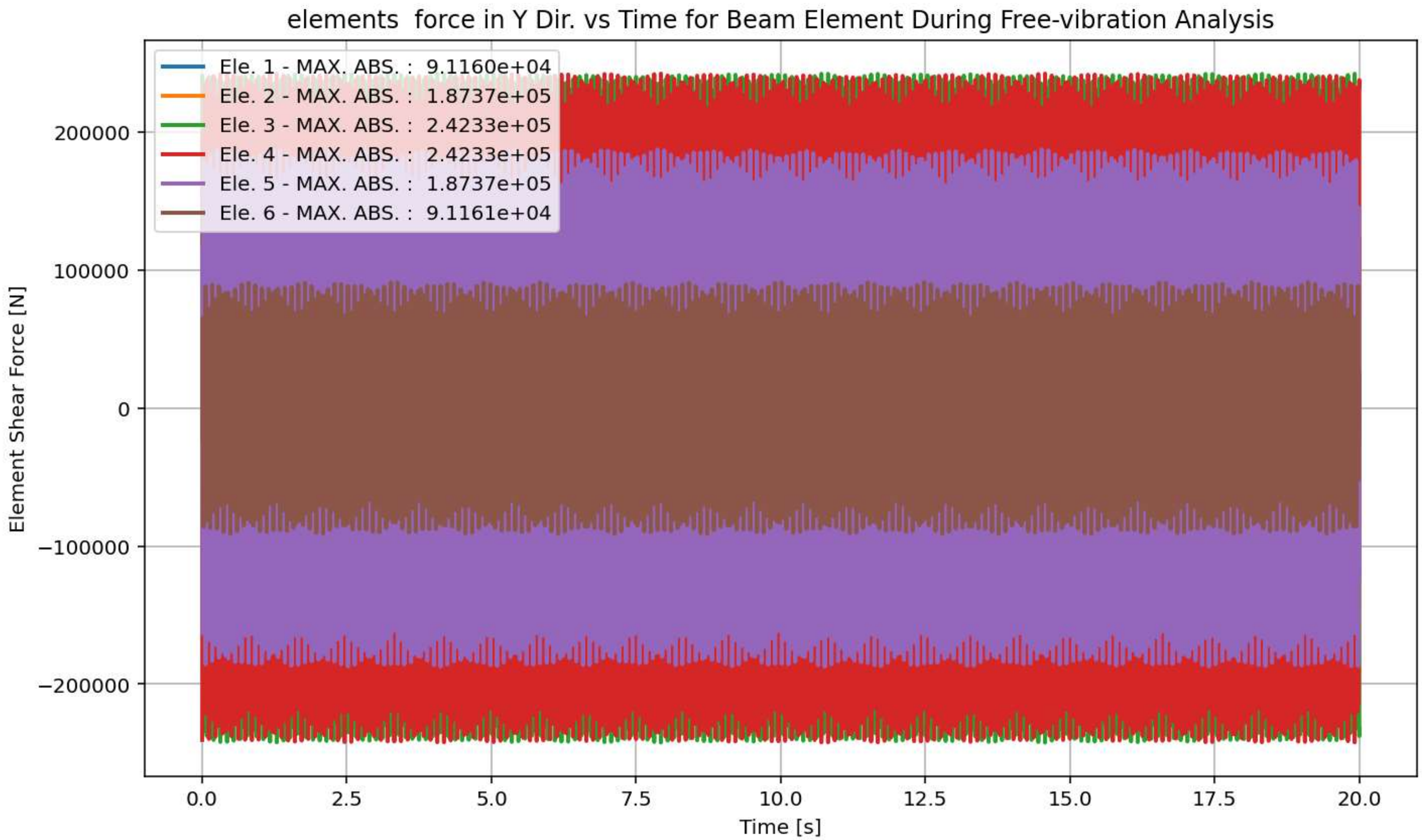


Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
200000
150000
100000
50000
0
−50000
−100000
−150000
−200000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement in Middle Span [mm]

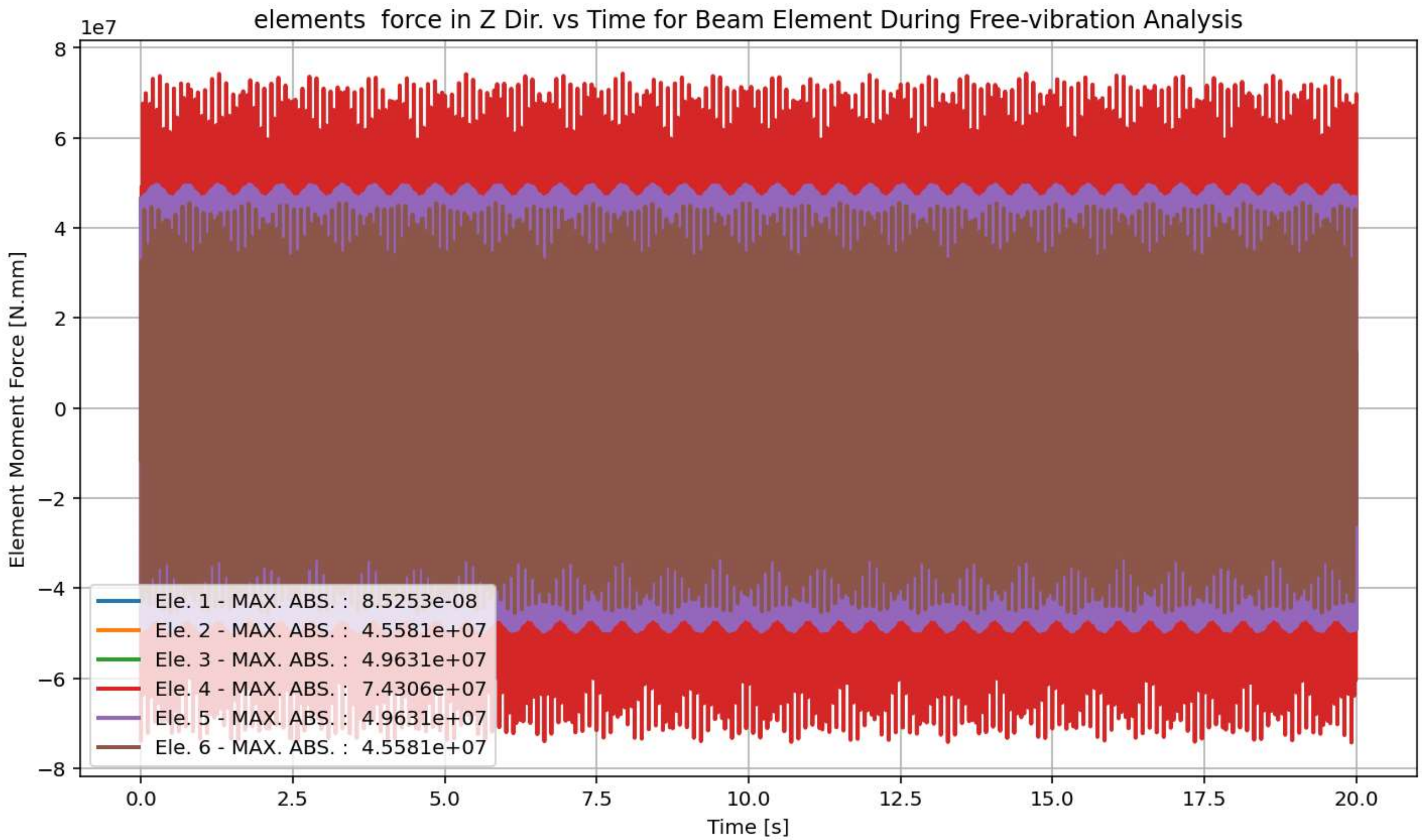


Reaction [N]
200000
100000
0
−100000
−200000
Mid Displacement [mm]
0.2
0.0
−0.2
Disp X [mm]
0.0005
0.0000
−0.0005
Disp Y [mm]
0.2
0.0
−0.2
Velocity X [mm/s]
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Velocity Y [mm/s]
40
20
0
−20
−40
Acceleration X [mm/s²]
100
50
0
−50
−100
Acceleration Y [mm/s²]
10000
5000
0
−5000
−10000
0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5 20.0
Time (s)

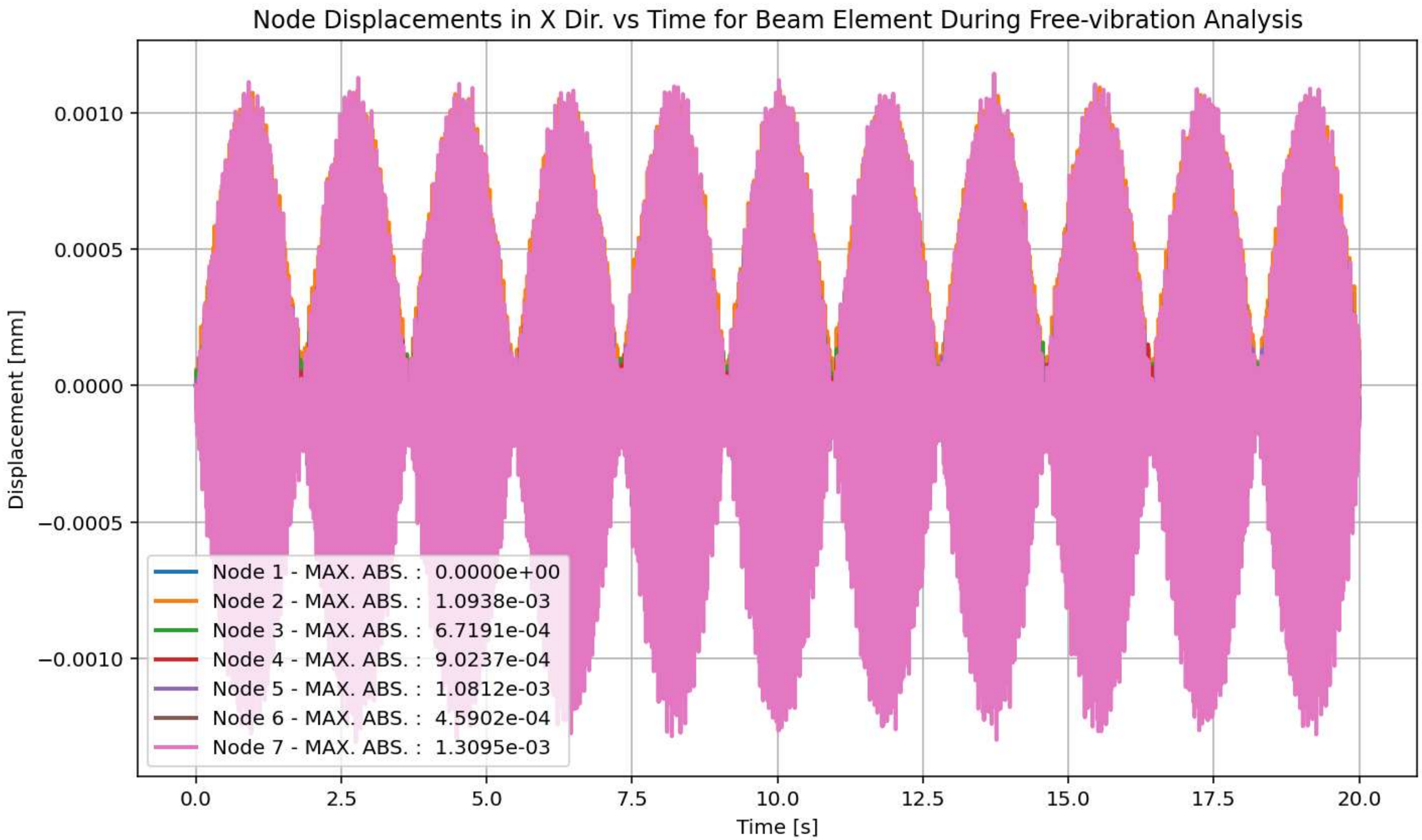

elements force in X Dir. vs Time for Beam Element During Free-vibration Analysis
6000
4000
2000
0
−2000
−4000
−6000
Element Axial Force [N]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
Ele. 1 - MAX. ABS. : 5.4084e+03
Ele. 2 - MAX. ABS. : 2.8044e+03
Ele. 3 - MAX. ABS. : 6.6529e+03
Ele. 4 - MAX. ABS. : 1.4250e+03
Ele. 5 - MAX. ABS. : 6.3123e+03
Ele. 6 - MAX. ABS. : 4.2490e+03

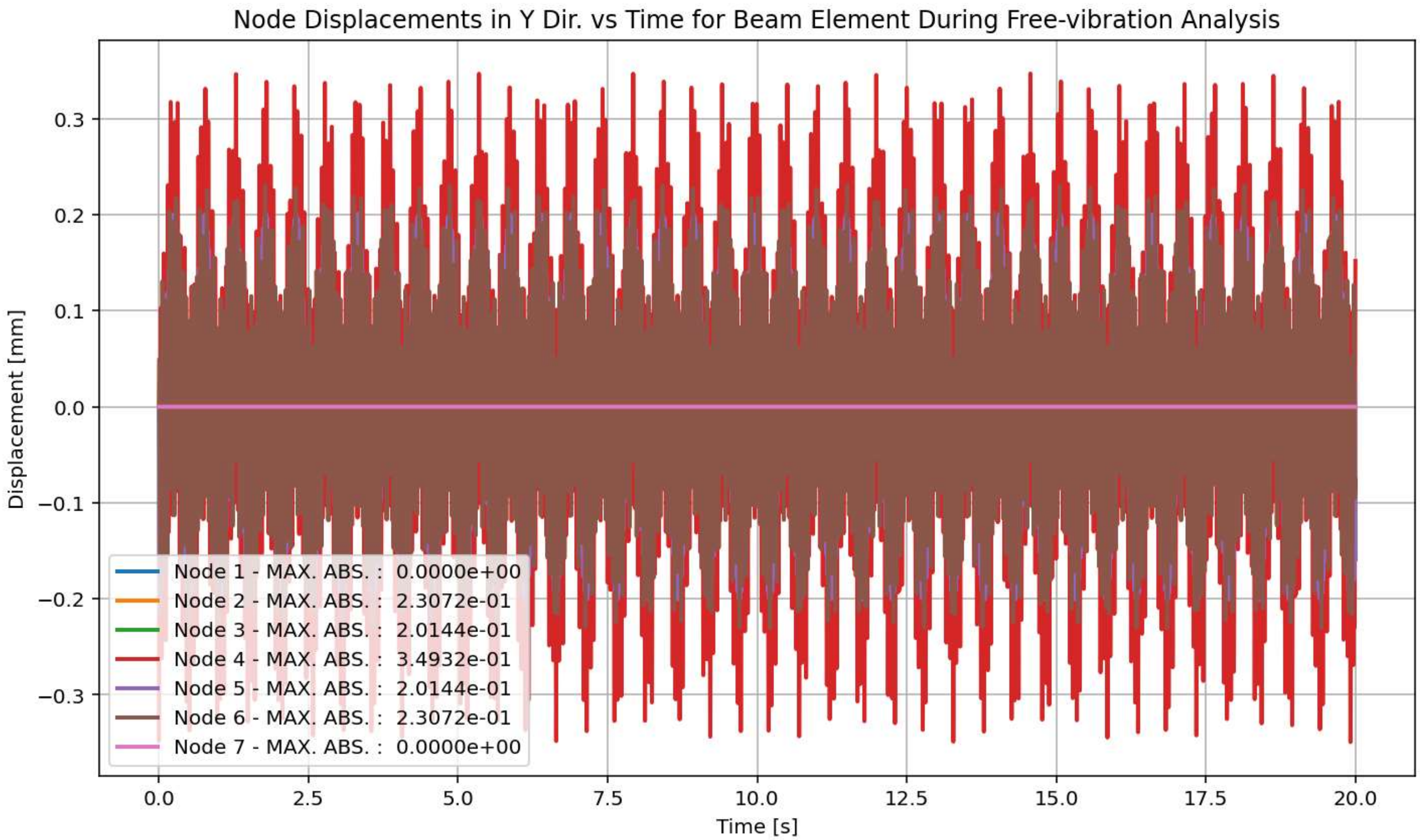


elements force in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Ele. 1 - MAX. ABS. : 9.1160e+04
Ele. 2 - MAX. ABS. : 1.8737e+05
Ele. 3 - MAX. ABS. : 2.4233e+05
Ele. 4 - MAX. ABS. : 2.4233e+05
Ele. 5 - MAX. ABS. : 1.8737e+05
Ele. 6 - MAX. ABS. : 9.1161e+04
Element Shear Force [N]
200000
100000
0
−100000
−200000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

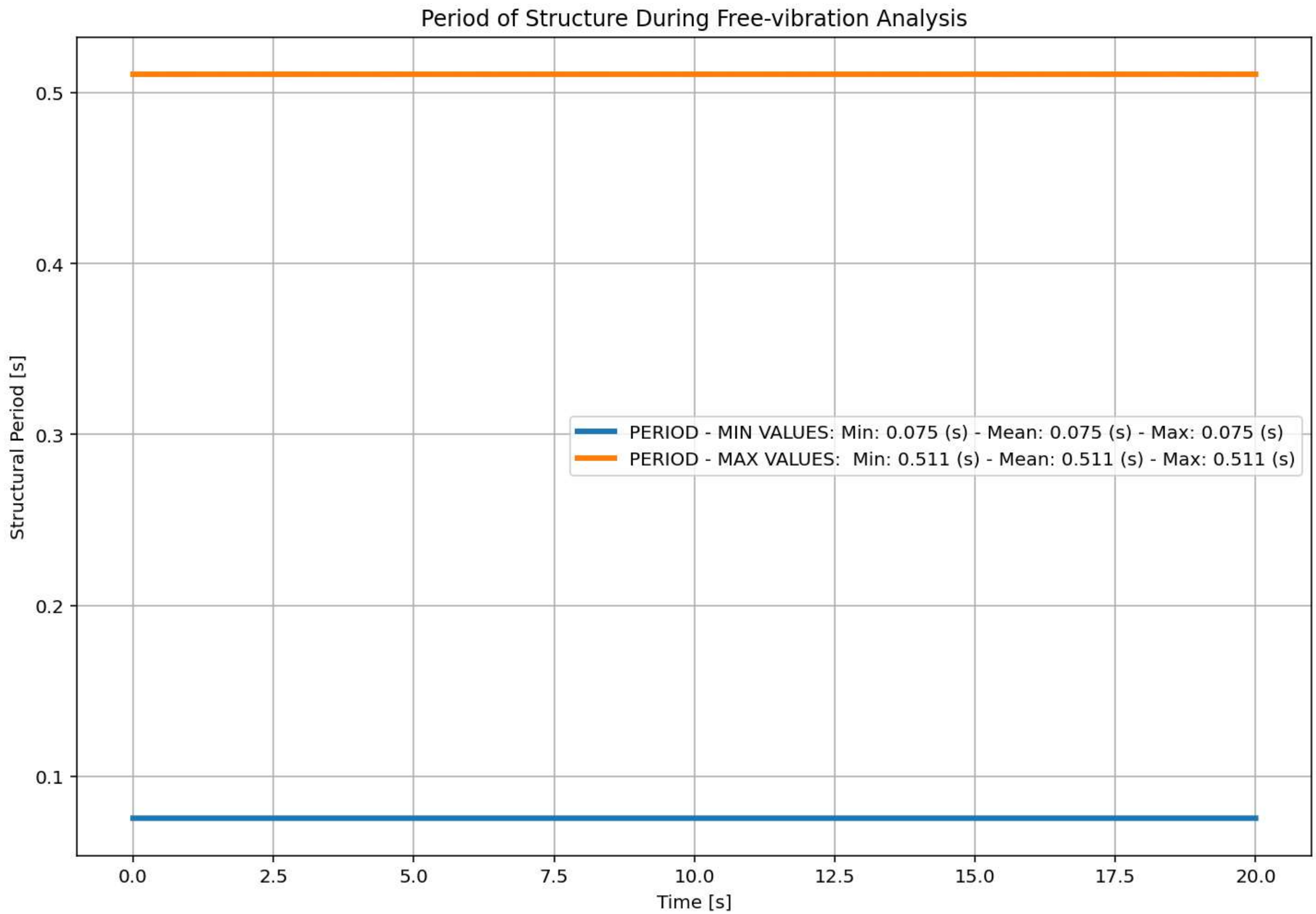


elements force in Z Dir. vs Time for Beam Element During Free-vibration Analysis
1e7
8
6
4
2
0
−2
−4
−6
−8
Element Moment Force [N.mm]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
Ele. 1 - MAX. ABS. : 8.5253e-08
Ele. 2 - MAX. ABS. : 4.5581e+07
Ele. 3 - MAX. ABS. : 4.9631e+07
Ele. 4 - MAX. ABS. : 7.4306e+07
Ele. 5 - MAX. ABS. : 4.9631e+07
Ele. 6 - MAX. ABS. : 4.5581e+07

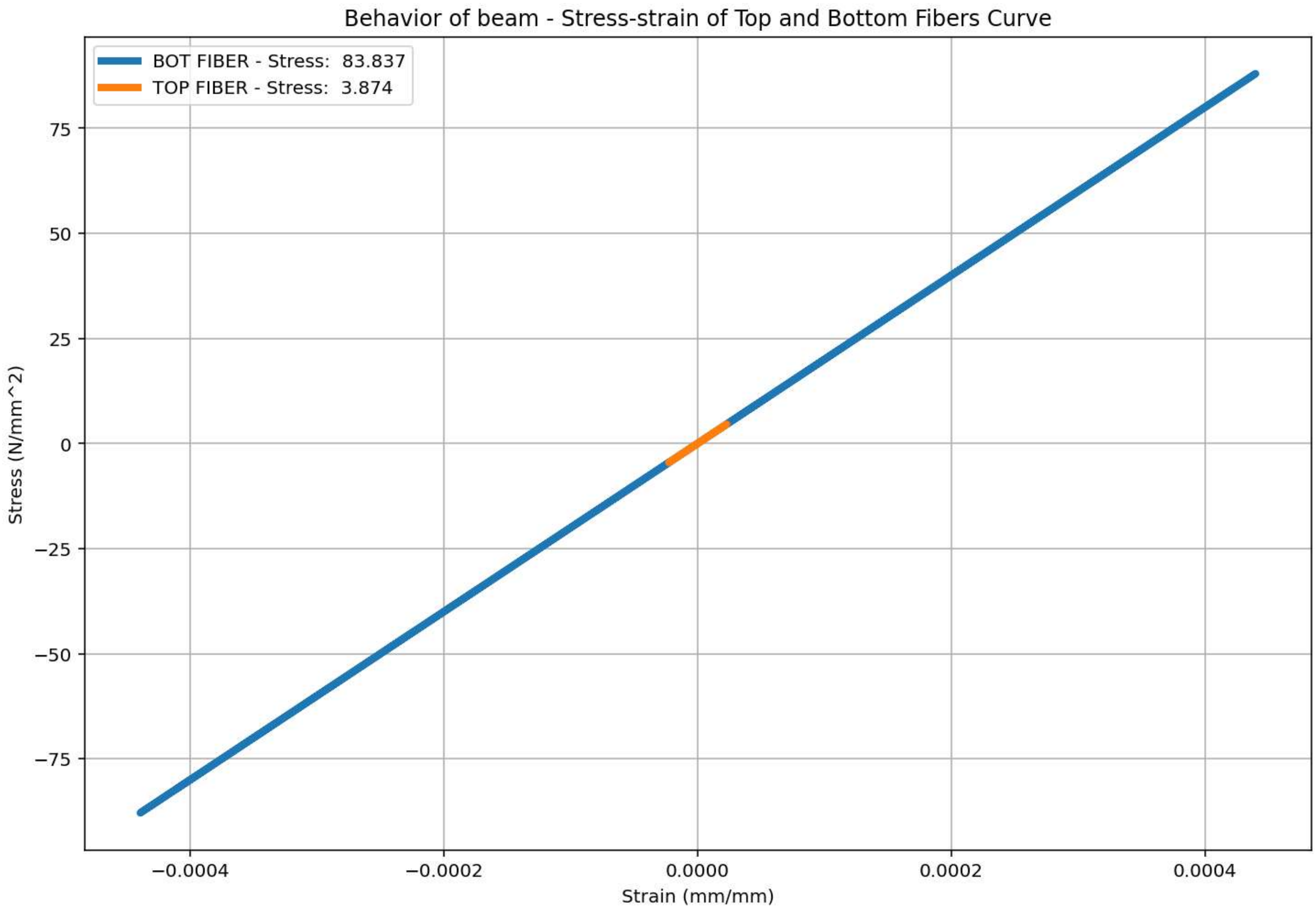

Node Displacements in X Dir. vs Time for Beam Element During Free-vibration Analysis
Displacement [mm]
Time [s]
0.0010
0.0005
0.0000
−0.0005
−0.0010
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 1.0938e-03
Node 3 - MAX. ABS. : 6.7191e-04
Node 4 - MAX. ABS. : 9.0237e-04
Node 5 - MAX. ABS. : 1.0812e-03
Node 6 - MAX. ABS. : 4.5902e-04
Node 7 - MAX. ABS. : 1.3095e-03

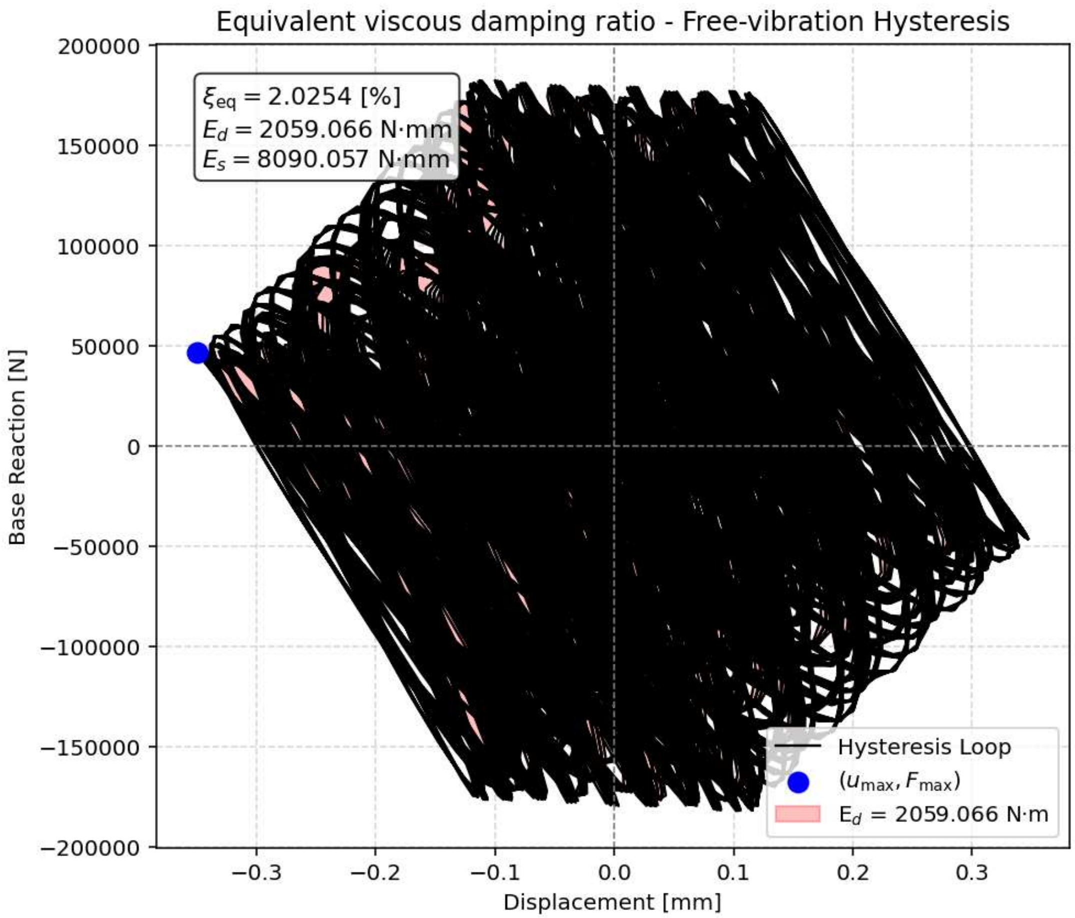


Node Displacements in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Displacement [mm]
Time [s]
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 2.3072e-01
Node 3 - MAX. ABS. : 2.0144e-01
Node 4 - MAX. ABS. : 3.4932e-01
Node 5 - MAX. ABS. : 2.0144e-01
Node 6 - MAX. ABS. : 2.3072e-01
Node 7 - MAX. ABS. : 0.0000e+00

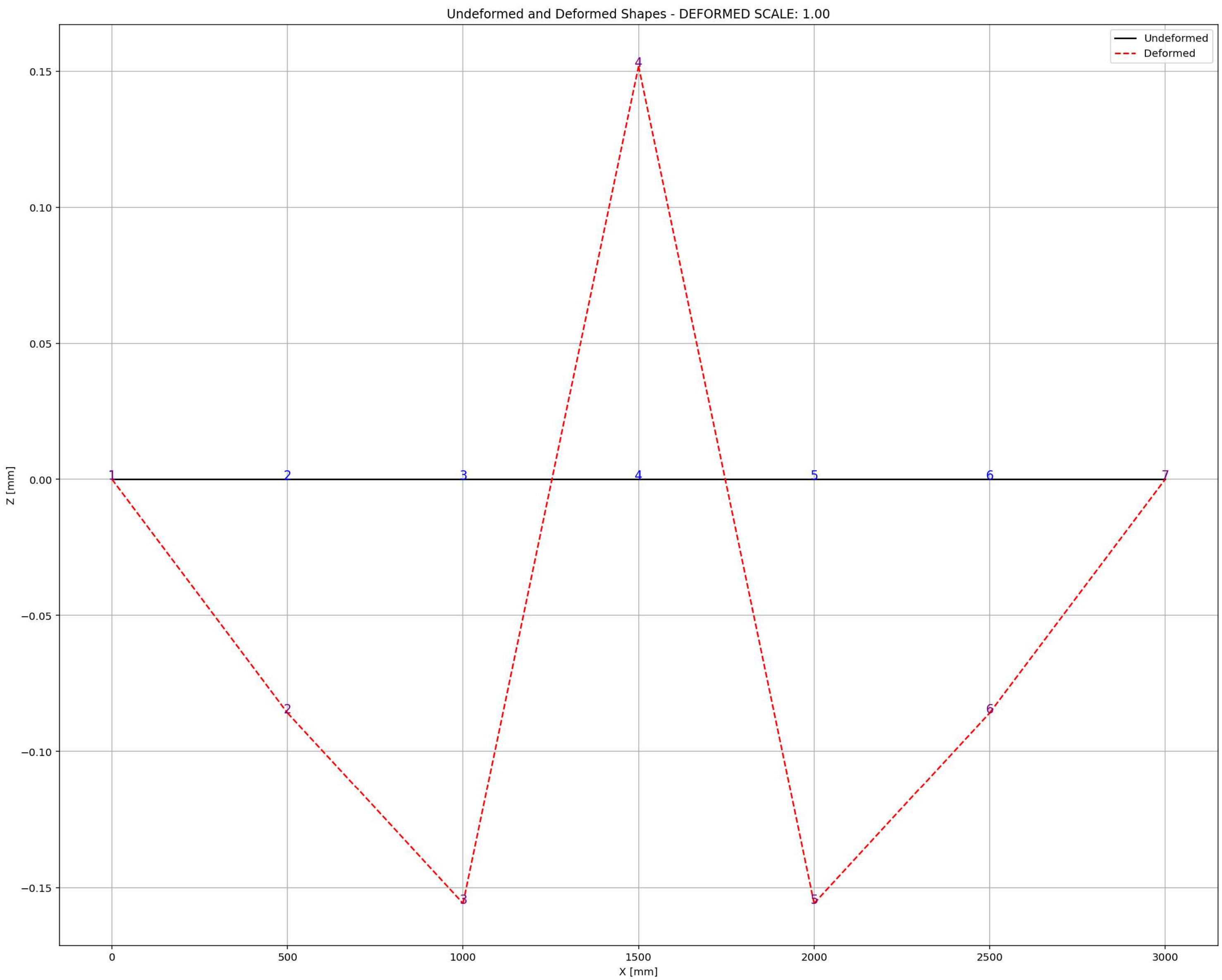


Period of Structure During Free-vibration Analysis
Structural Period [s]
0.1
0.2
0.3
0.4
0.5
Time [s]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
PERIOD - MIN VALUES: Min: 0.075 (s) - Mean: 0.075 (s) - Max: 0.075 (s)
PERIOD - MAX VALUES: Min: 0.511 (s) - Mean: 0.511 (s) - Max: 0.511 (s)


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 83.837
TOP FIBER - Stress: 3.874
75
50
25
0
−25
−50
−75
Stress (N/mm^2)
−0.0004
−0.0002
0.0000
0.0002
0.0004
Strain (mm/mm)

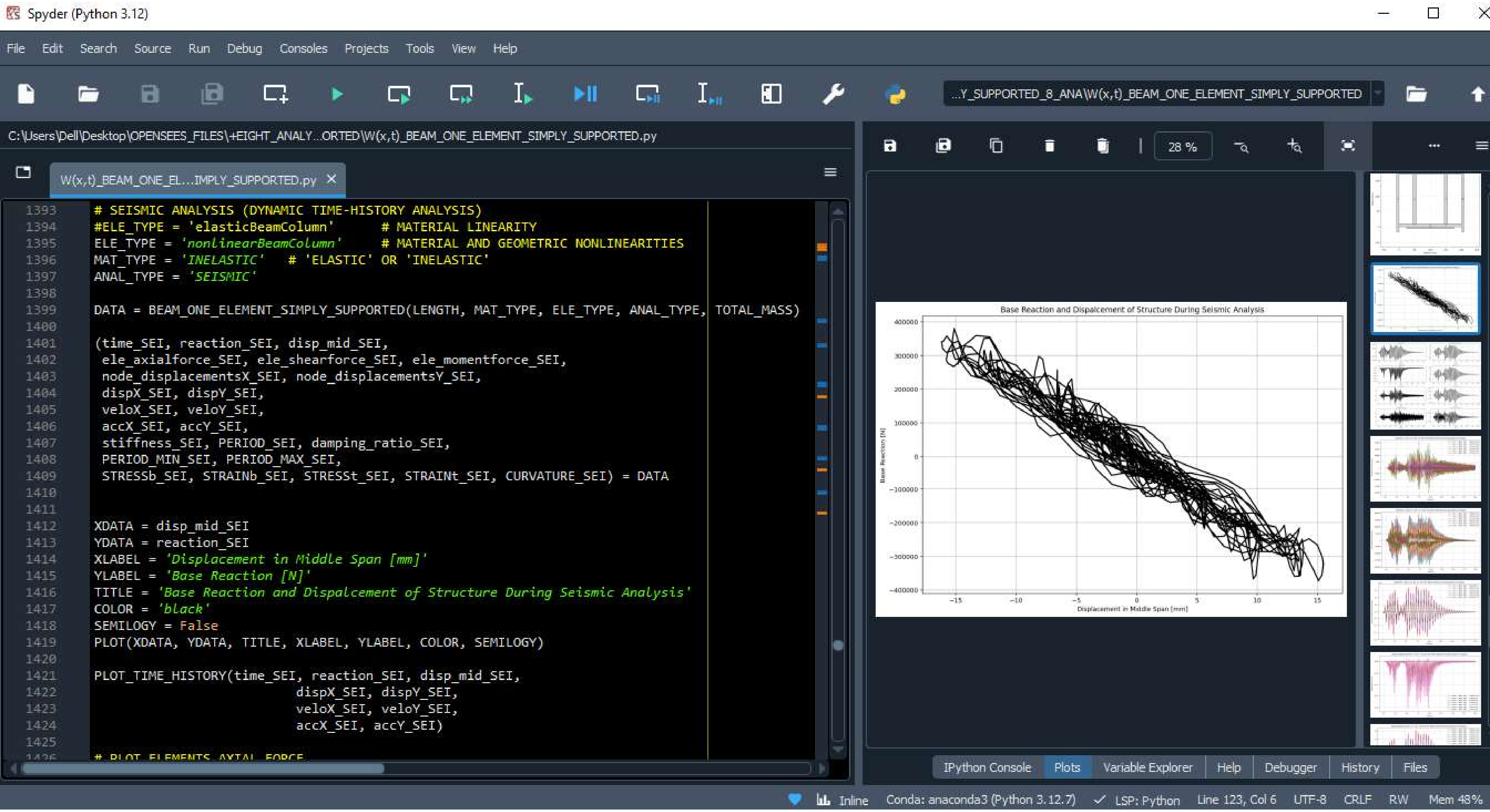


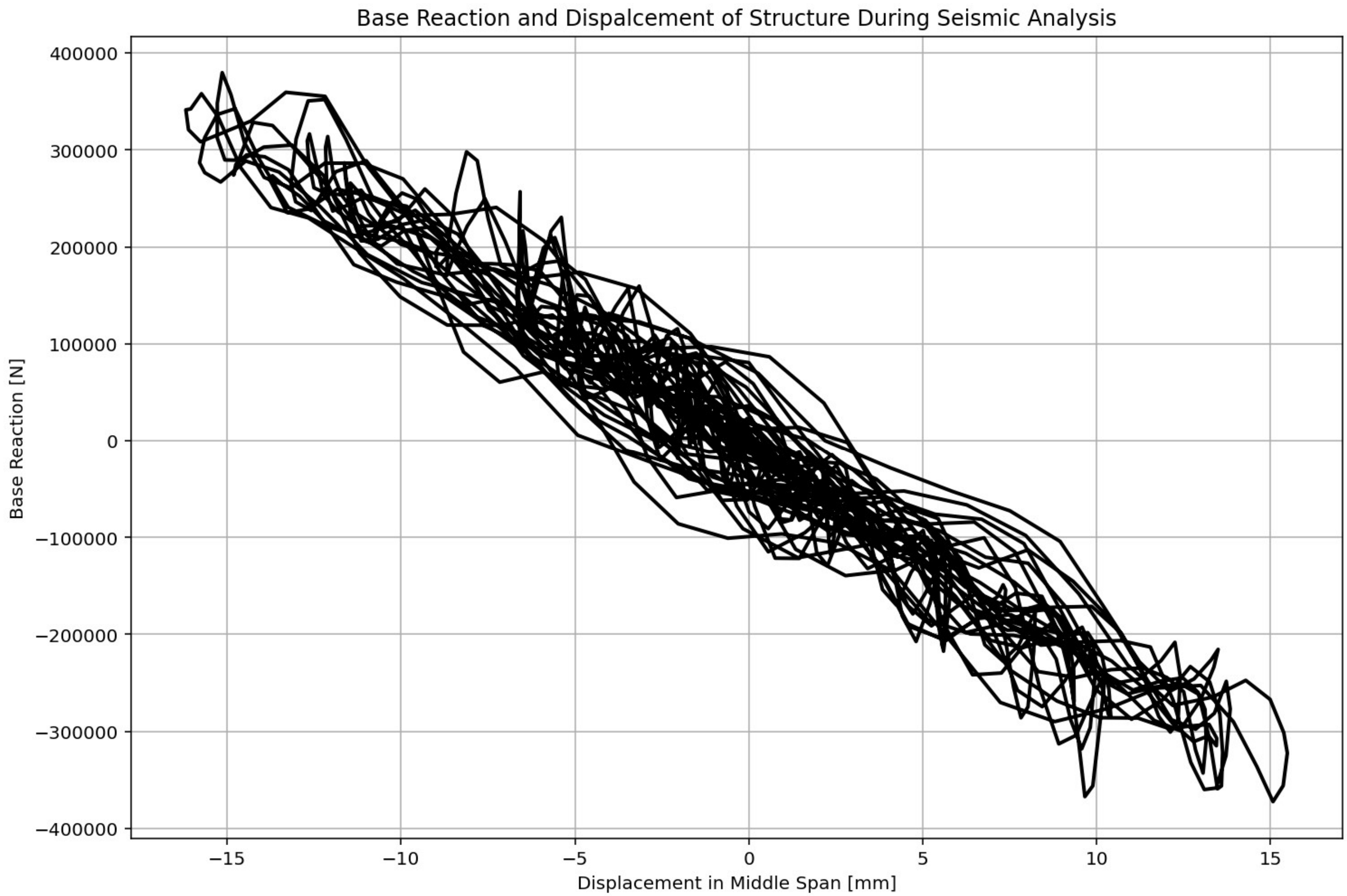
Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 2.0254$ [%]
$E_d = 2059.066$ N·mm
$E_s = 8090.057$ N·mm
Hysteresis Loop
(u_{max}, F_{max})
E_d = 2059.066 N·m
Base Reaction [N]
Displacement [mm]
200000
150000
100000
50000
0
−50000
−100000
−150000
−200000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3



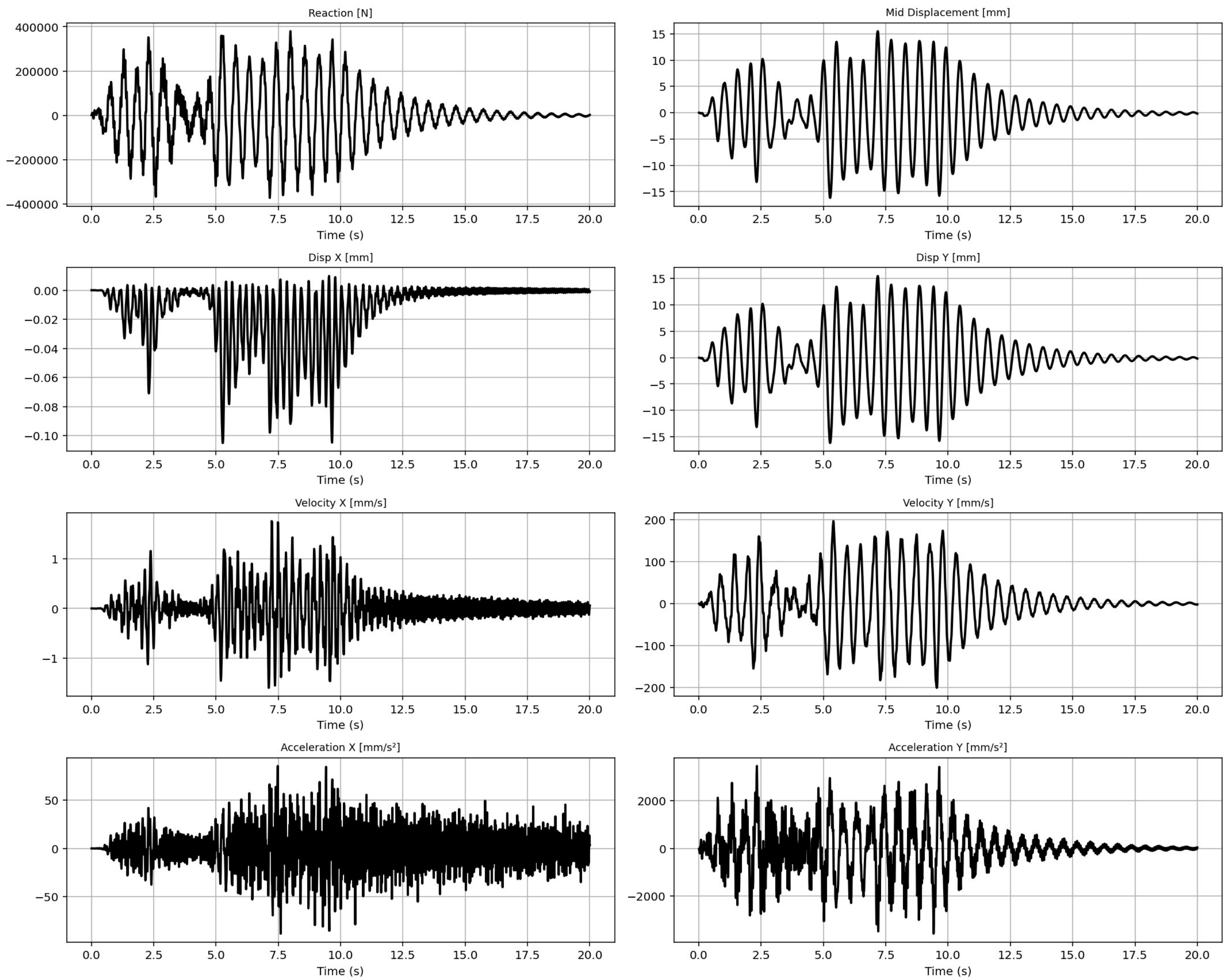
Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
500
1000
1500
2000
2500
3000
1
2
3
4
5
6
7

SEISMIC ANALYSIS

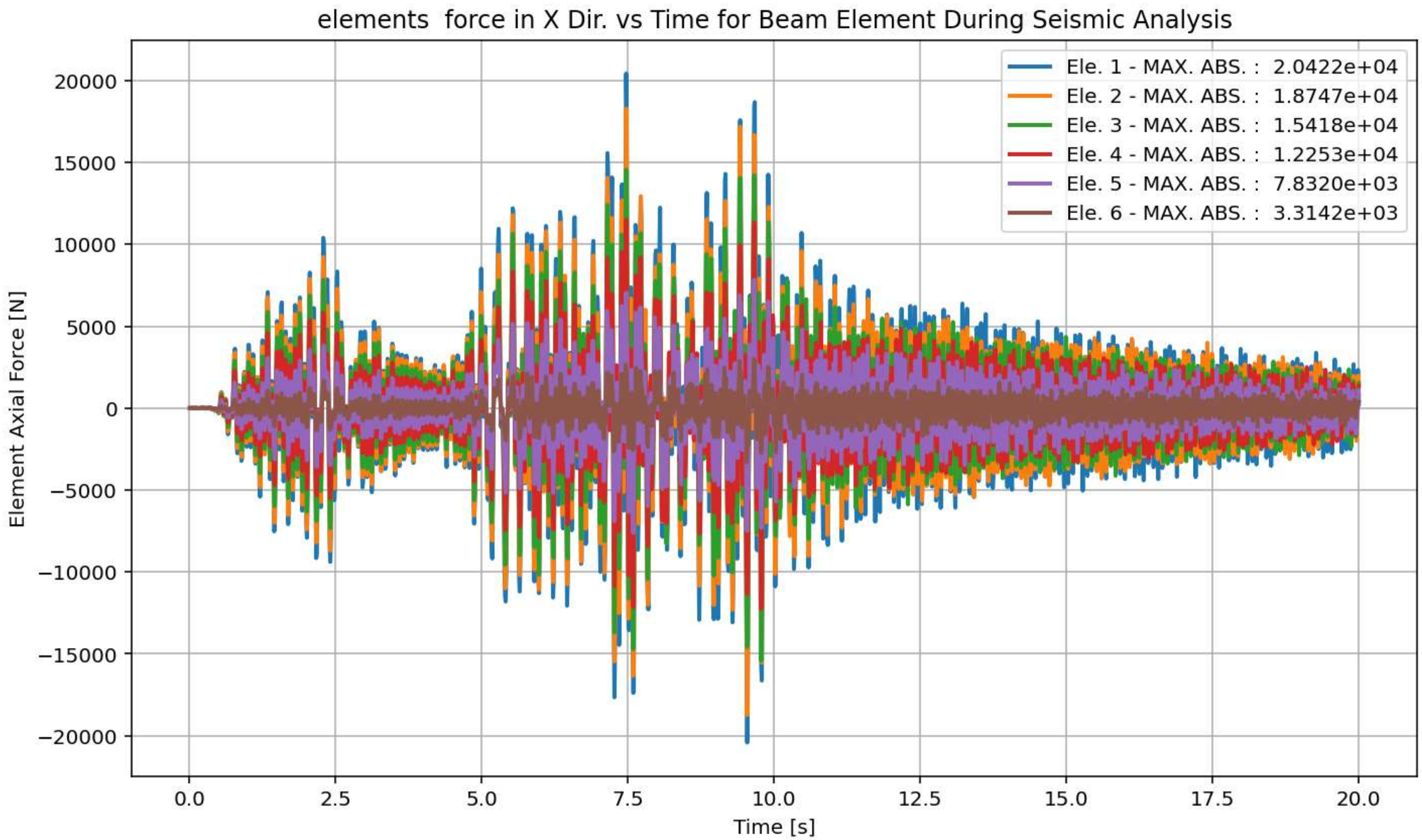

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...Y_SUPPORTED_8_ANA\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...ORTED\W(x,t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED.py
W(x,t)_BEAM_ONE_EL...IMPLY_SUPPORTED.py
SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'elasticBeamColumn'   # MATERIAL LINEARITY
ELE_TYPE = 'nonlinearBeamColumn'   # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'SEISMIC'
DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(time_SEI, reaction_SEI, disp_mid_SEI,
ele_axialforce_SEI, ele_shearforce_SEI, ele_momentforce_SEI,
node_displacementsX_SEI, node_displacementsY_SEI,
dispX_SEI, dispY_SEI,
veloX_SEI, veloY_SEI,
accX_SEI, accY_SEI,
stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
PERIOD_MIN_SEI, PERIOD_MAX_SEI,
STRESSb_SEI, STRAINb_SEI, STRESSt_SEI, STRAINt_SEI, CURVATURE_SEI) = DATA
XDATA = disp_mid_SEI
YDATA = reaction_SEI
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Dispalcement of Structure During Seismic Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_SEI, reaction_SEI, disp_mid_SEI,
dispX_SEI, dispY_SEI,
veloX_SEI, veloY_SEI,
accX_SEI, accY_SEI)
PLOT ELEMENTS AXIAL FORCE
28 %
Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 123, Col 6 UTF-8 CRLF RW Mem 48%



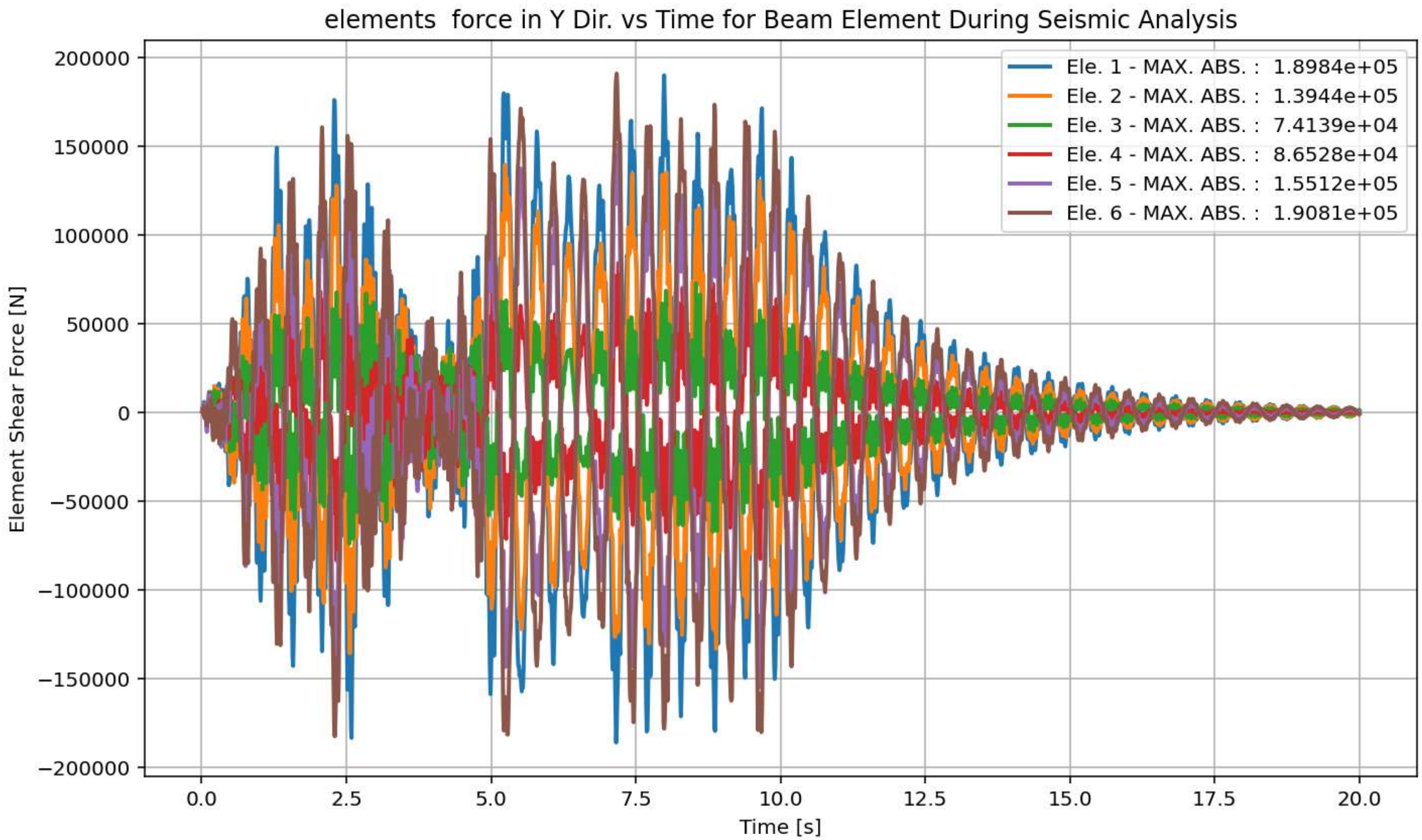
Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
400000
300000
200000
100000
0
−100000
−200000
−300000
−400000
−15
−10
−5
0
5
10
15
Displacement in Middle Span [mm]



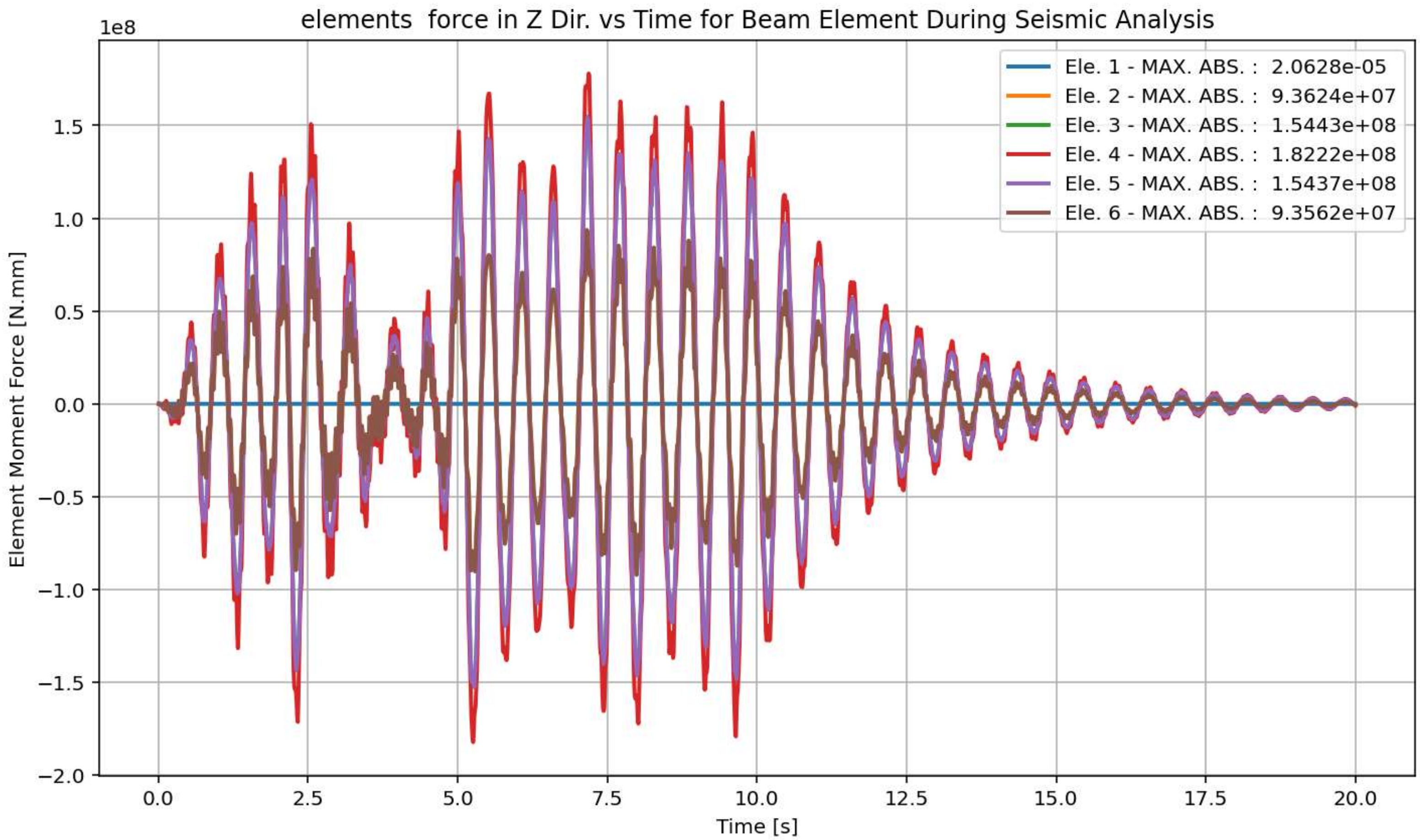
Reaction [N]
Mid Displacement [mm]
Disp X [mm]
Disp Y [mm]
Velocity X [mm/s]
Velocity Y [mm/s]
Acceleration X [mm/s²]
Acceleration Y [mm/s²]
Time (s)

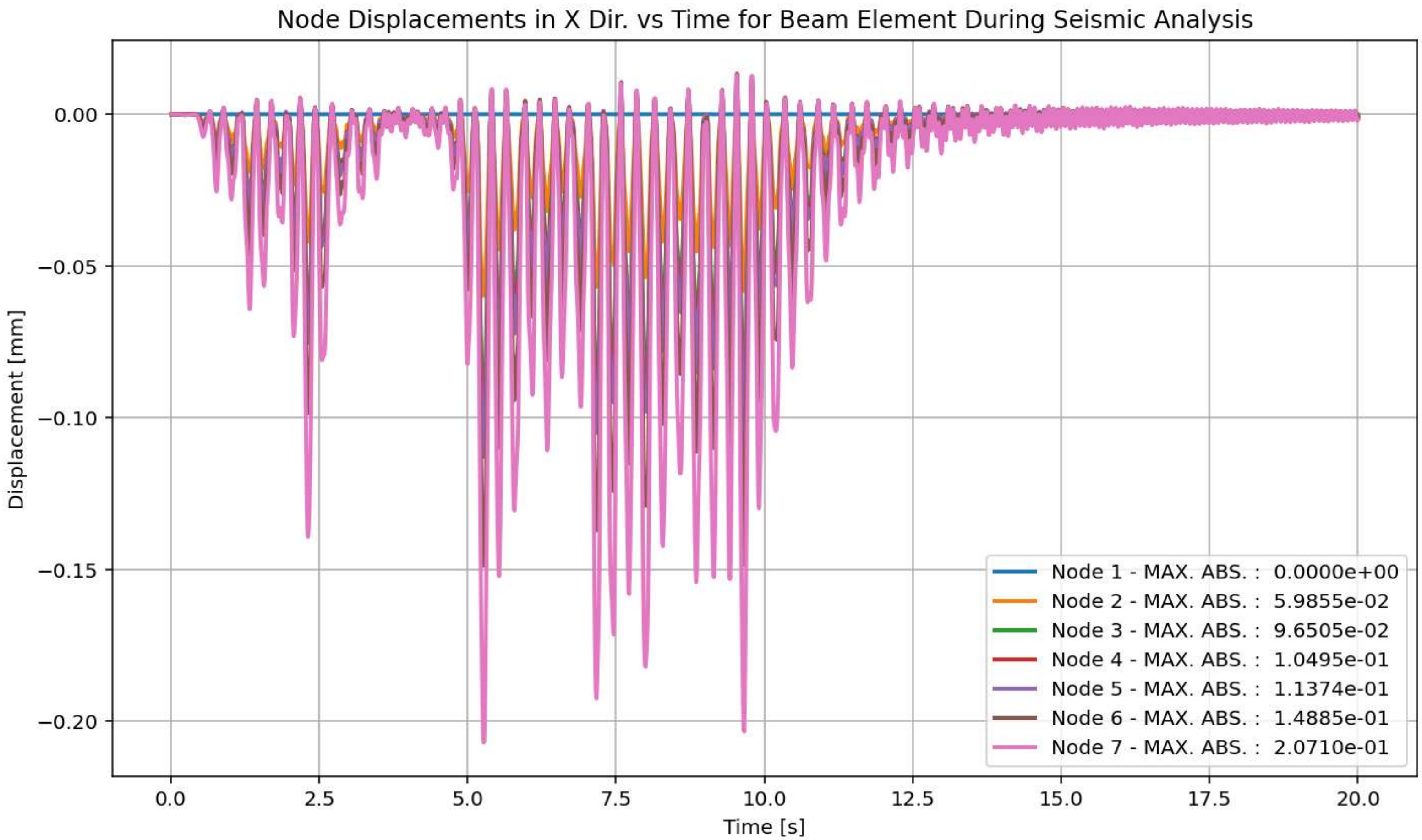


elements force in X Dir. vs Time for Beam Element During Seismic Analysis
Ele. 1 - MAX. ABS. : 2.0422e+04
Ele. 2 - MAX. ABS. : 1.8747e+04
Ele. 3 - MAX. ABS. : 1.5418e+04
Ele. 4 - MAX. ABS. : 1.2253e+04
Ele. 5 - MAX. ABS. : 7.8320e+03
Ele. 6 - MAX. ABS. : 3.3142e+03
Element Axial Force [N]
Time [s]
20000
15000
10000
5000
0
−5000
−10000
−15000
−20000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

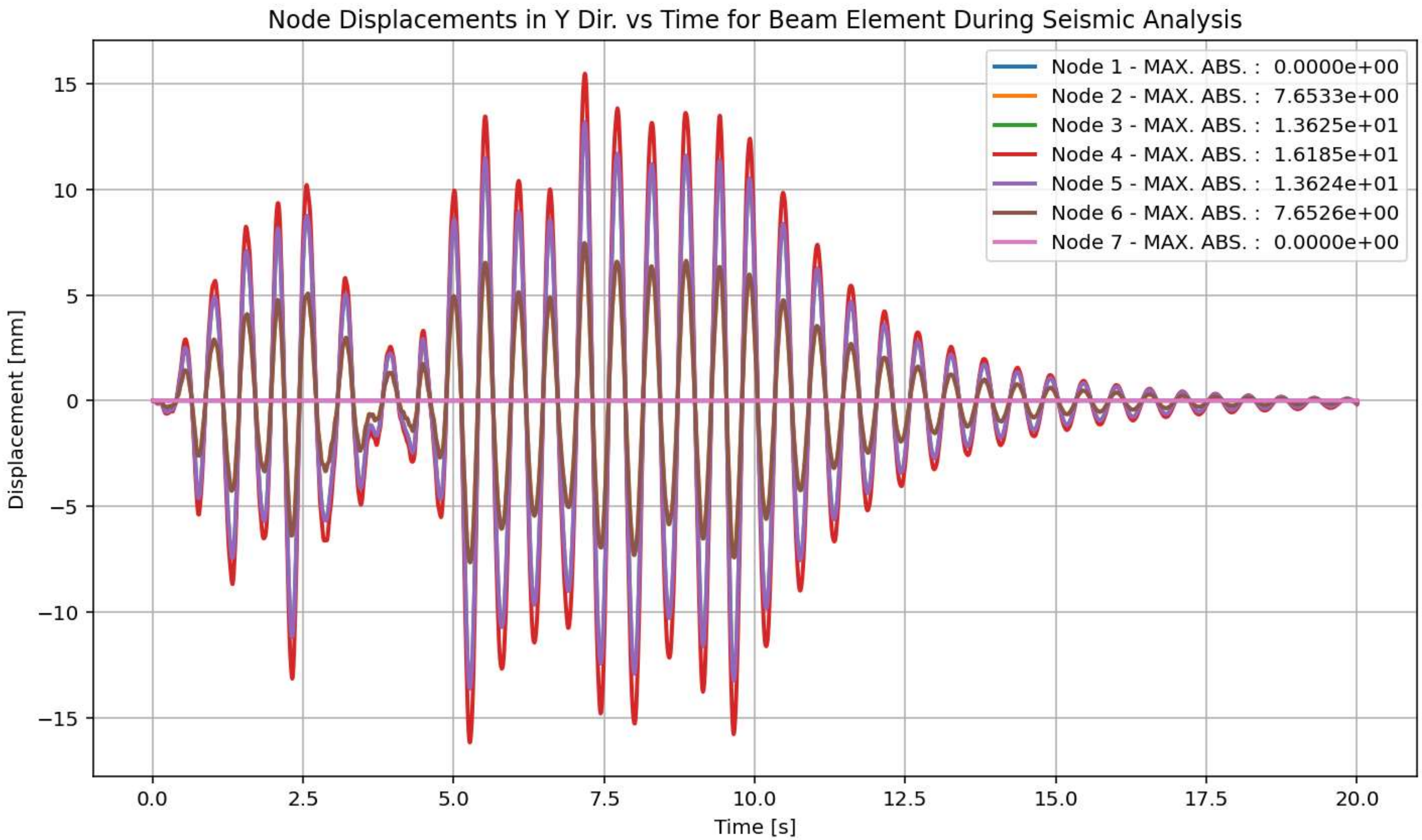


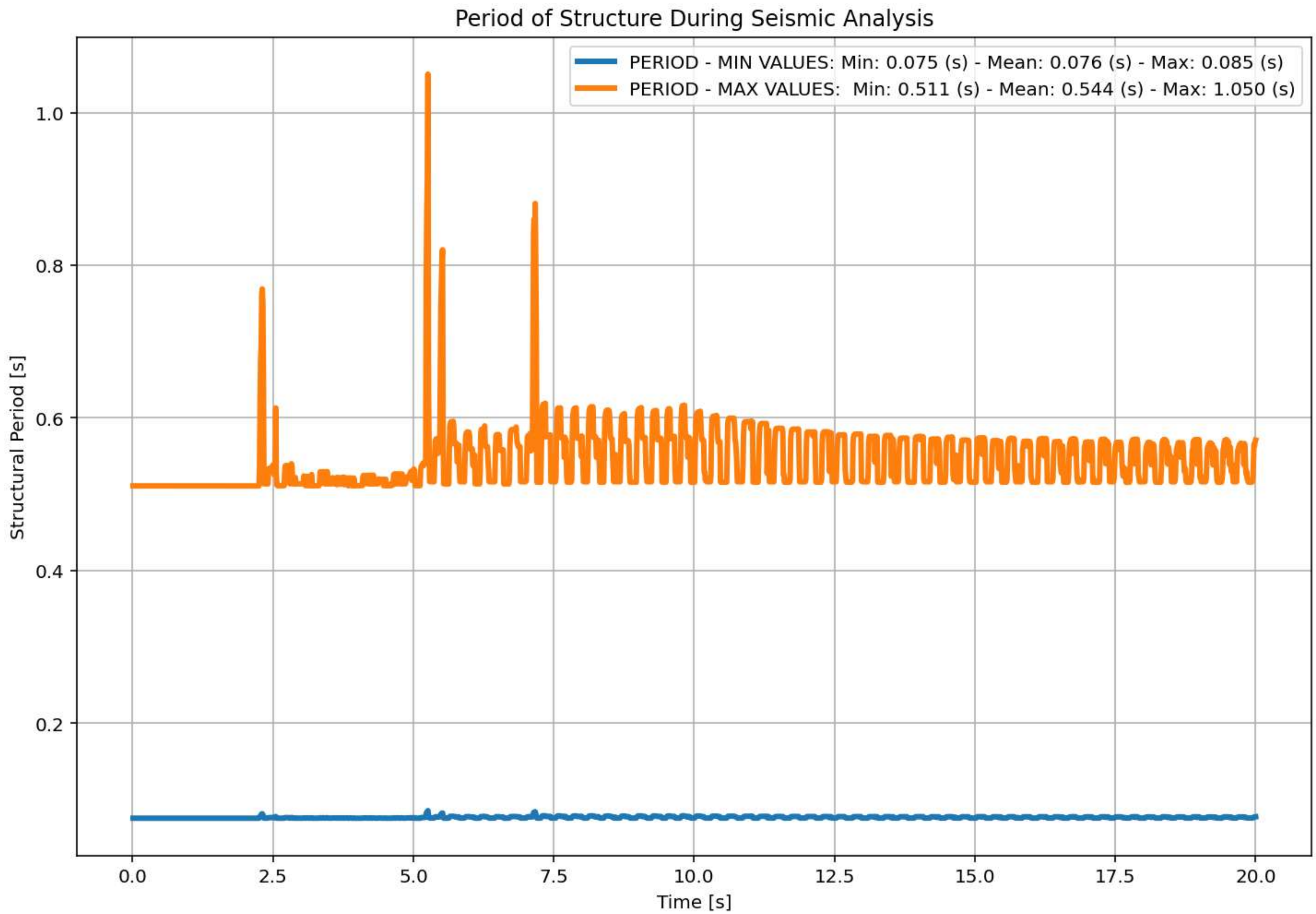
elements force in Y Dir. vs Time for Beam Element During Seismic Analysis
Ele. 1 - MAX. ABS. : 1.8984e+05
Ele. 2 - MAX. ABS. : 1.3944e+05
Ele. 3 - MAX. ABS. : 7.4139e+04
Ele. 4 - MAX. ABS. : 8.6528e+04
Ele. 5 - MAX. ABS. : 1.5512e+05
Ele. 6 - MAX. ABS. : 1.9081e+05
Element Shear Force [N]
Time [s]
200000
150000
100000
50000
0
−50000
−100000
−150000
−200000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

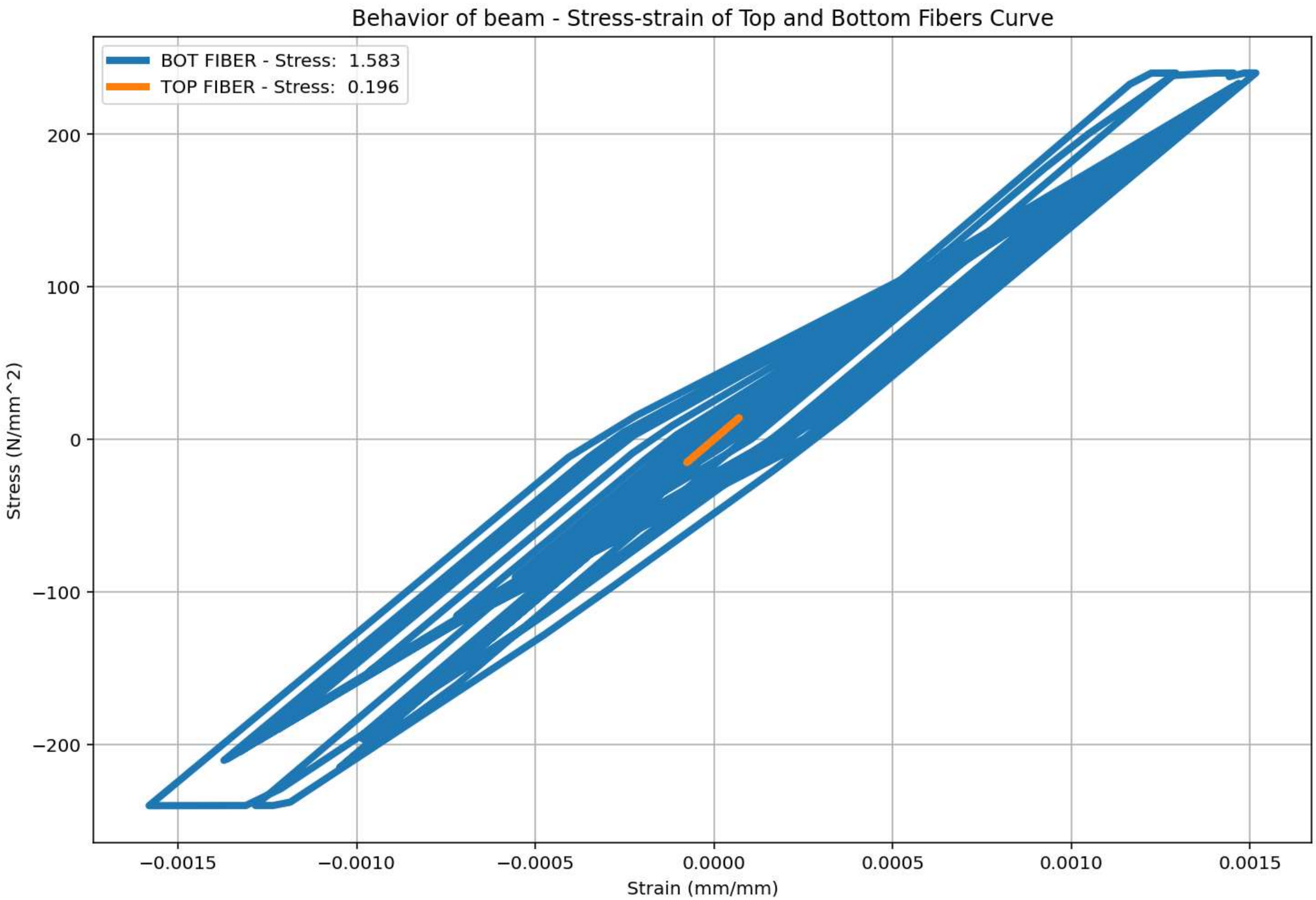

elements force in Z Dir. vs Time for Beam Element During Seismic Analysis
1e8
Ele. 1 - MAX. ABS. : 2.0628e-05
Ele. 2 - MAX. ABS. : 9.3624e+07
Ele. 3 - MAX. ABS. : 1.5443e+08
Ele. 4 - MAX. ABS. : 1.8222e+08
Ele. 5 - MAX. ABS. : 1.5437e+08
Ele. 6 - MAX. ABS. : 9.3562e+07
Element Moment Force [N.mm]
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

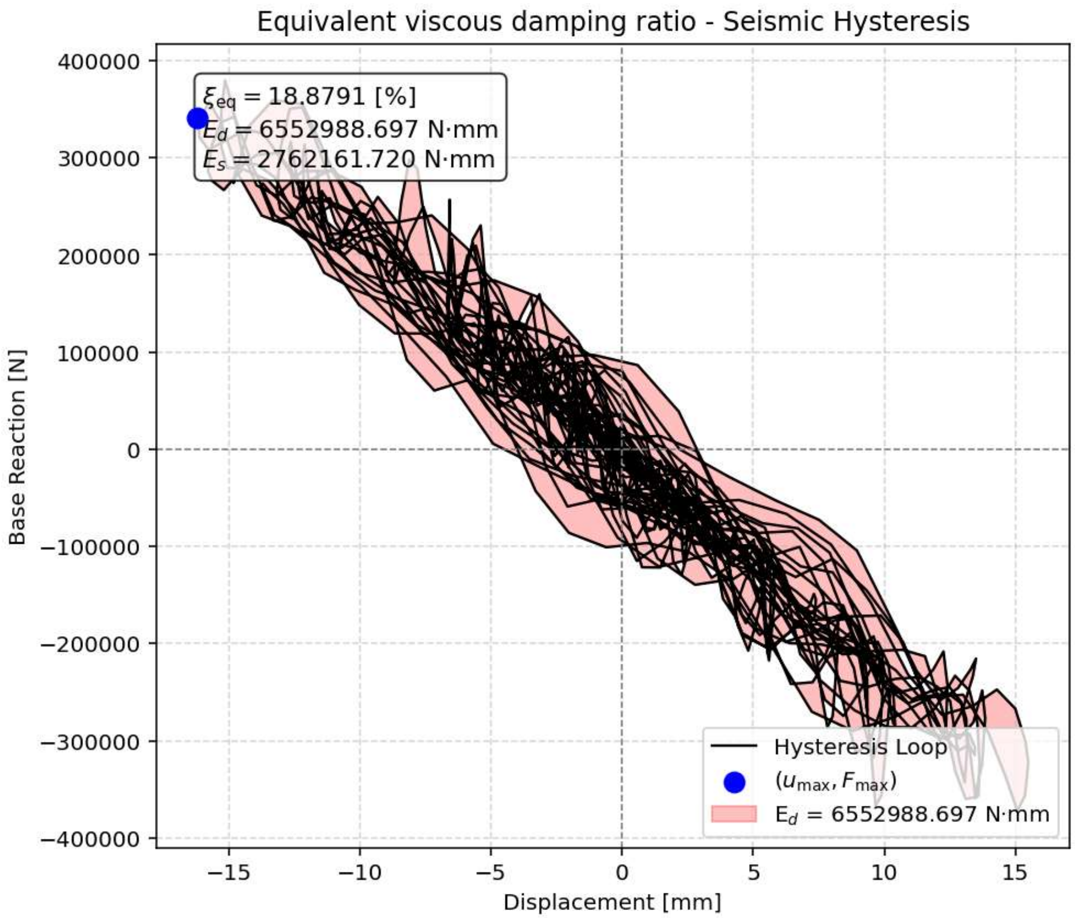


Node Displacements in X Dir. vs Time for Beam Element During Seismic Analysis
Displacement [mm]
Time [s]
0.00
−0.05
−0.10
−0.15
−0.20
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 5.9855e-02
Node 3 - MAX. ABS. : 9.6505e-02
Node 4 - MAX. ABS. : 1.0495e-01
Node 5 - MAX. ABS. : 1.1374e-01
Node 6 - MAX. ABS. : 1.4885e-01
Node 7 - MAX. ABS. : 2.0710e-01

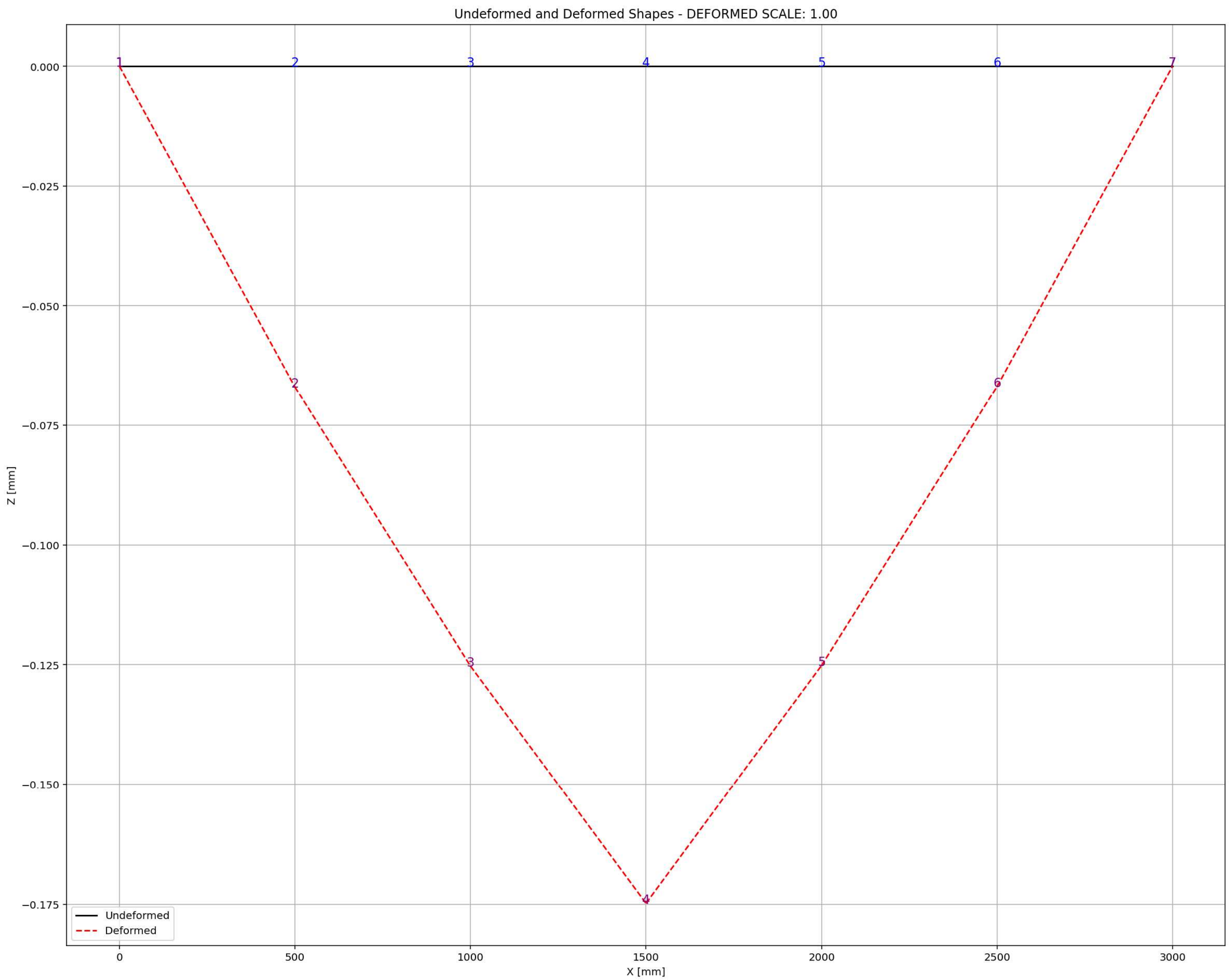

Node Displacements in Y Dir. vs Time for Beam Element During Seismic Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 7.6533e+00
Node 3 - MAX. ABS. : 1.3625e+01
Node 4 - MAX. ABS. : 1.6185e+01
Node 5 - MAX. ABS. : 1.3624e+01
Node 6 - MAX. ABS. : 7.6526e+00
Node 7 - MAX. ABS. : 0.0000e+00
Displacement [mm]
Time [s]
15
10
5
0
−5
−10
−15
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Period of Structure During Seismic Analysis
PERIOD - MIN VALUES: Min: 0.075 (s) - Mean: 0.076 (s) - Max: 0.085 (s)
PERIOD - MAX VALUES: Min: 0.511 (s) - Mean: 0.544 (s) - Max: 1.050 (s)
Structural Period [s]
Time [s]
0.2
0.4
0.6
0.8
1.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

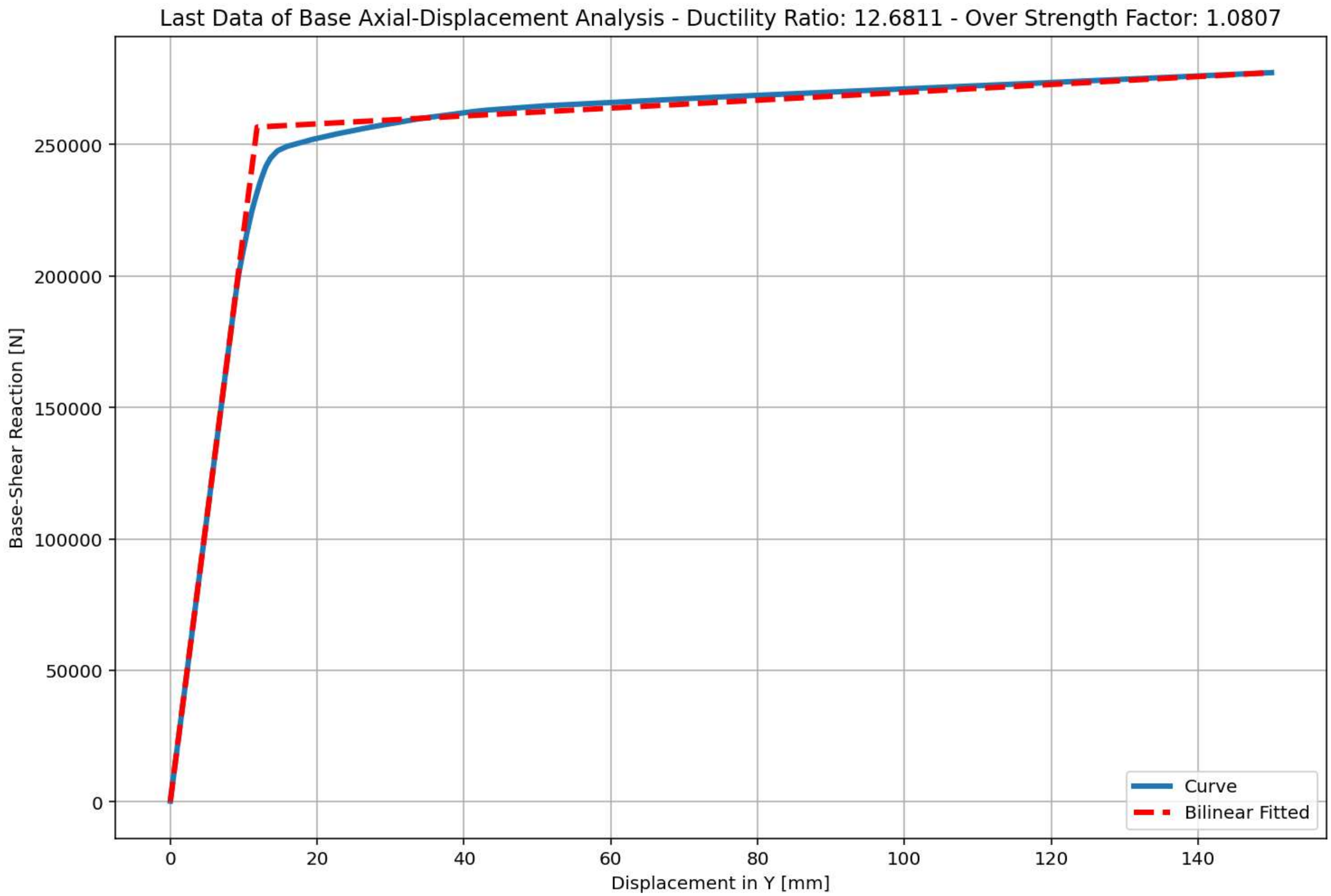

Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 1.583
TOP FIBER - Stress: 0.196
200
100
0
−100
−200
Stress (N/mm^2)
−0.0015
−0.0010
−0.0005
0.0000
0.0005
0.0010
0.0015
Strain (mm/mm)



Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 18.8791$ [%]
$E_d = 6552988.697$ N·mm
$E_s = 2762161.720$ N·mm
400000
300000
200000
100000
0
−100000
−200000
−300000
−400000
Base Reaction [N]
−15
−10
−5
0
5
10
15
Displacement [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 6552988.697 N·mm

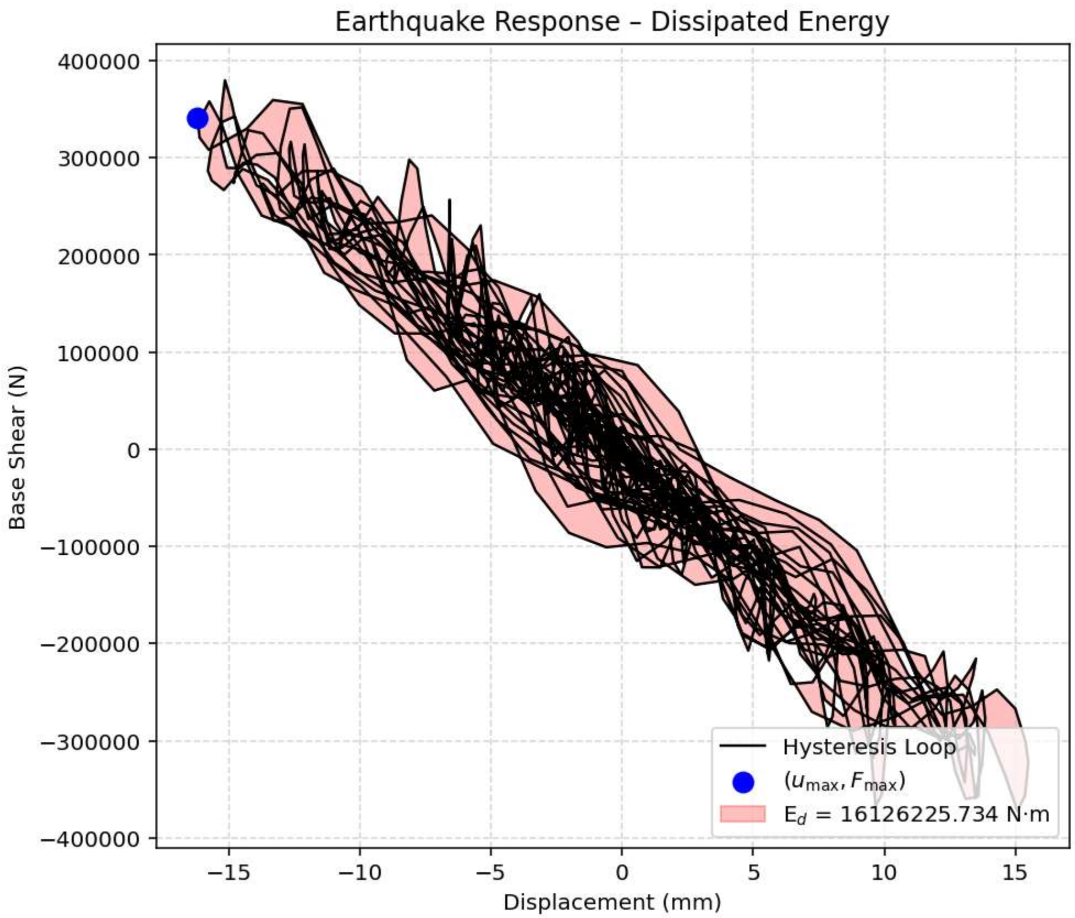


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
−0.150
−0.175
Z [mm]
0
500
1000
1500
2000
2500
3000
X [mm]
1
2
3
4
5
6
7
Undeformed
Deformed

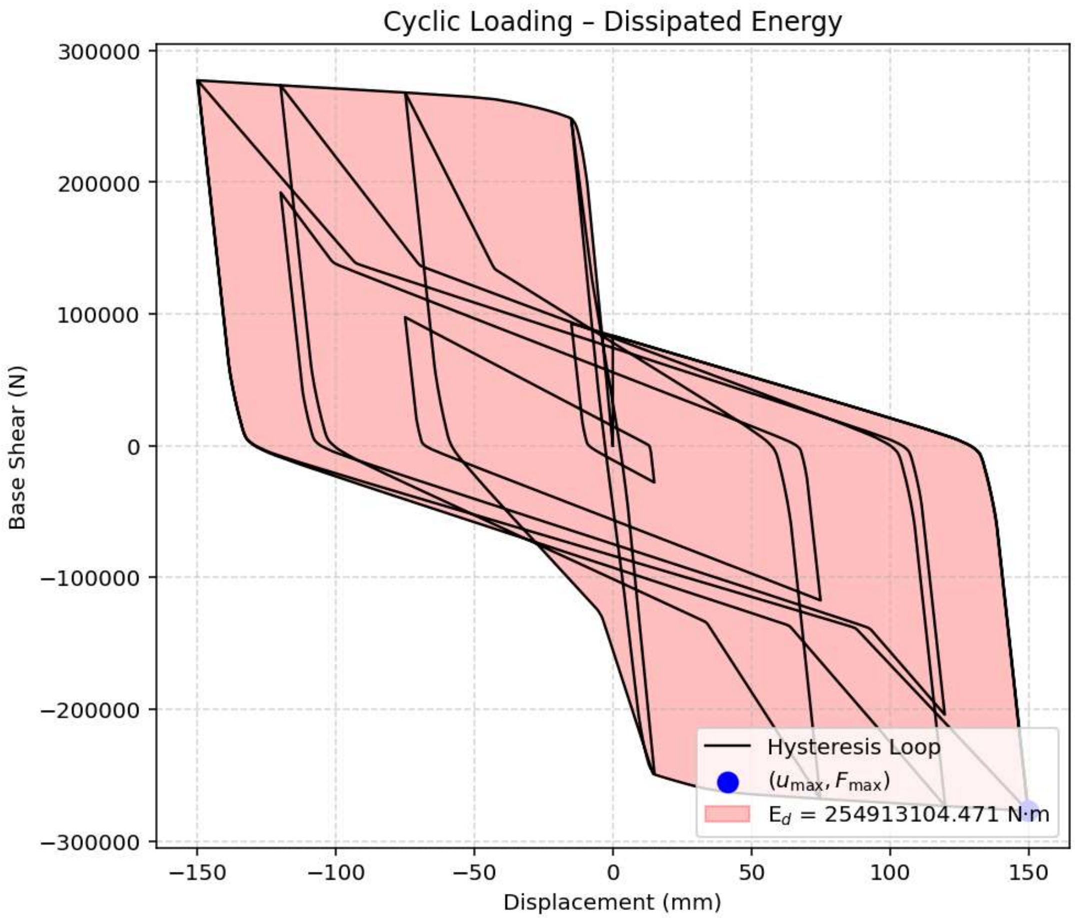


Last Data of Base Axial-Displacement Analysis - Ductility Ratio: 12.6811 - Over Strength Factor: 1.0807
Base-Shear Reaction [N]
0
50000
100000
150000
200000
250000
Displacement in Y [mm]
0
20
40
60
80
100
120
140
Curve
Bilinear Fitted

```
+=========================+
=   Analysis curve fitted =
    Disp       Base Shear
----------------------------
[[0.00000000e+00 0.00000000e+00]
 [1.18286124e+01 2.56663651e+05]
 [1.50000000e+02 2.77371080e+05]]
+=========================+
+----------------------------------------------------+
 Structure Elastic Stiffness :     21698.54
 Structure Plastic Stiffness :     1849.14
 Structure Tangent Stiffness :     149.87
 Structure Ductility Ratio :       12.68
 Structure Over Strength Factor:   1.08
+----------------------------------------------------+
Over Strength Coefficient (Ω0):      1.0807
Displacement Ductility Ratio (μ):    12.6811
Ductility Coefficient (Rμ):          12.6811
Structural Behavior Coefficient (R): 13.7042
Structural Ductility Damage Index in Y Direction: 3.1530 (%)
```

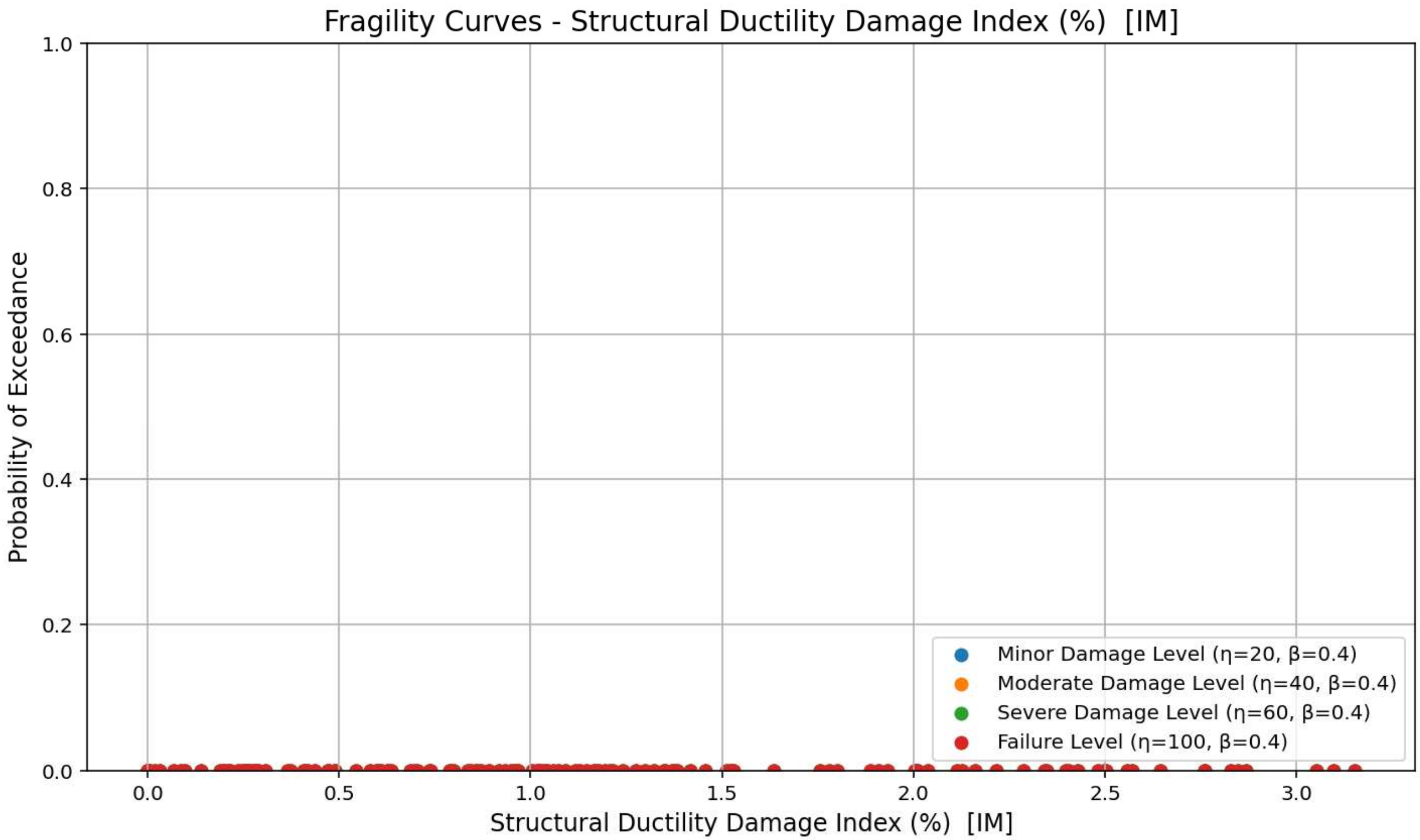


Earthquake Response – Dissipated Energy
400000
300000
200000
100000
0
−100000
−200000
−300000
−400000
Base Shear (N)
−15
−10
−5
0
5
10
15
Displacement (mm)
Hysteresis Loop
$(u_{\max}, F_{\max})$
E_d = 16126225.734 N·m


Cyclic Loading – Dissipated Energy
Base Shear (N)
Displacement (mm)
300000
200000
100000
0
−100000
−200000
−300000
−150
−100
−50
0
50
100
150
Hysteresis Loop
$(u_{\mathrm{max}}, F_{\mathrm{max}})$
E_d = 254913104.471 N·m

Dissipated Energy from Earthquake= 16126225.73 N·mm
Dissipated Energy from Cyclic Displacement= 254913104.47 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 6.326 [%]
ZONE 1: VERY MINOR DAMAGE



Fragility Curves - Structural Ductility Damage Index (%) [IM]
Probability of Exceedance
Structural Ductility Damage Index (%) [IM]
Minor Damage Level (η=20, β=0.4)
Moderate Damage Level (η=40, β=0.4)
Severe Damage Level (η=60, β=0.4)
Failure Level (η=100, β=0.4)
1.0
0.8
0.6
0.4
0.2
0.0
0.5
1.0
1.5
2.0
2.5
3.0